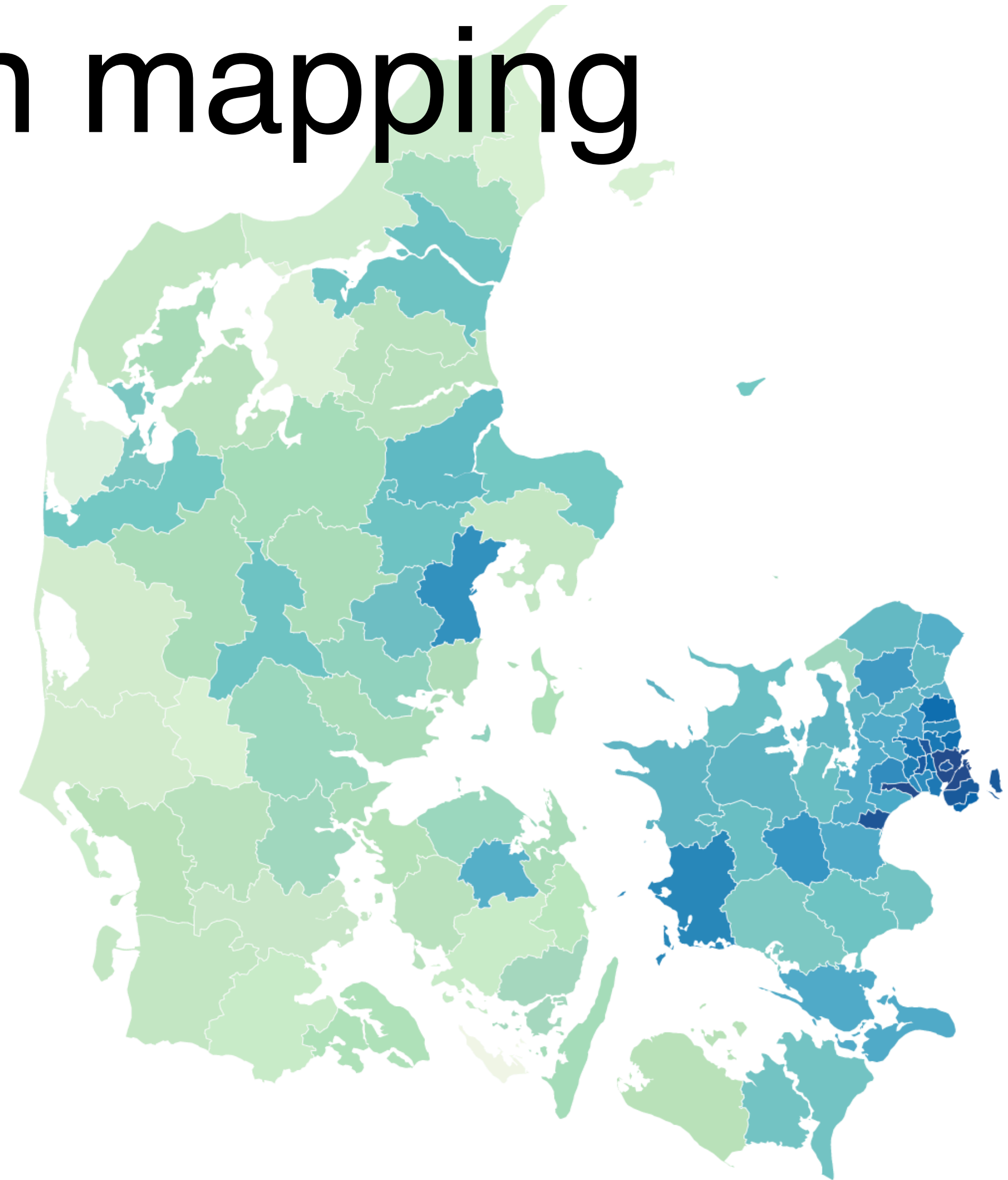


Lecture 3: Choropleth mapping

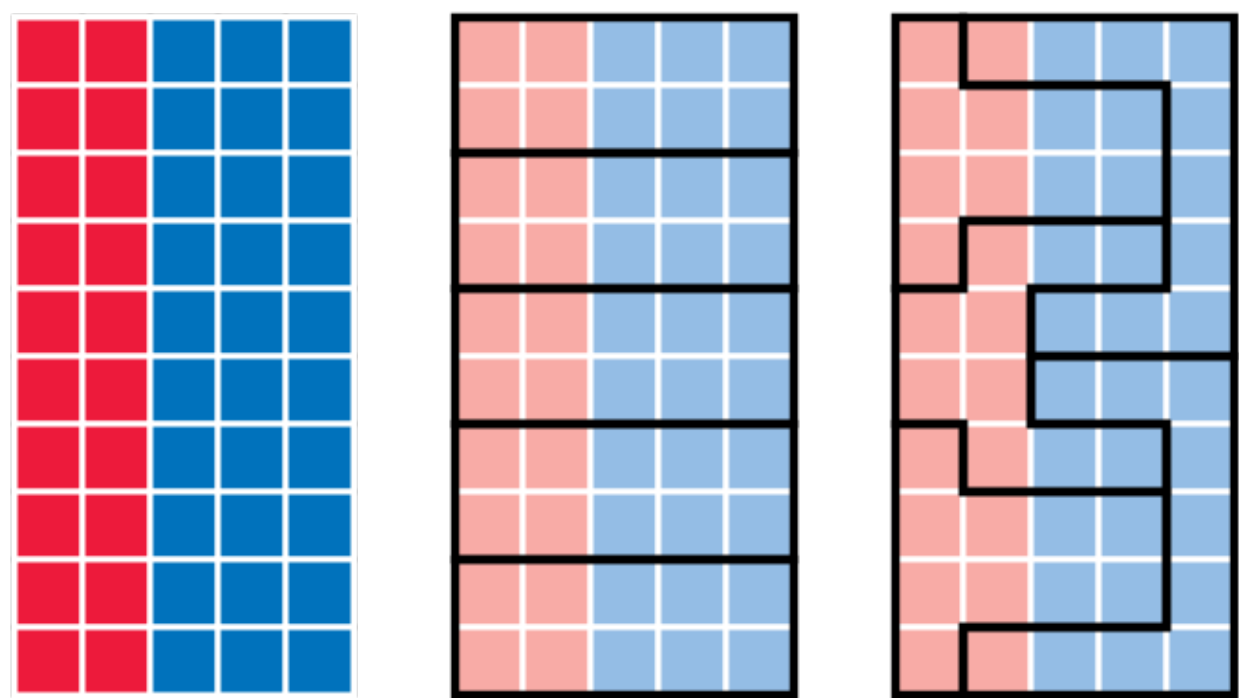
Instructor: Michael Szell

Feb 11, 2026



Today you will learn about MAUP and Choropleths

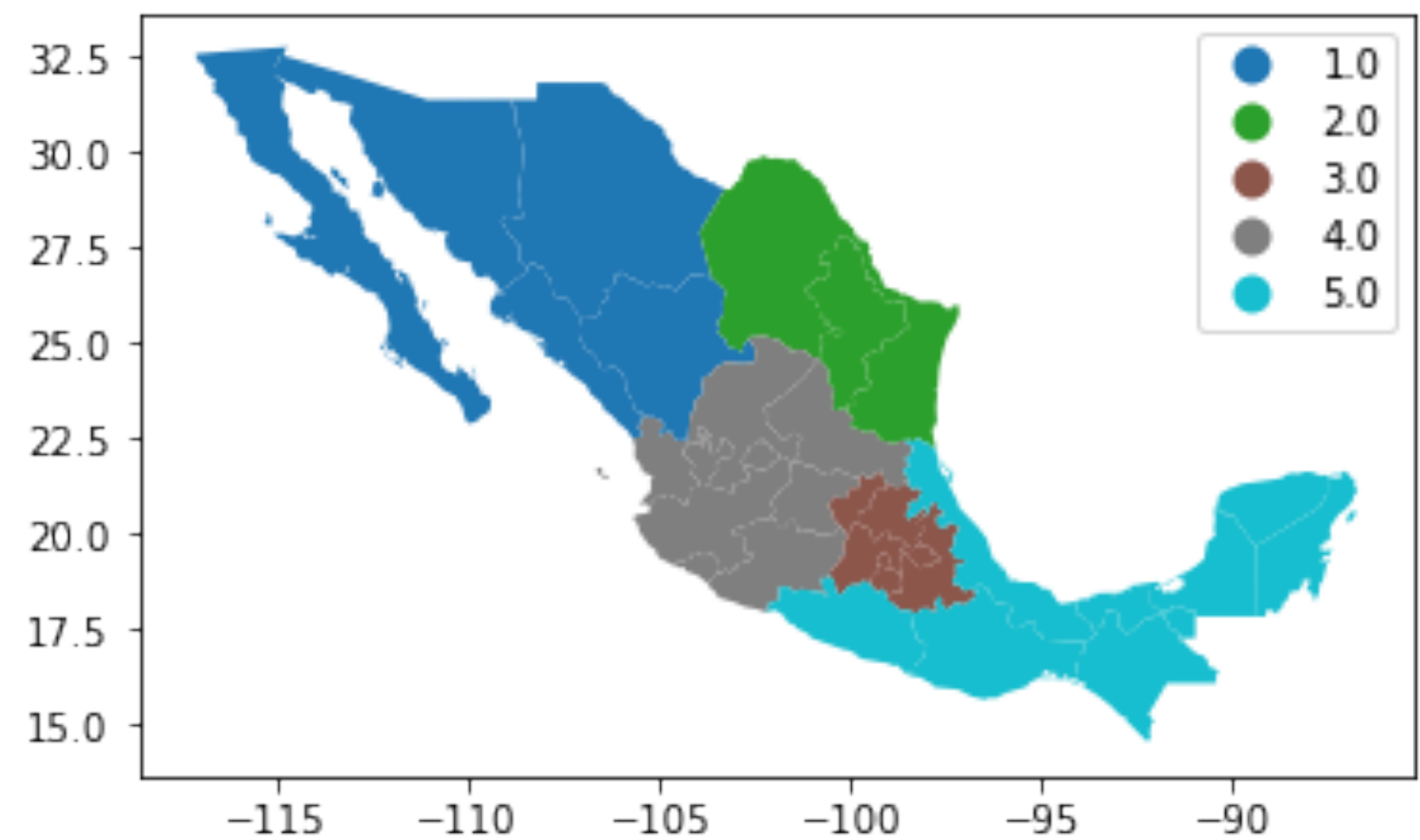
Biases from spatial aggregation (MAUP)



Classification schemes

$$c_j < y_i \leq c_{j+1} \quad \forall y_i \in C_j$$

Choropleths in Python



The most common steps in Data Preprocessing are:

Aggregation

Sampling

Dimensionality reduction

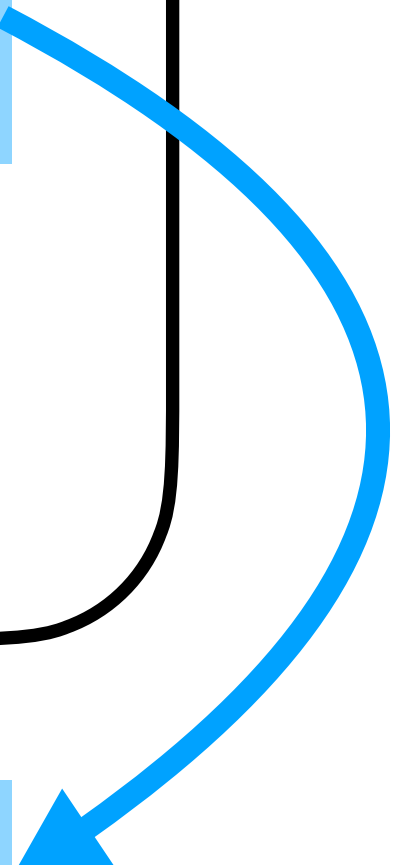
Discretization / Classification

Variable transformation

Aggregation = Combining objects into a single one

Student ID	Year	Grade Point Average (GPA)	...
⬢ 1034262	Senior	3.24	...
1052663	Sophomore	3.51	...
1082246	Freshman	3.62	...
⋮	⋮		

NULL	Non-Freshman	3.375	
------	--------------	-------	--



Aggregation = Combining objects into a single one

Examples:

GPS coordinate → Zip Code → City → Country

Second → Minute → Hour → Day → Week → Month → Year

Advantages: Data reduction, easier to process, high-level view, smaller statistical fluctuations

Disadvantages: Loss of details, introducing biases

Choropleth maps visualize geospatial data with colors

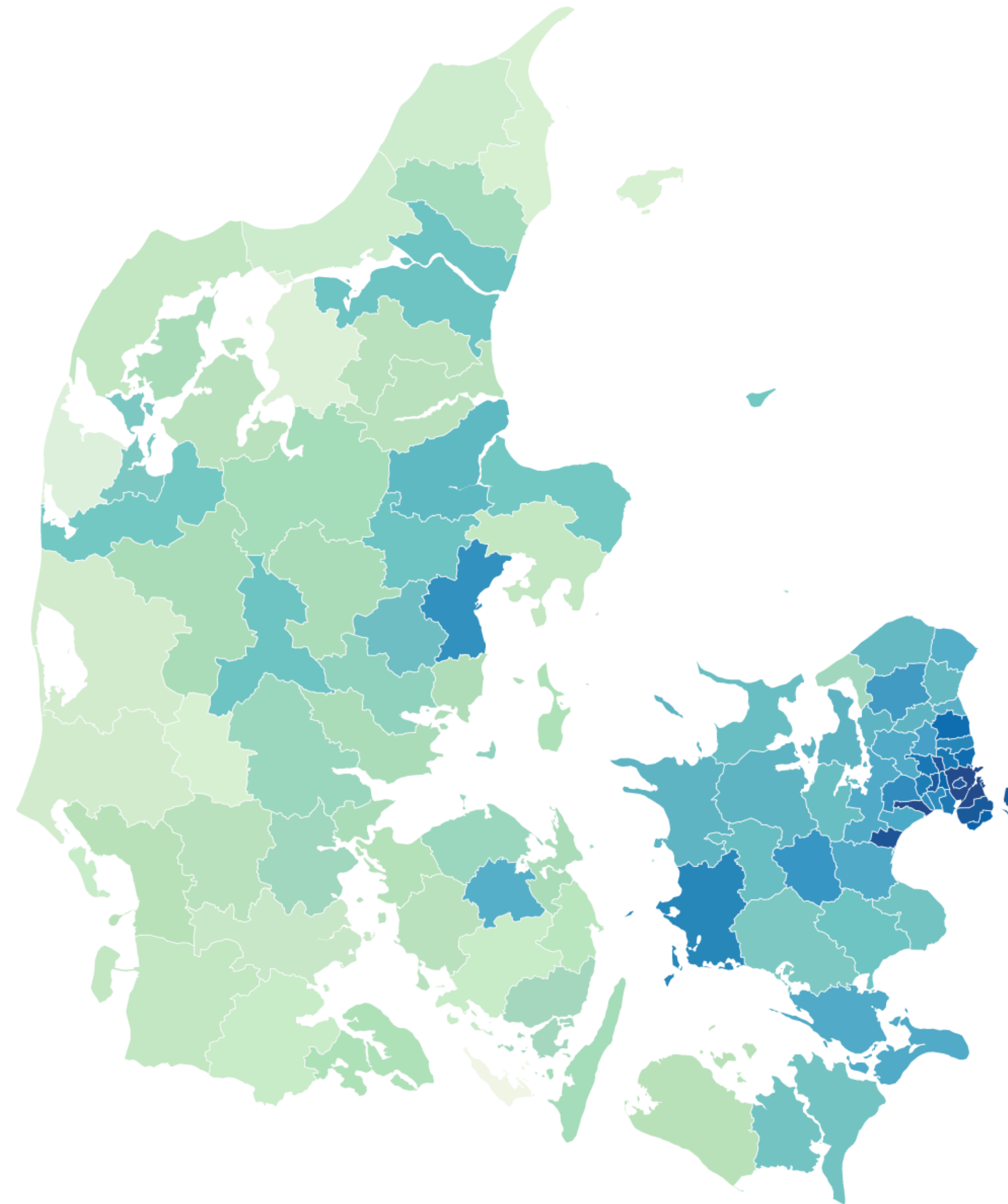
Choro = region
pleth = multiple

Combine: Polygons + Attribute values

Values are often grouped into classes to reduce complexity

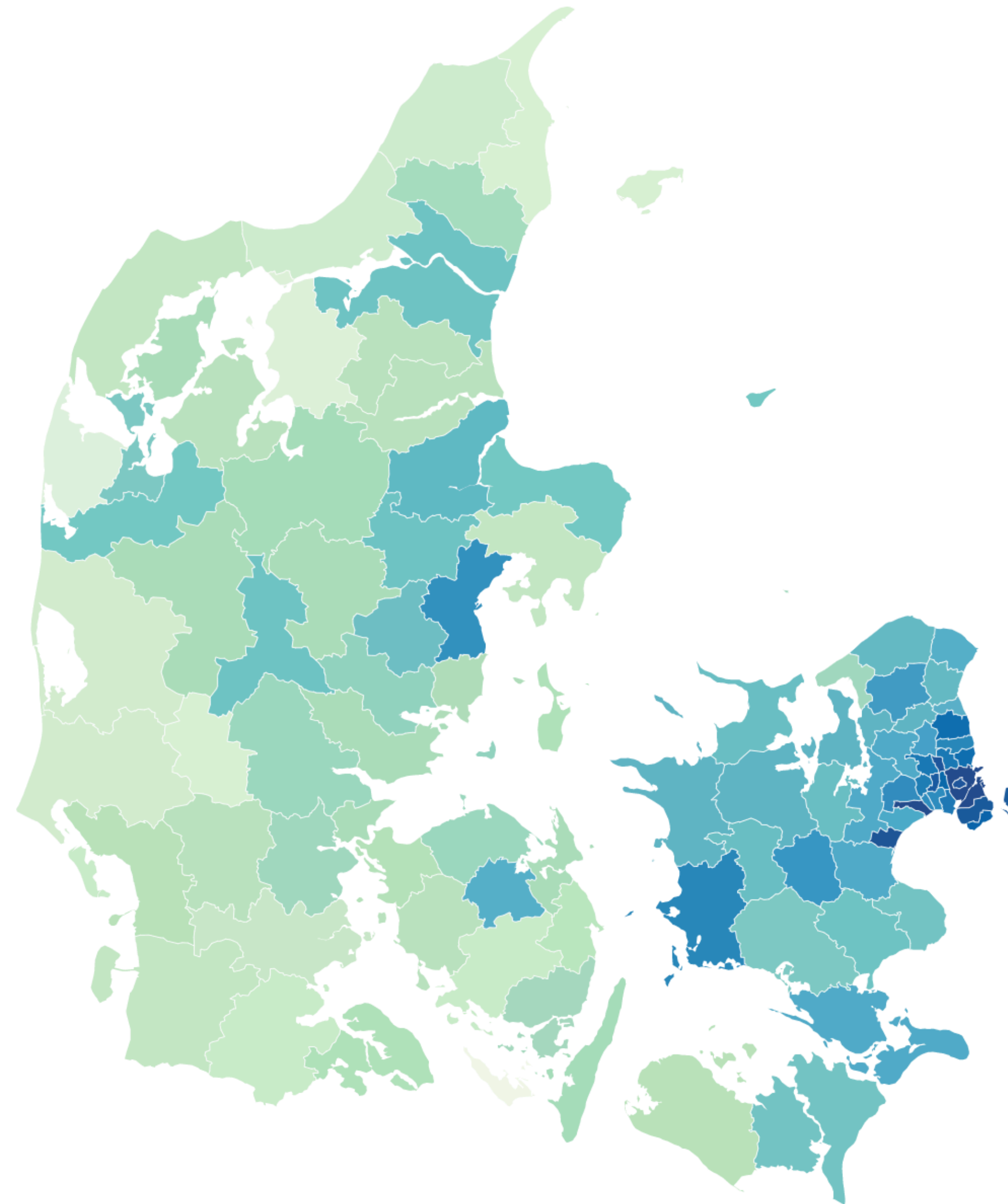
Denmark's coronavirus hotspots (by municipality), December 14th

Coronavirus cases per 100,000 residents over past 7 days as at December 14th (Source: SSI)



Denmark's coronavirus hotspots (by municipality), December 14th

Coronavirus cases per 100,000 residents over past 7 days as at December 14th (Source: SSI)

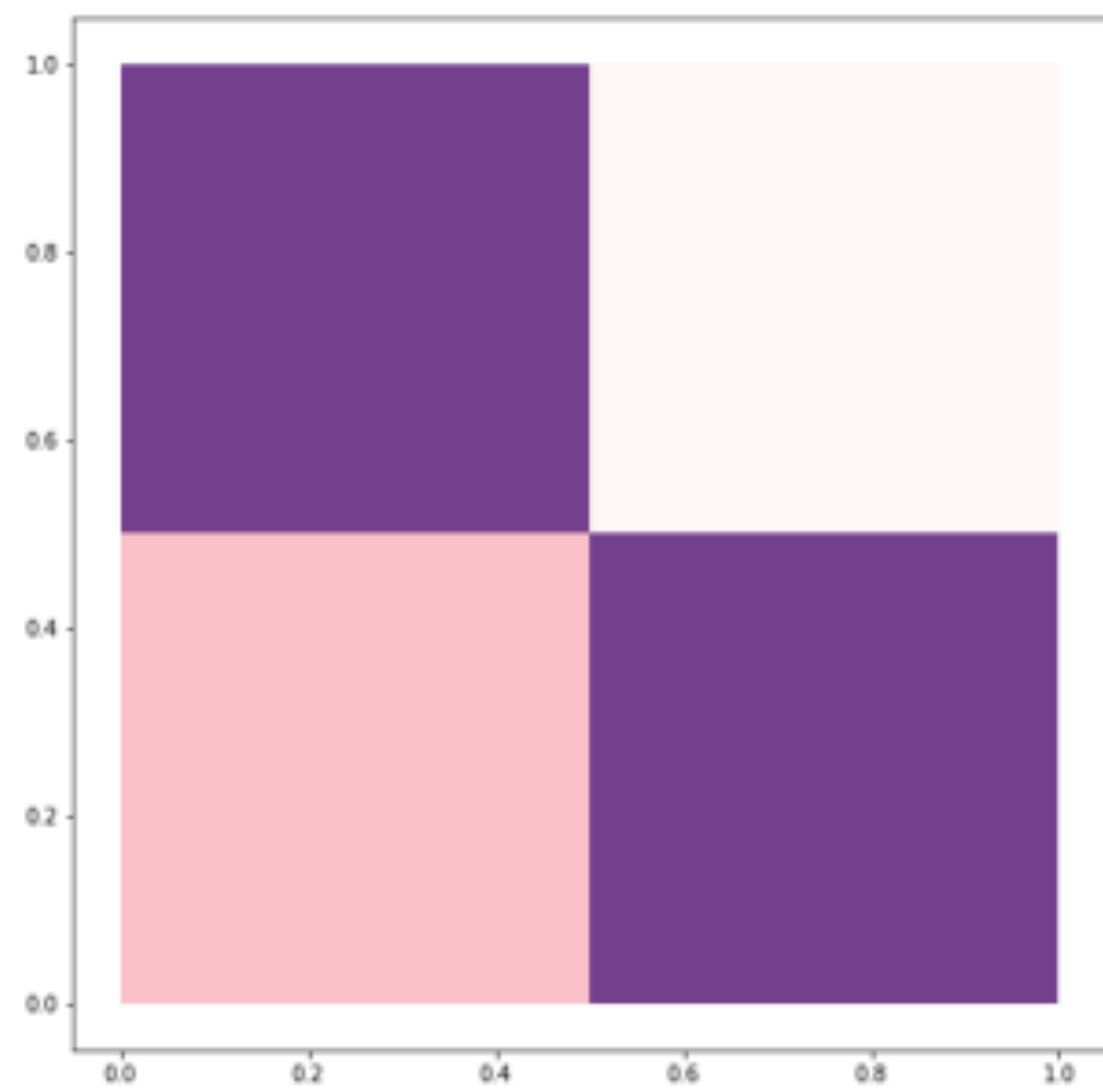


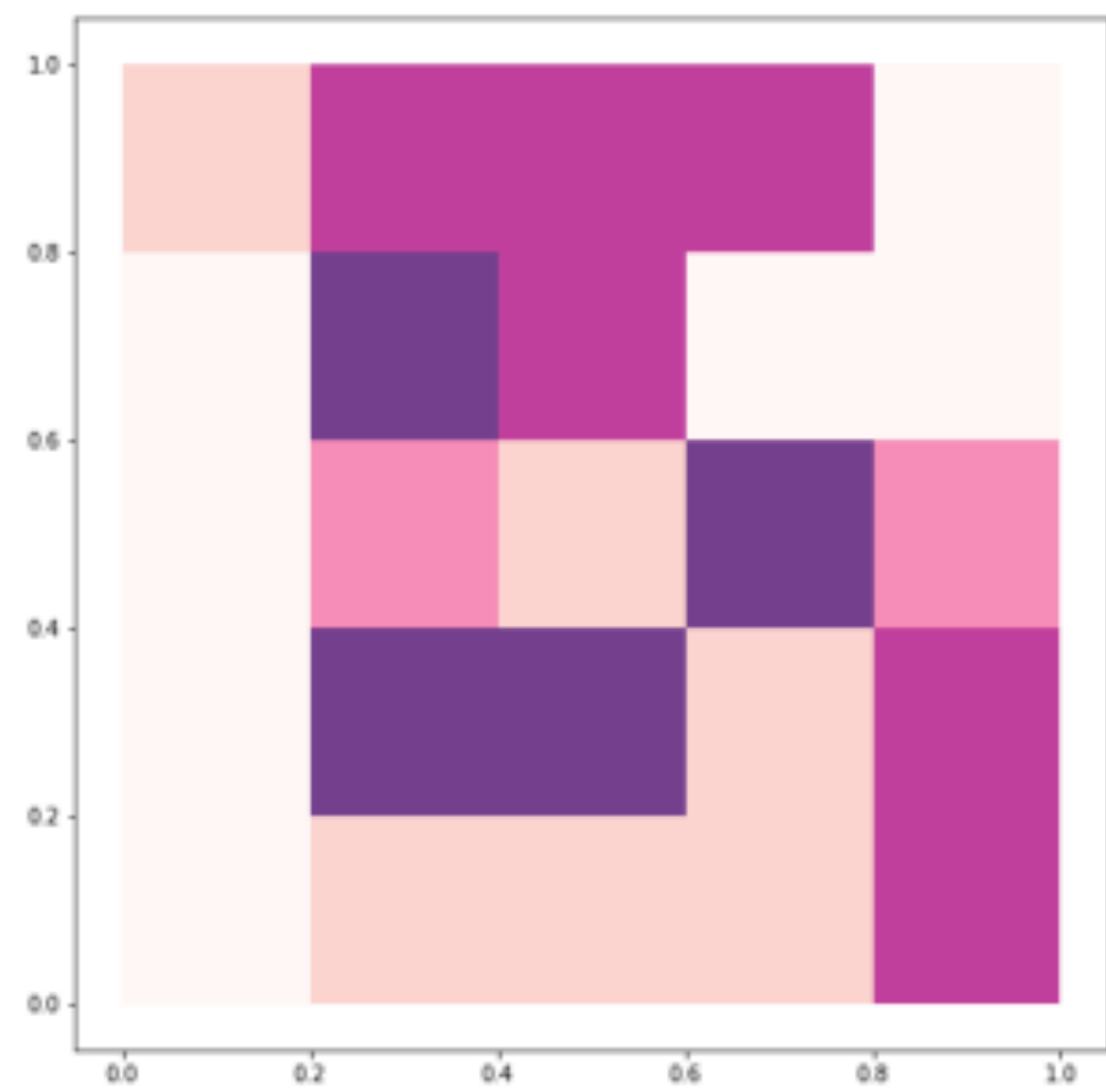
3 choices:
1) spatial units
2) colors
3) classes

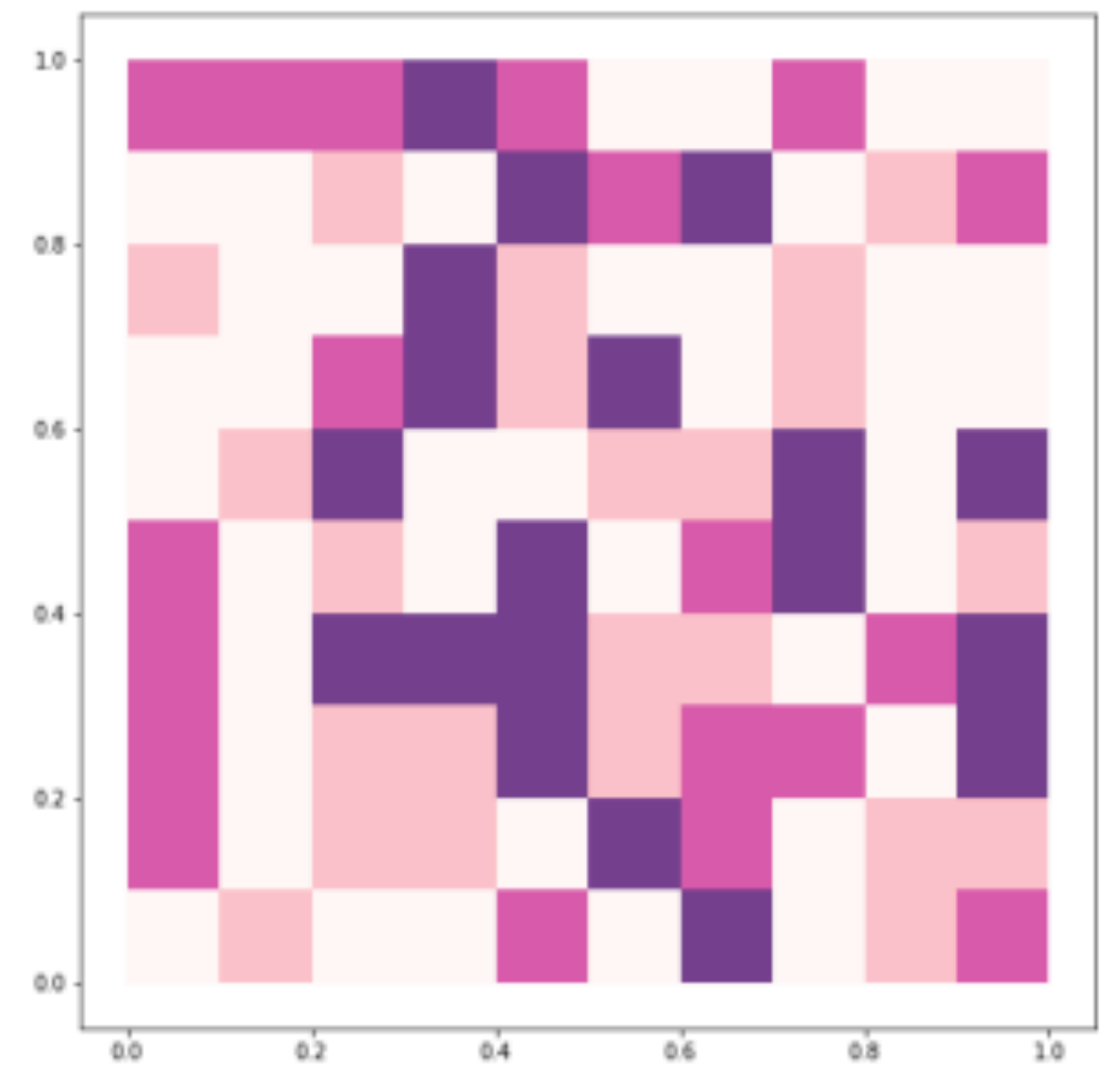
Population density (people aggregated in space)

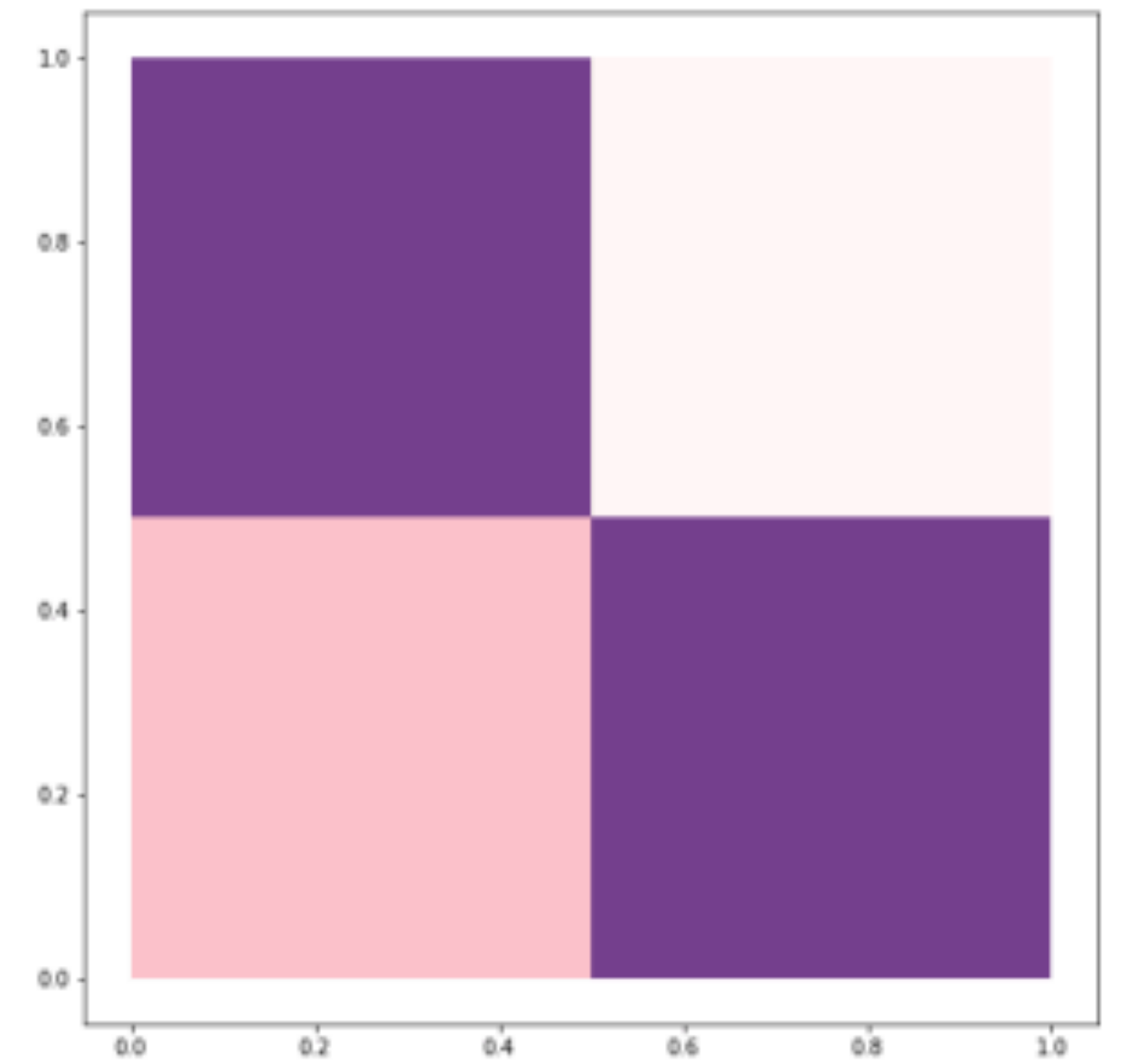
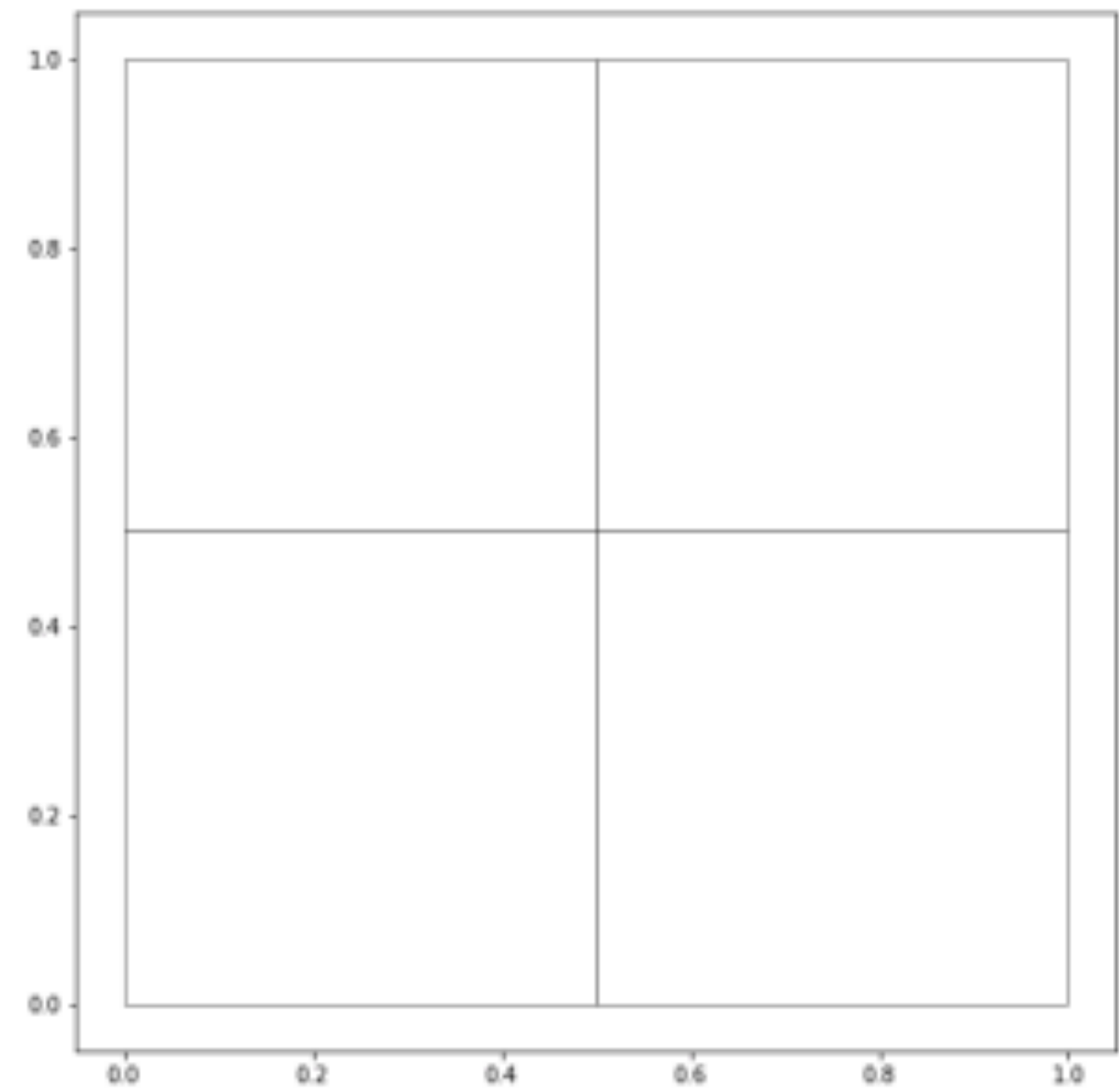
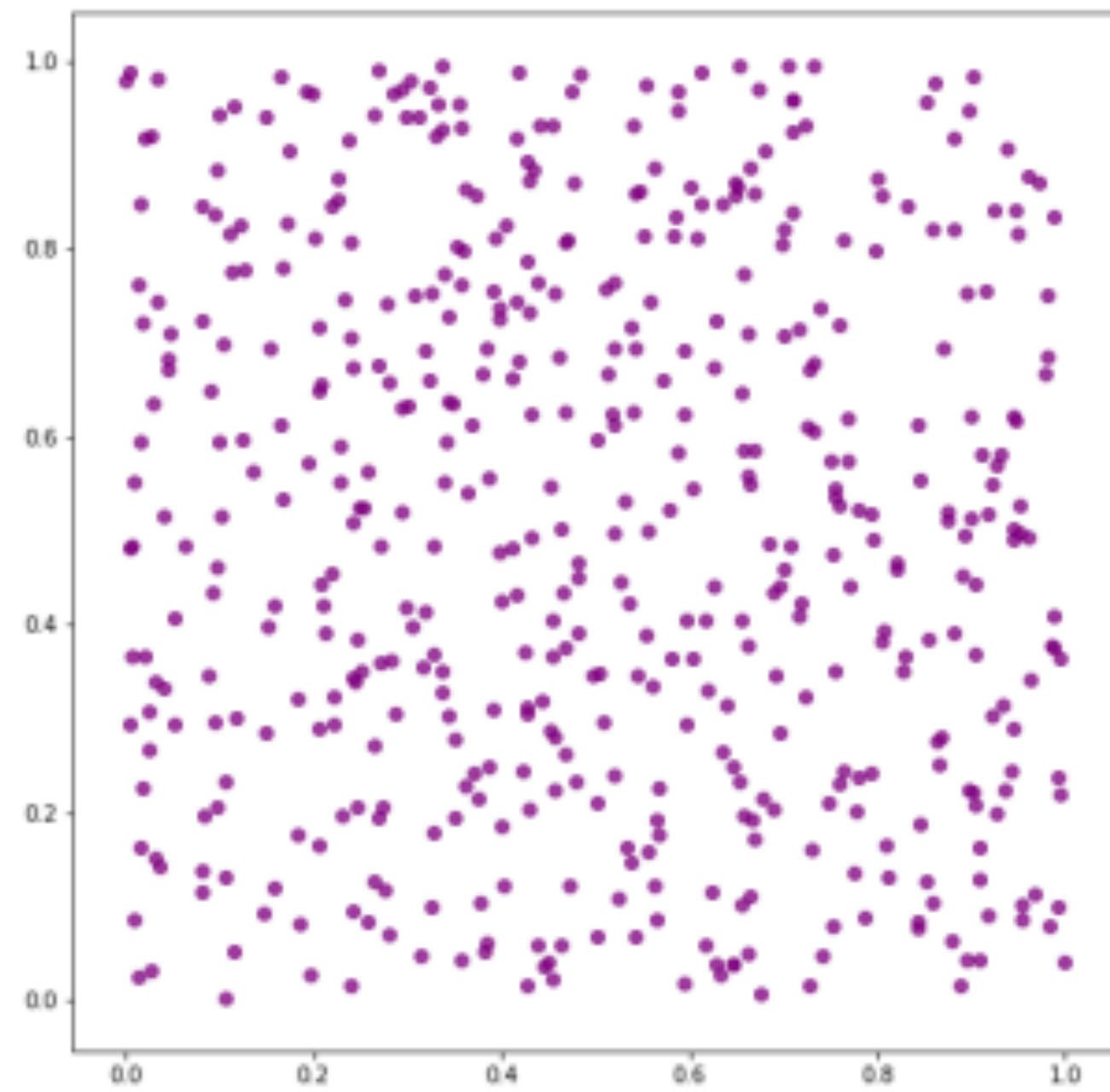
Group experiment

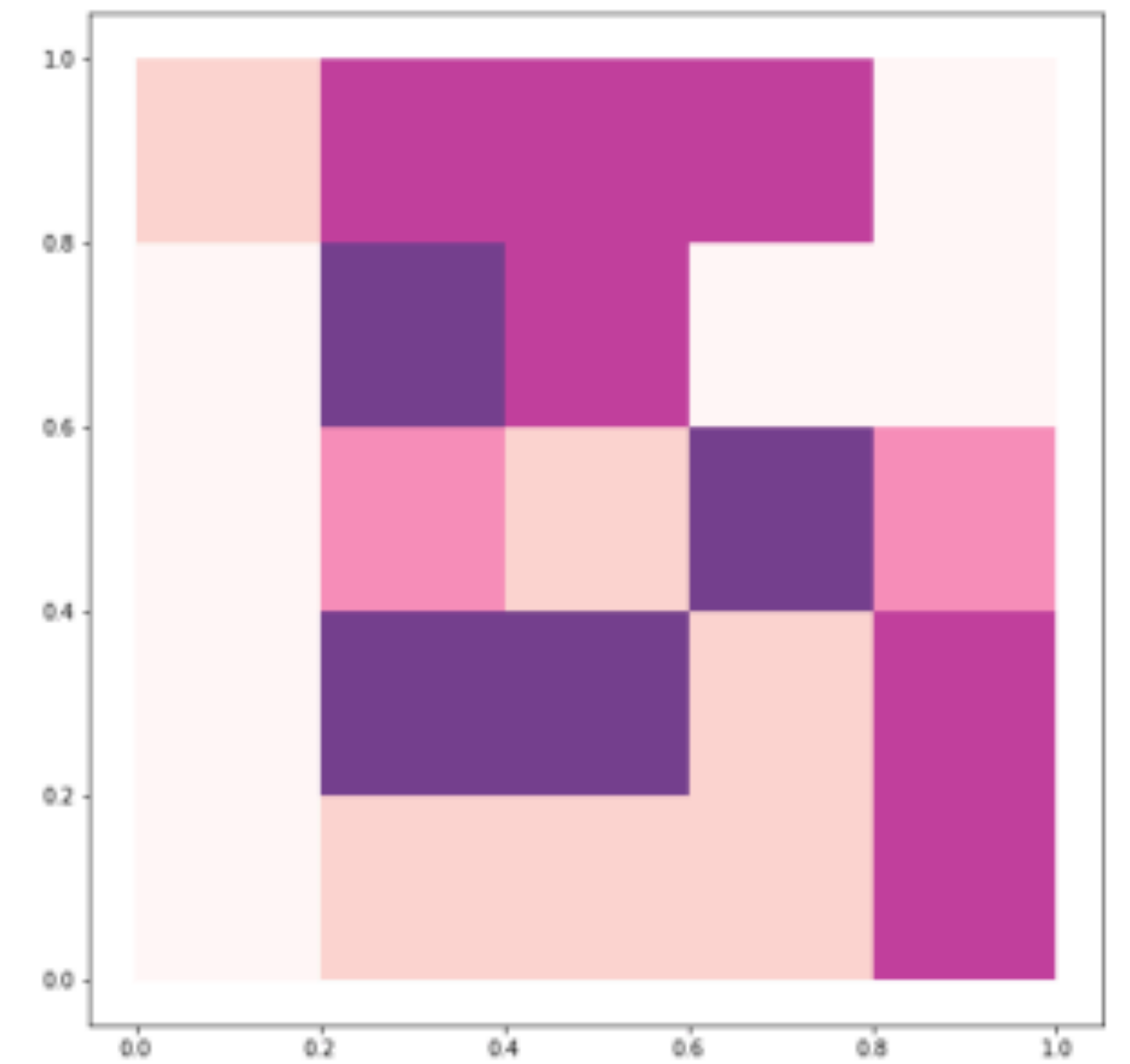
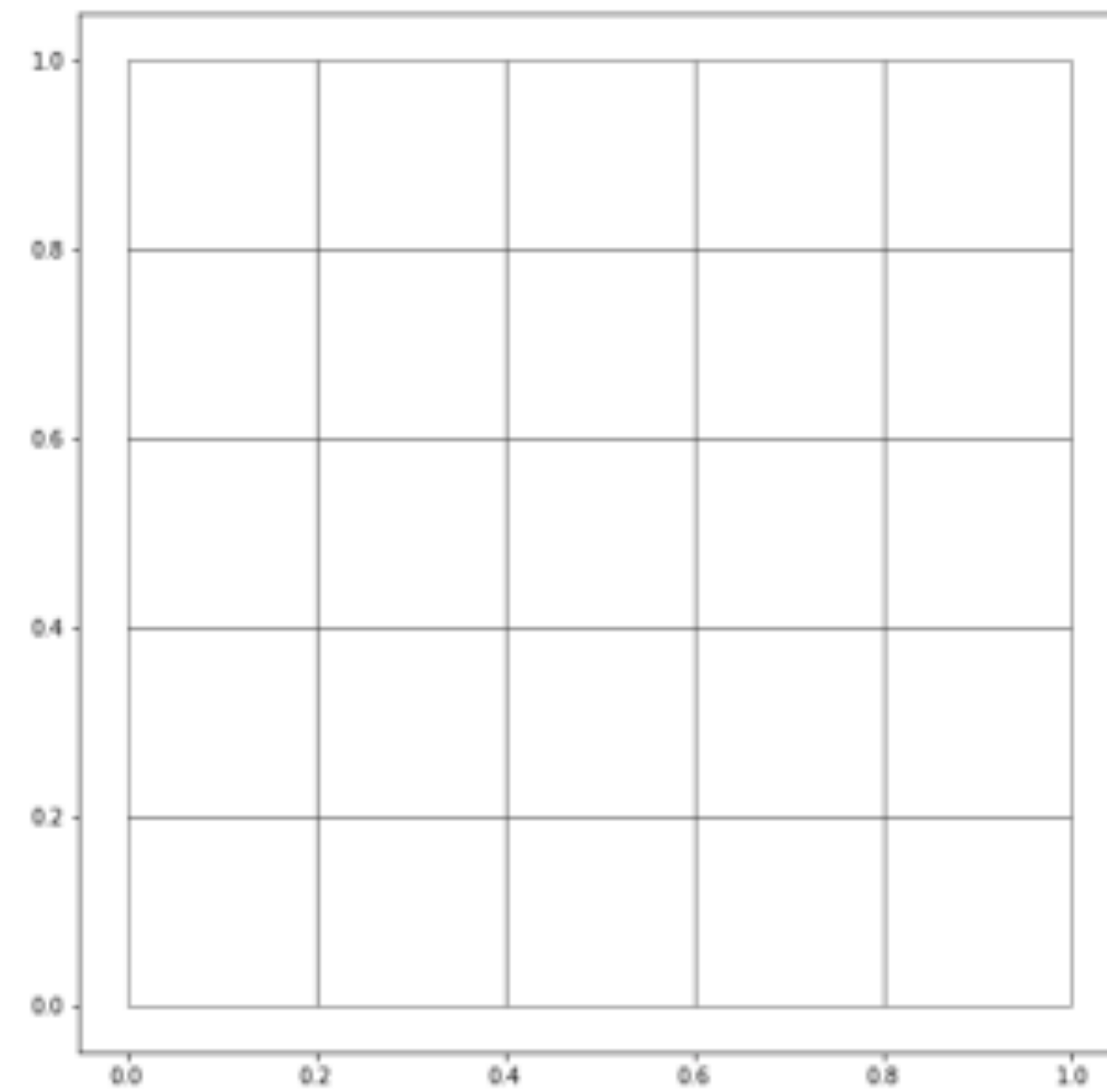
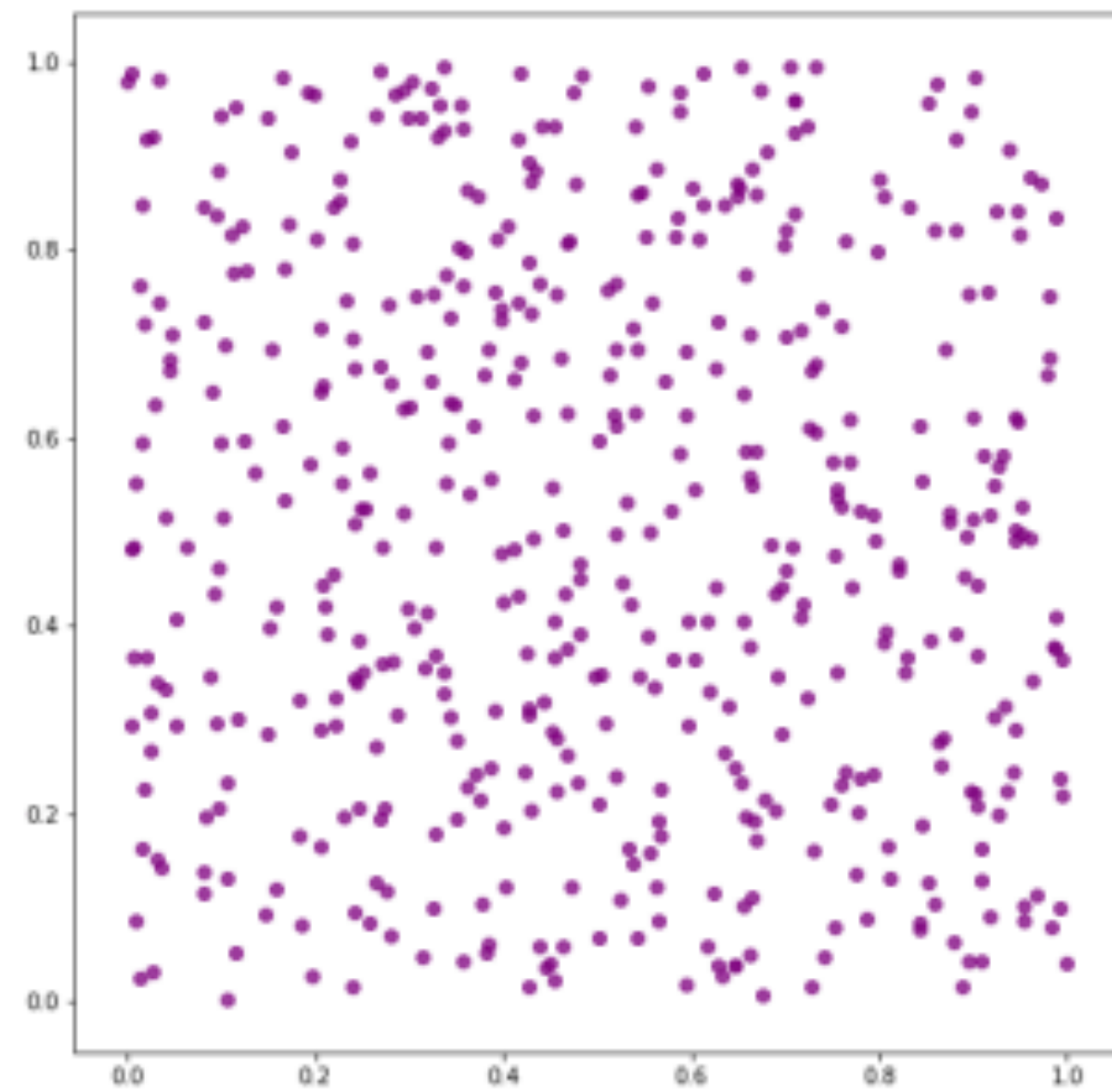
Each of 3 groups describes their
picture in 1 sentence

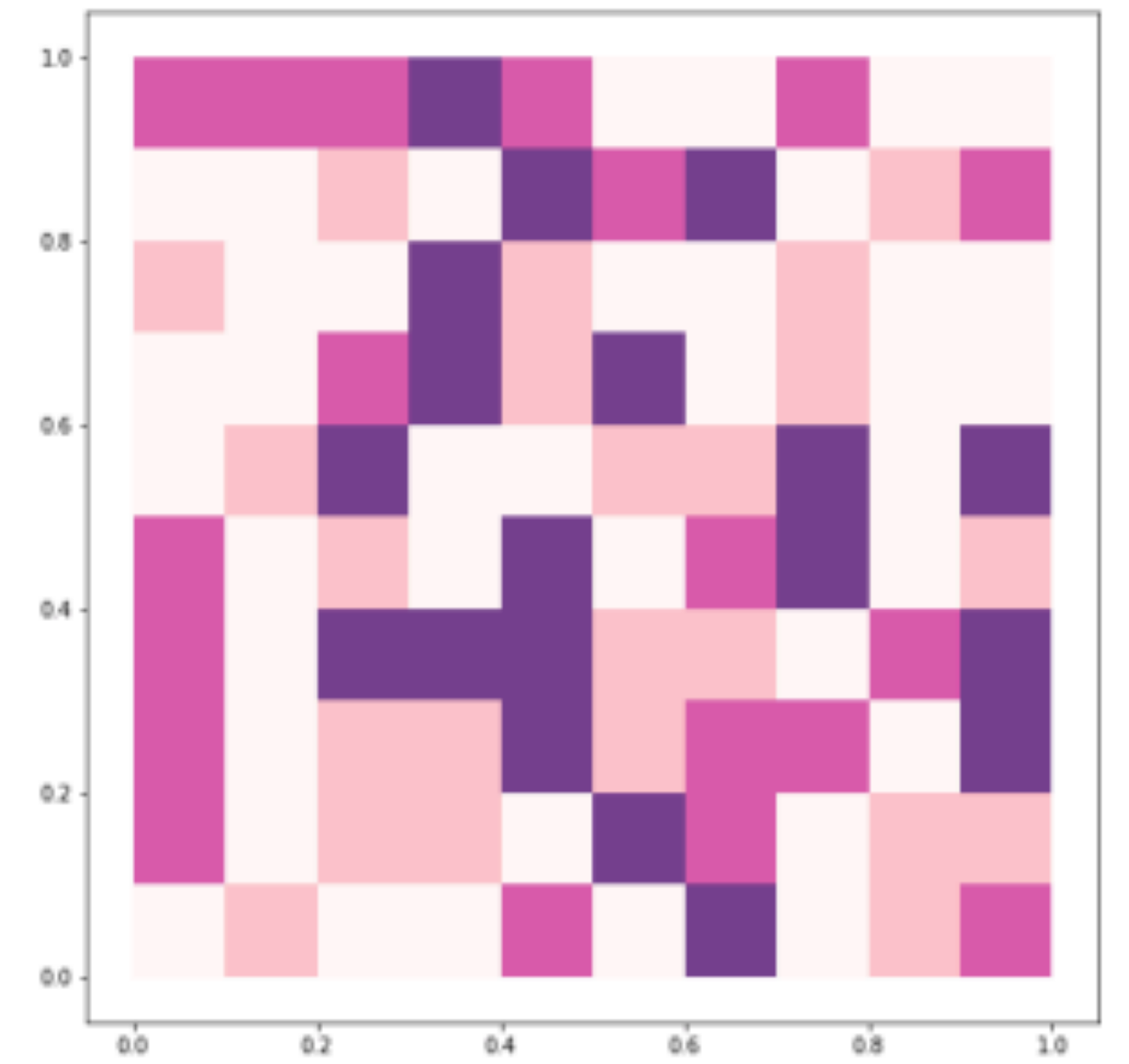
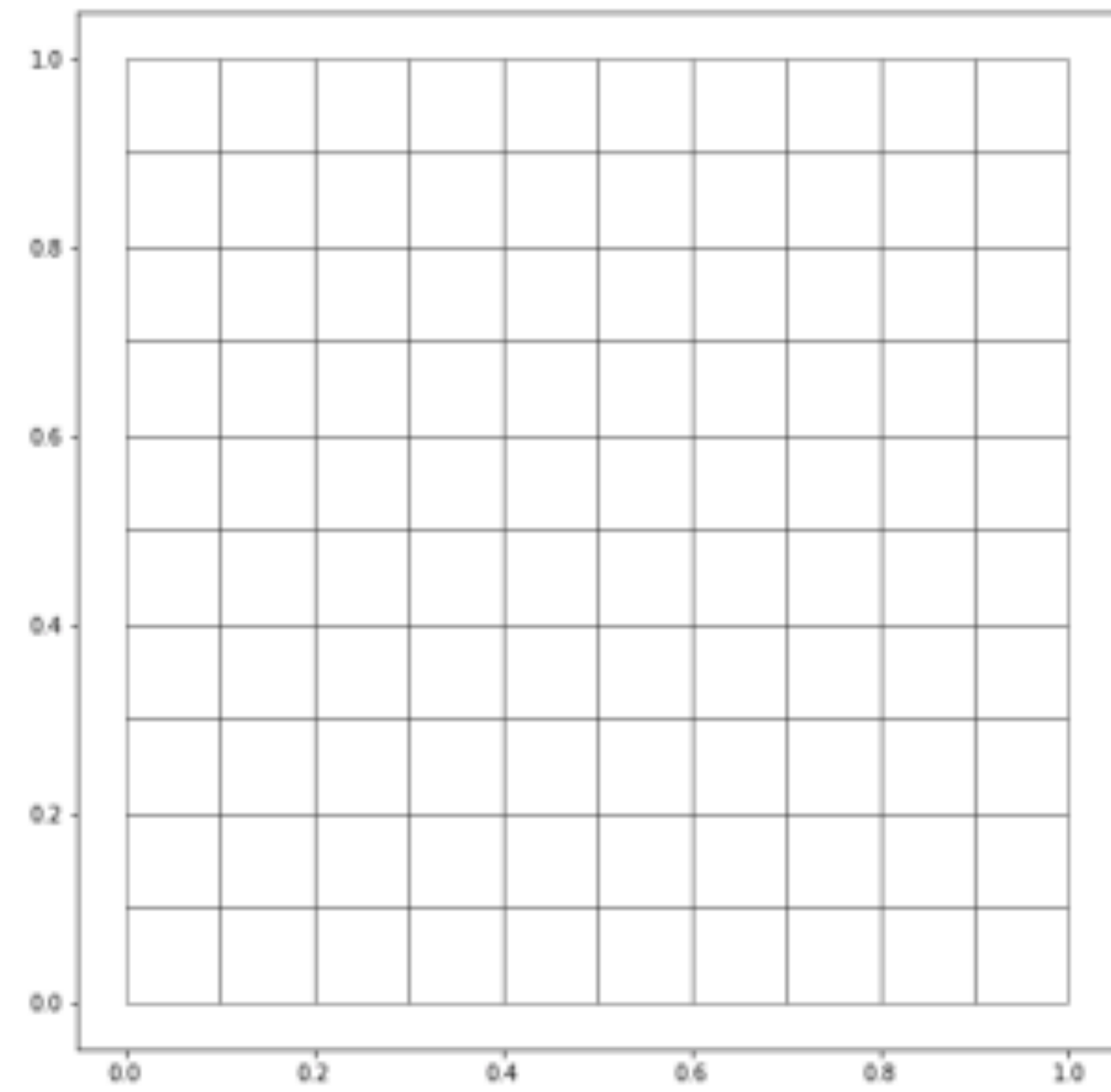
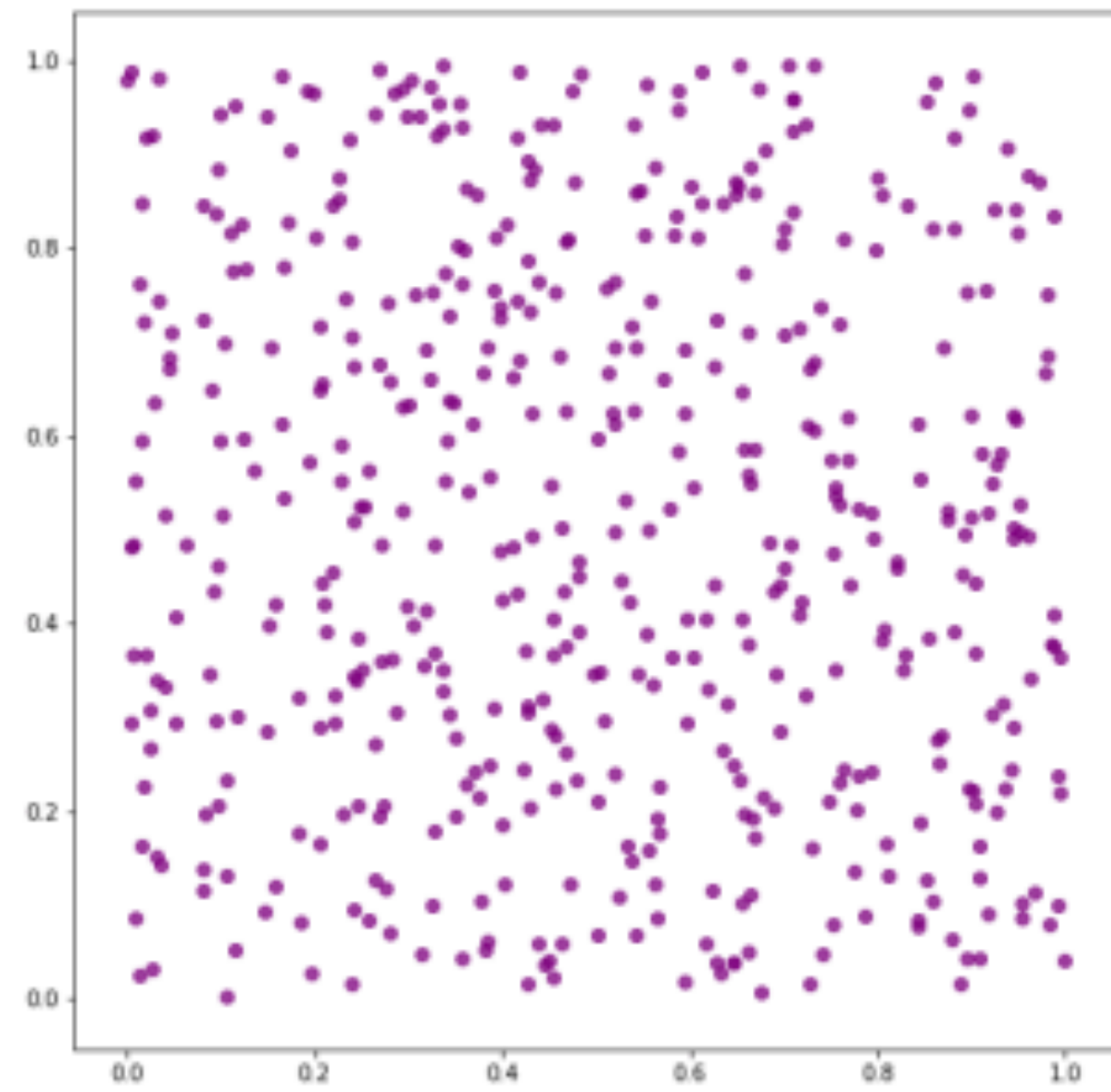




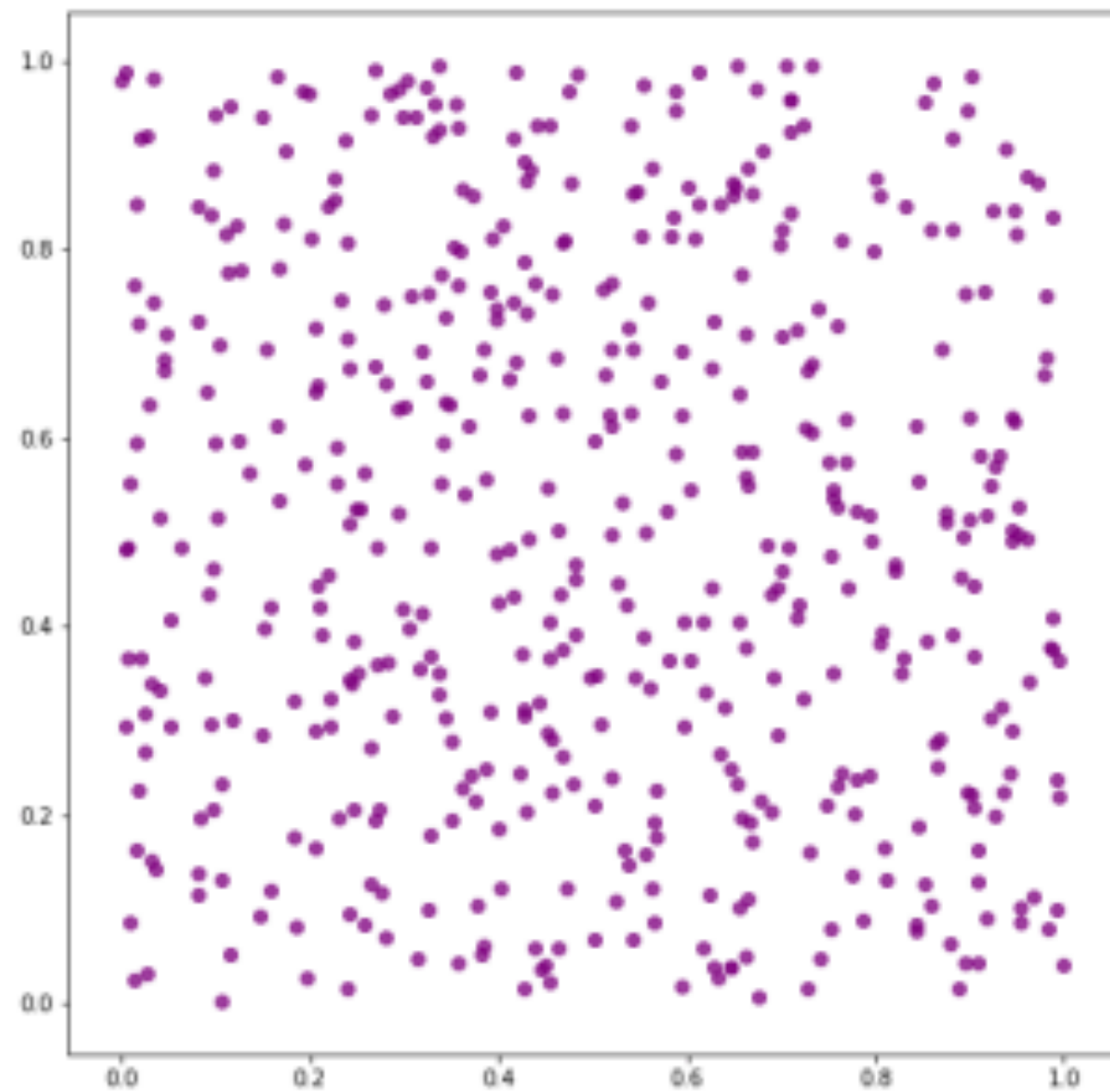




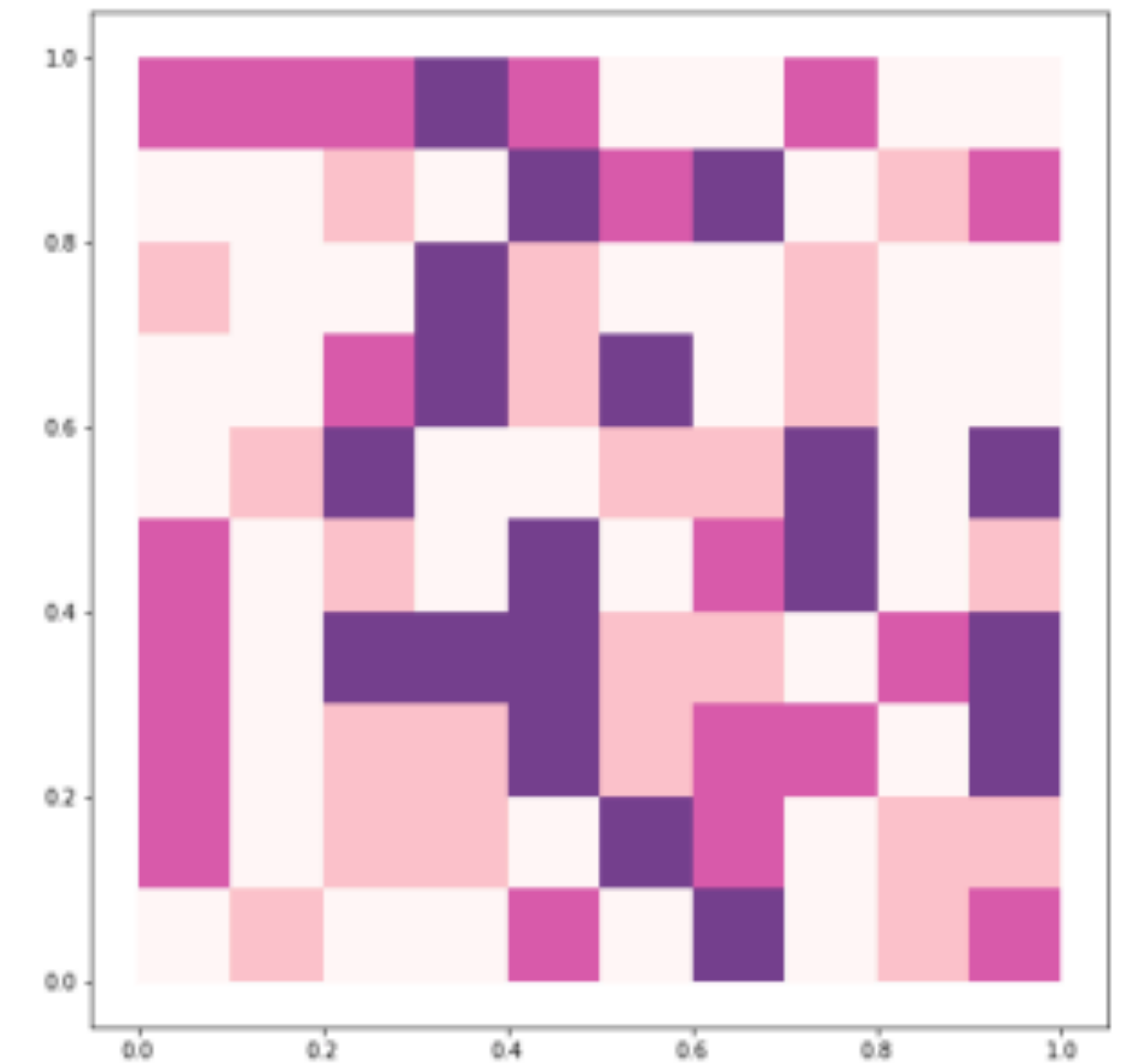
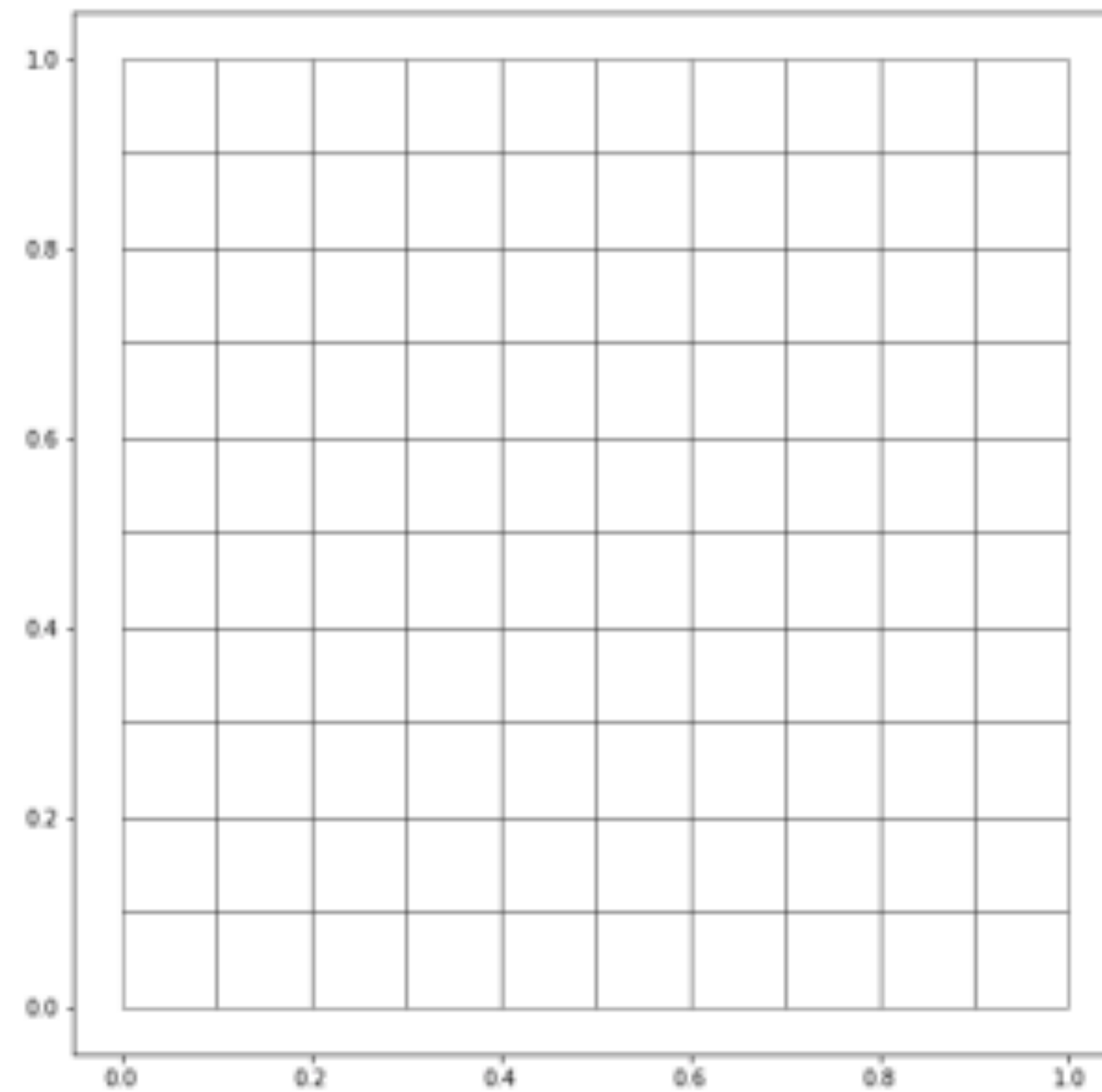




Same RANDOM data

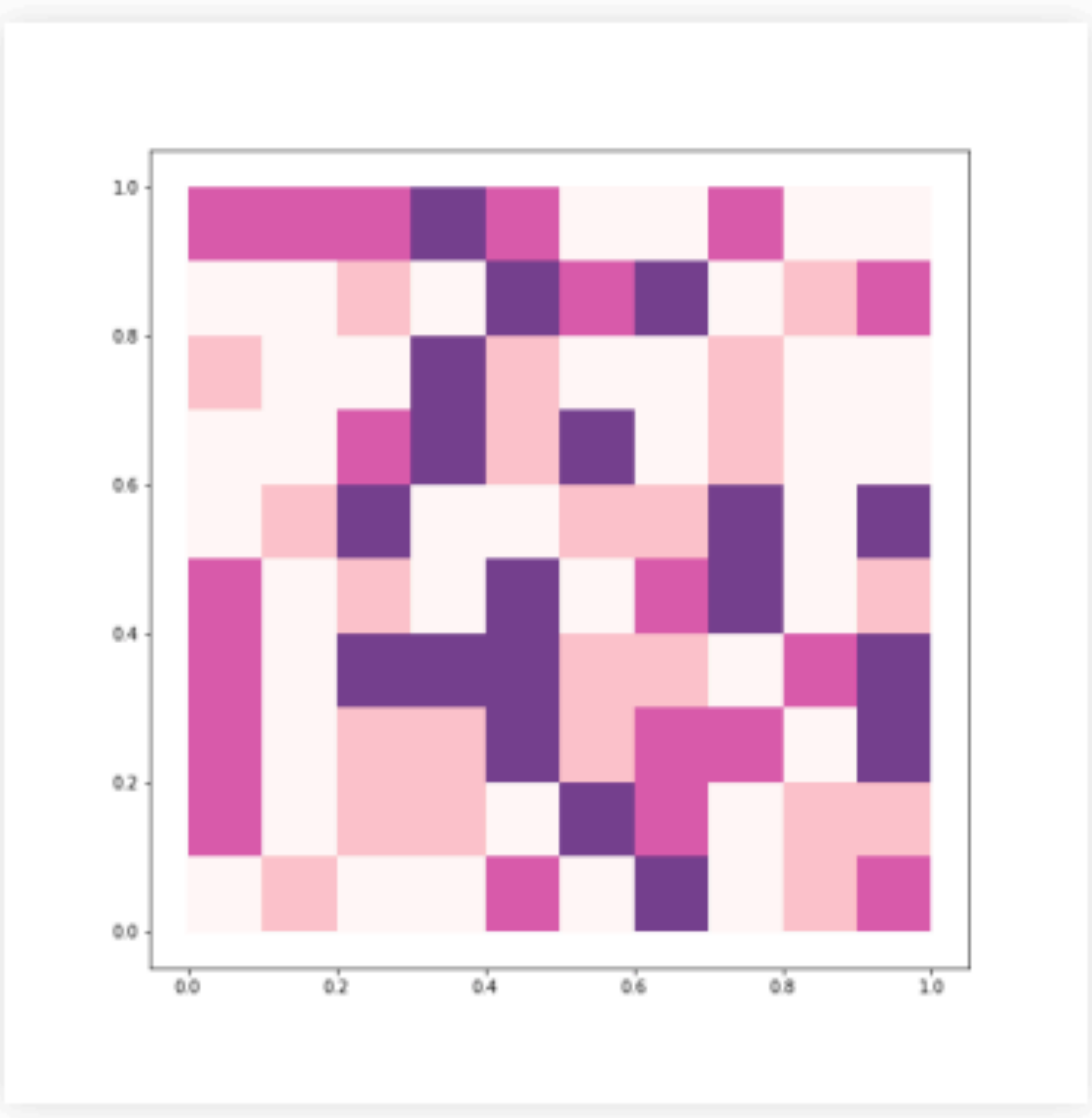
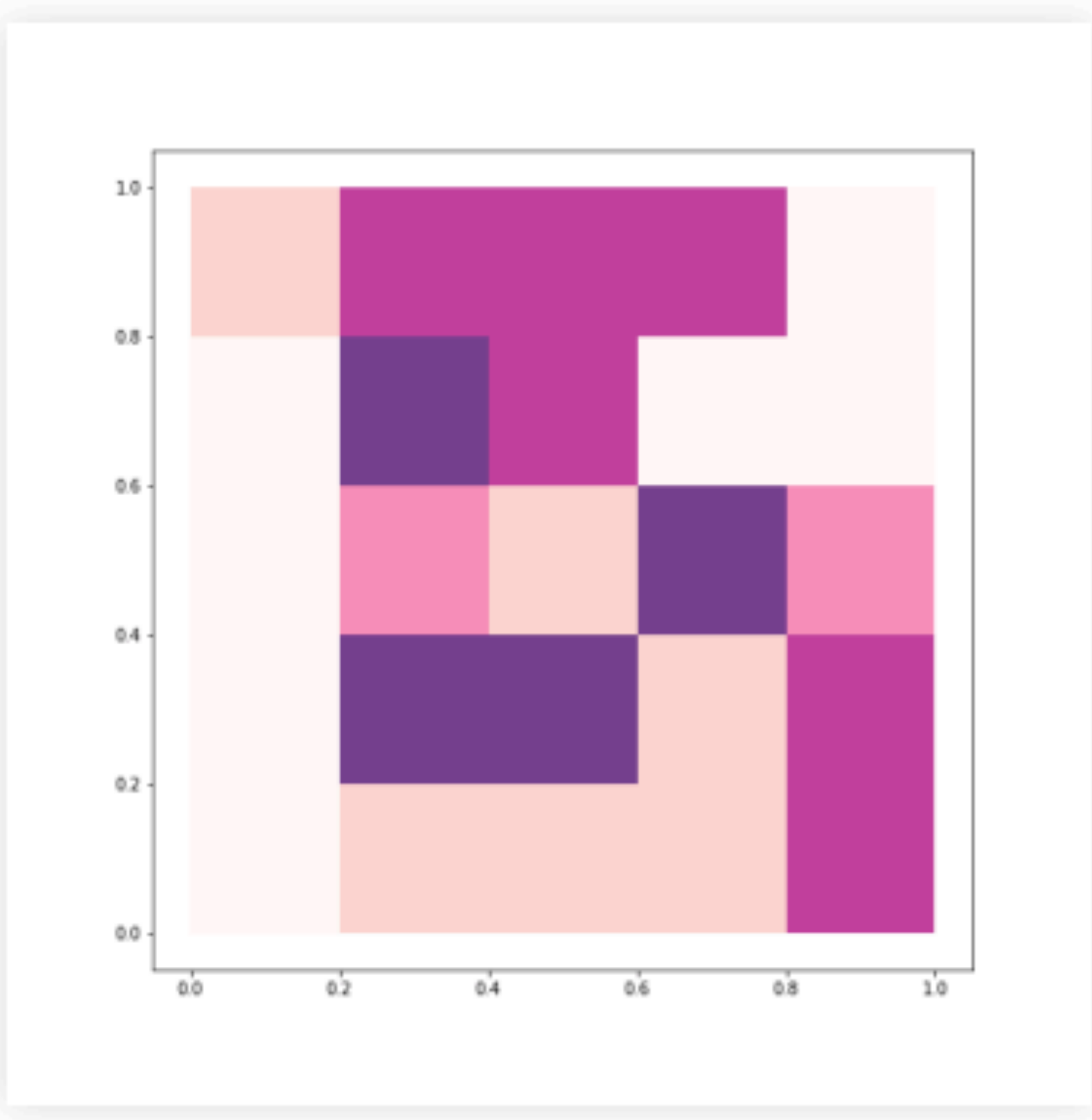
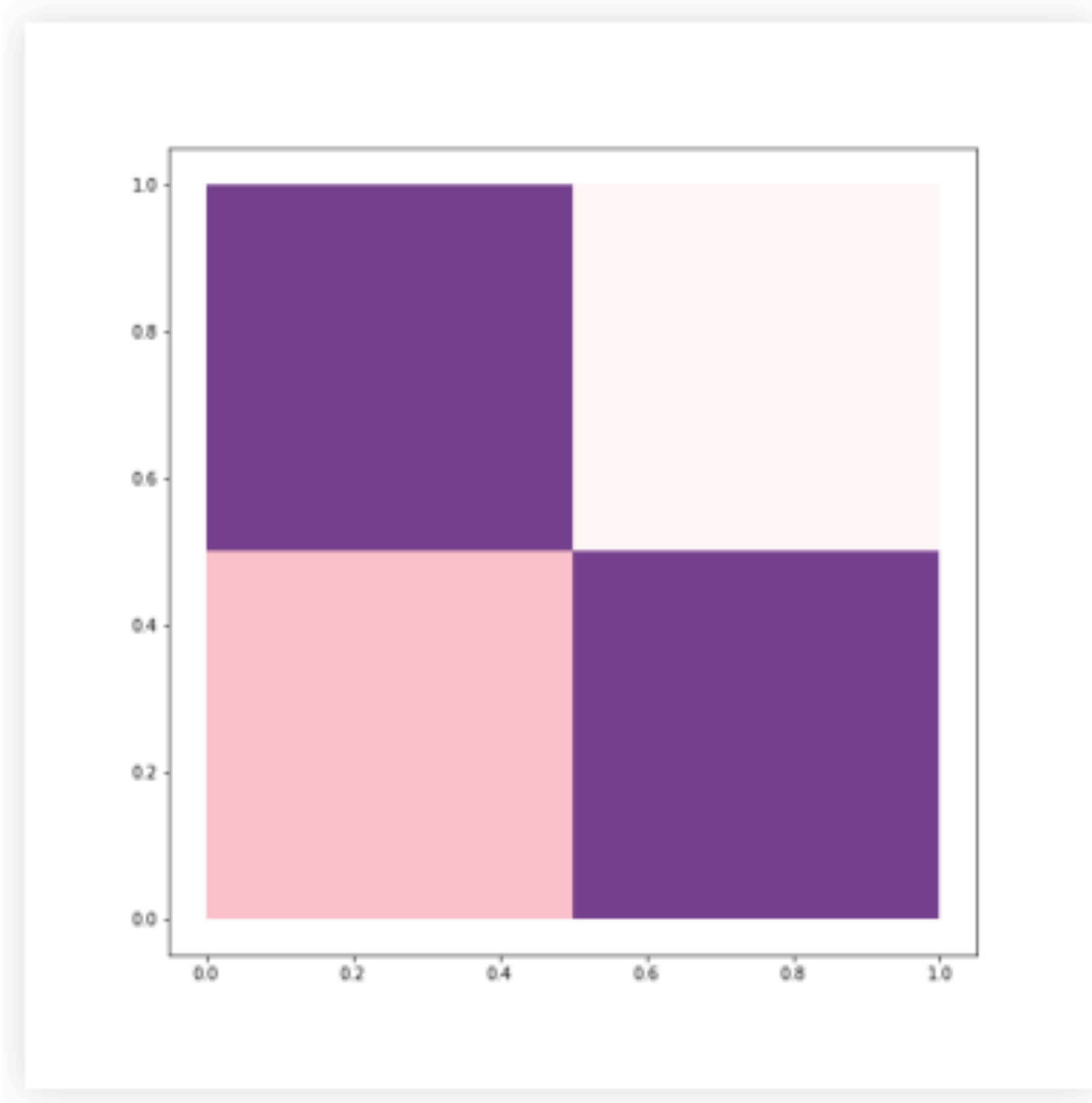


Same colors
Same classification



Different spatial units

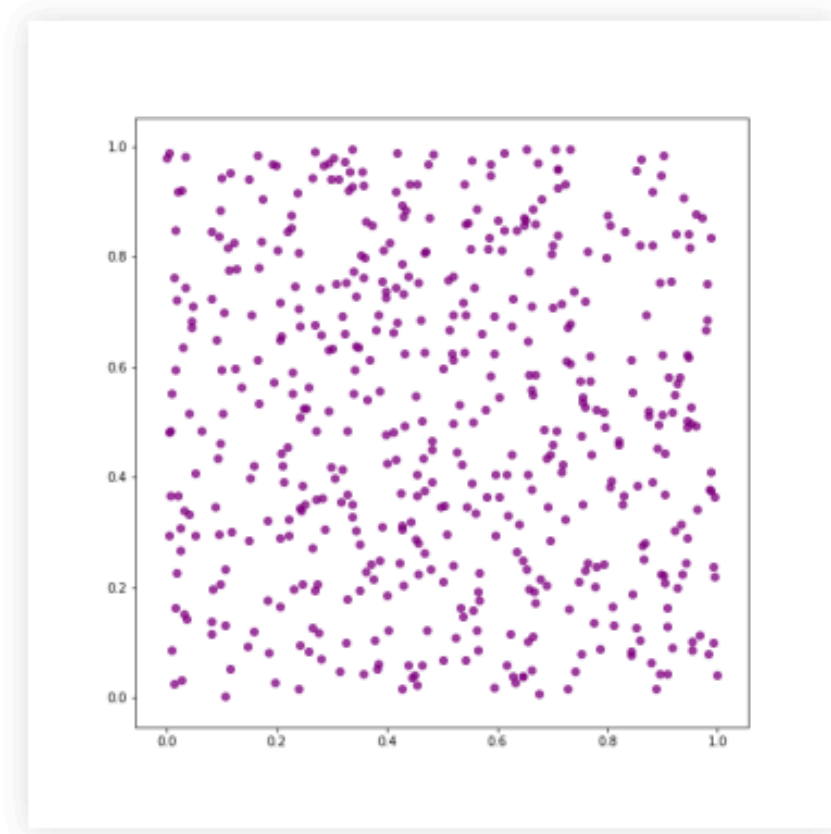
A common source of bias in spatial aggregation: MAUP



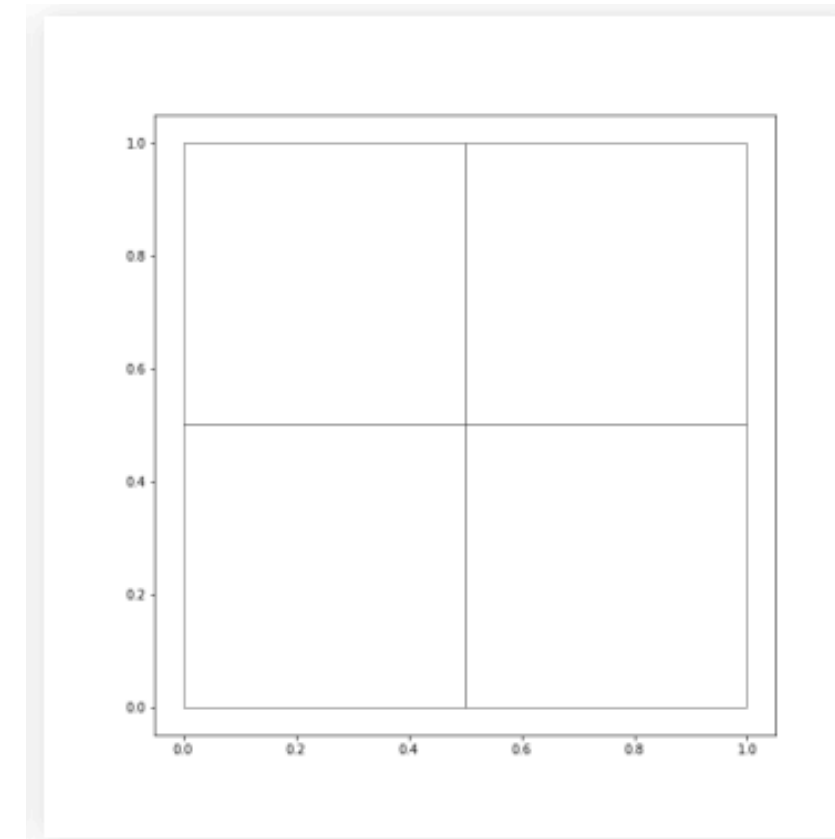
A common source of bias in spatial aggregation: MAUP

The **MAUP (Modifiable Areal Unit Problem)** is a scale and delineation mismatch between:

Underlying entities $\longleftrightarrow$ Unit of measurement

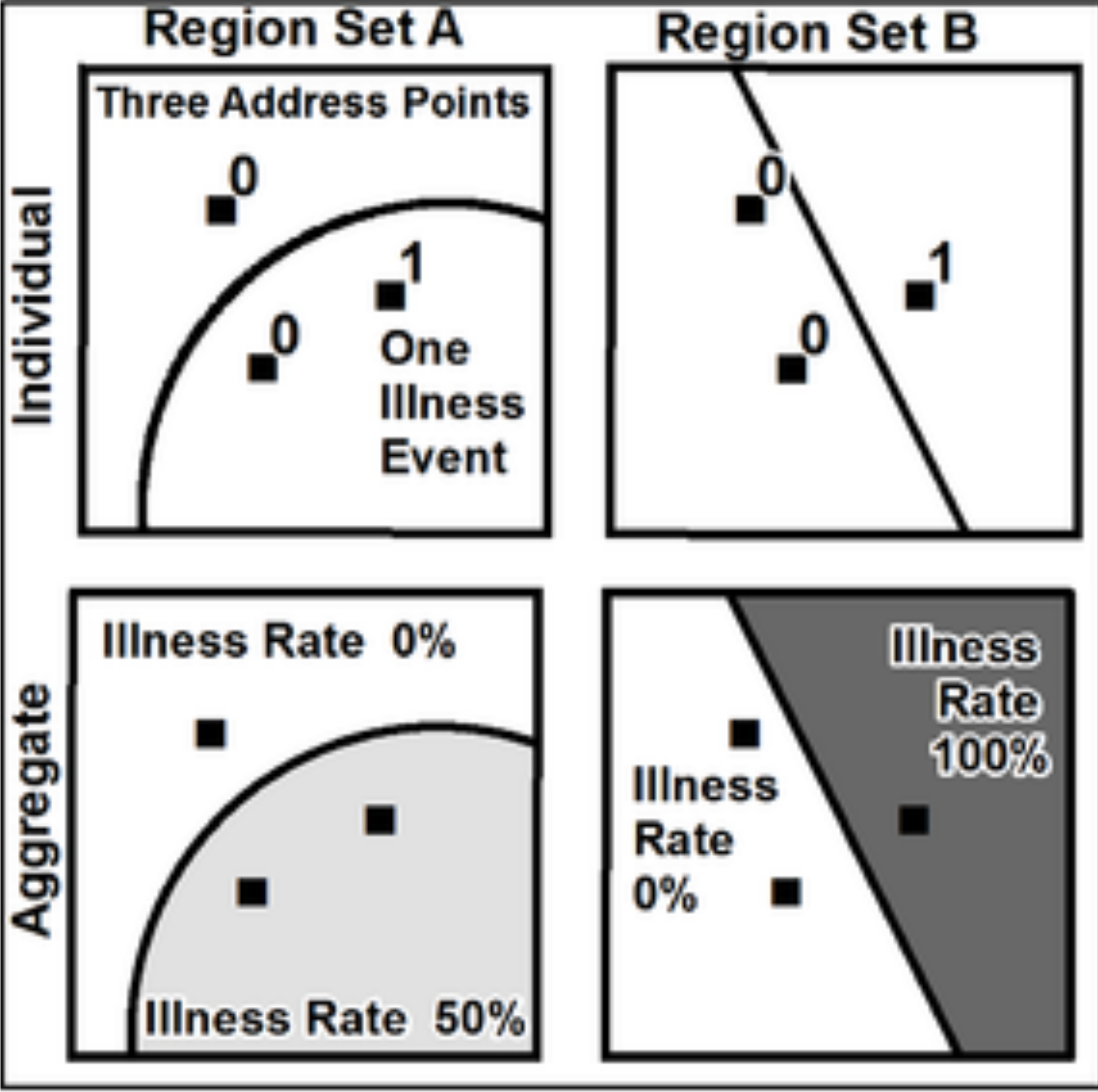


Individuals, shops, ..



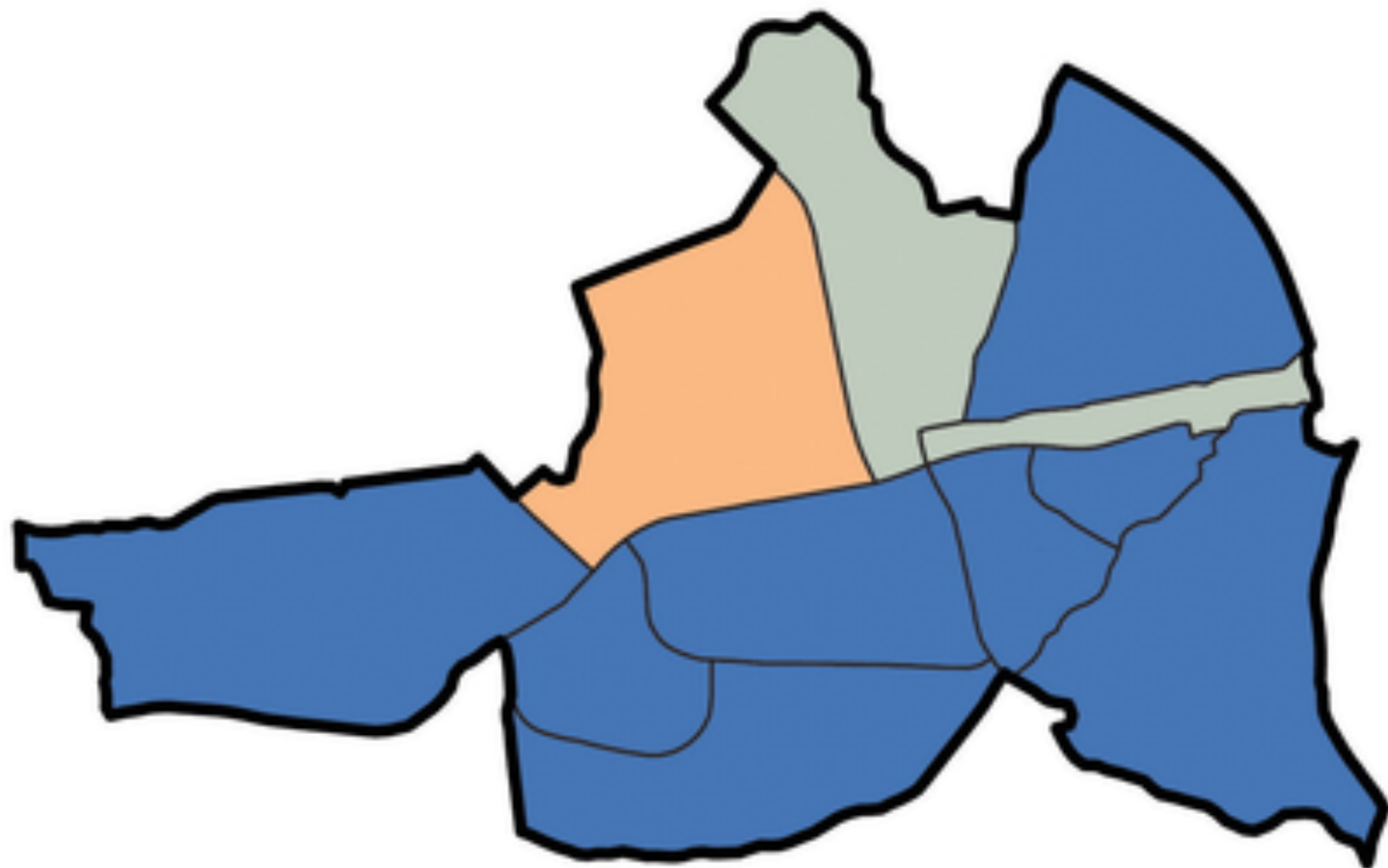
Districts, regions, resolution, ..

A common source of bias in spatial aggregation: MAUP

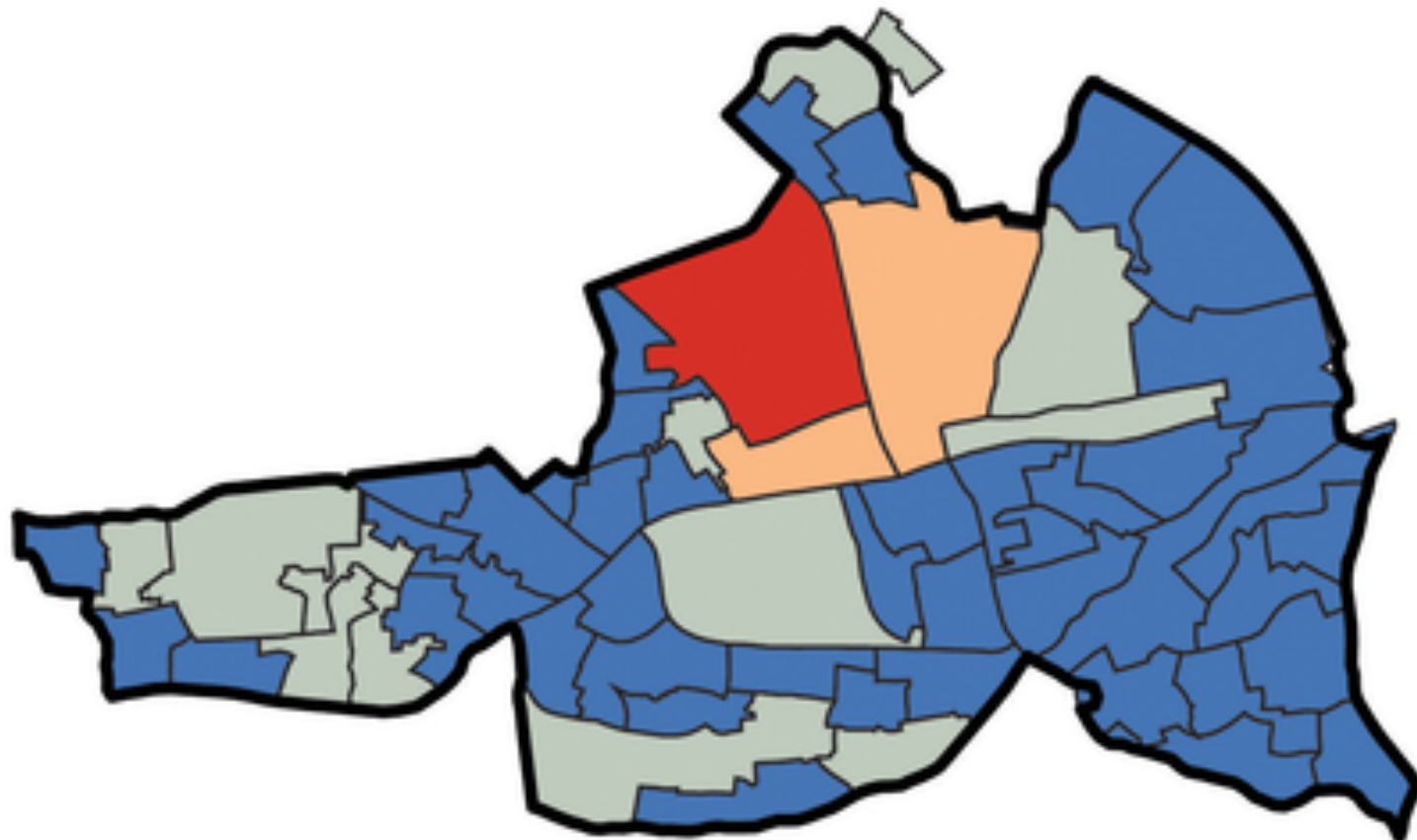


A common source of bias in spatial aggregation: MAUP

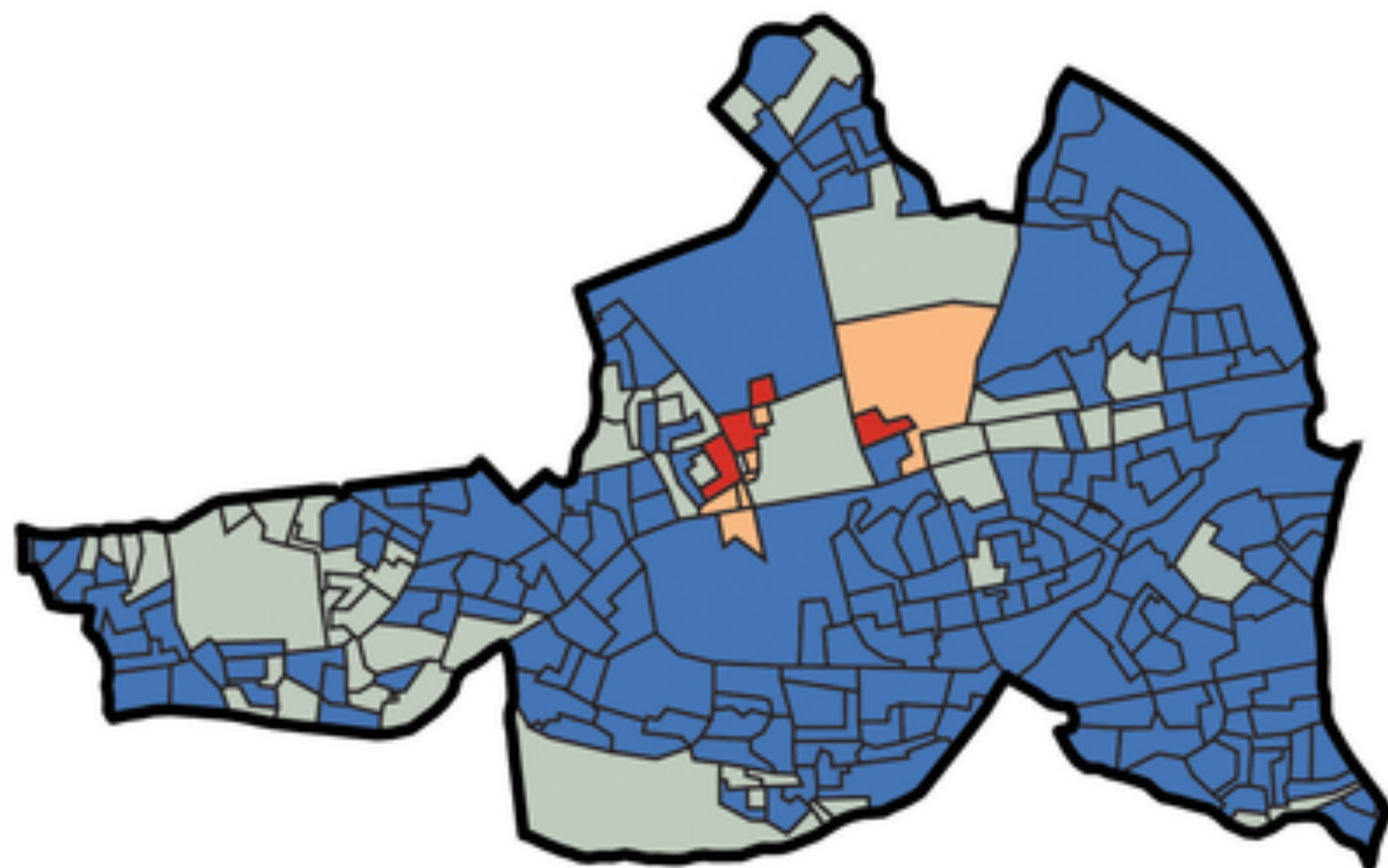
1. Electoral Divisions (EDs), 11 in Study Area



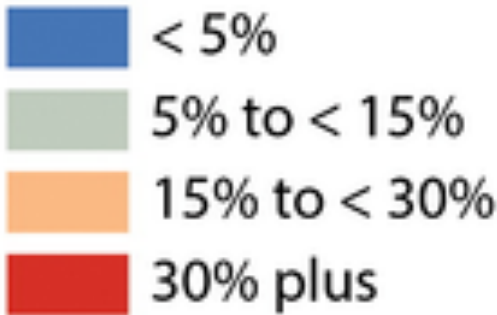
2. Enumerator Area (EAs), 56 in Study Area



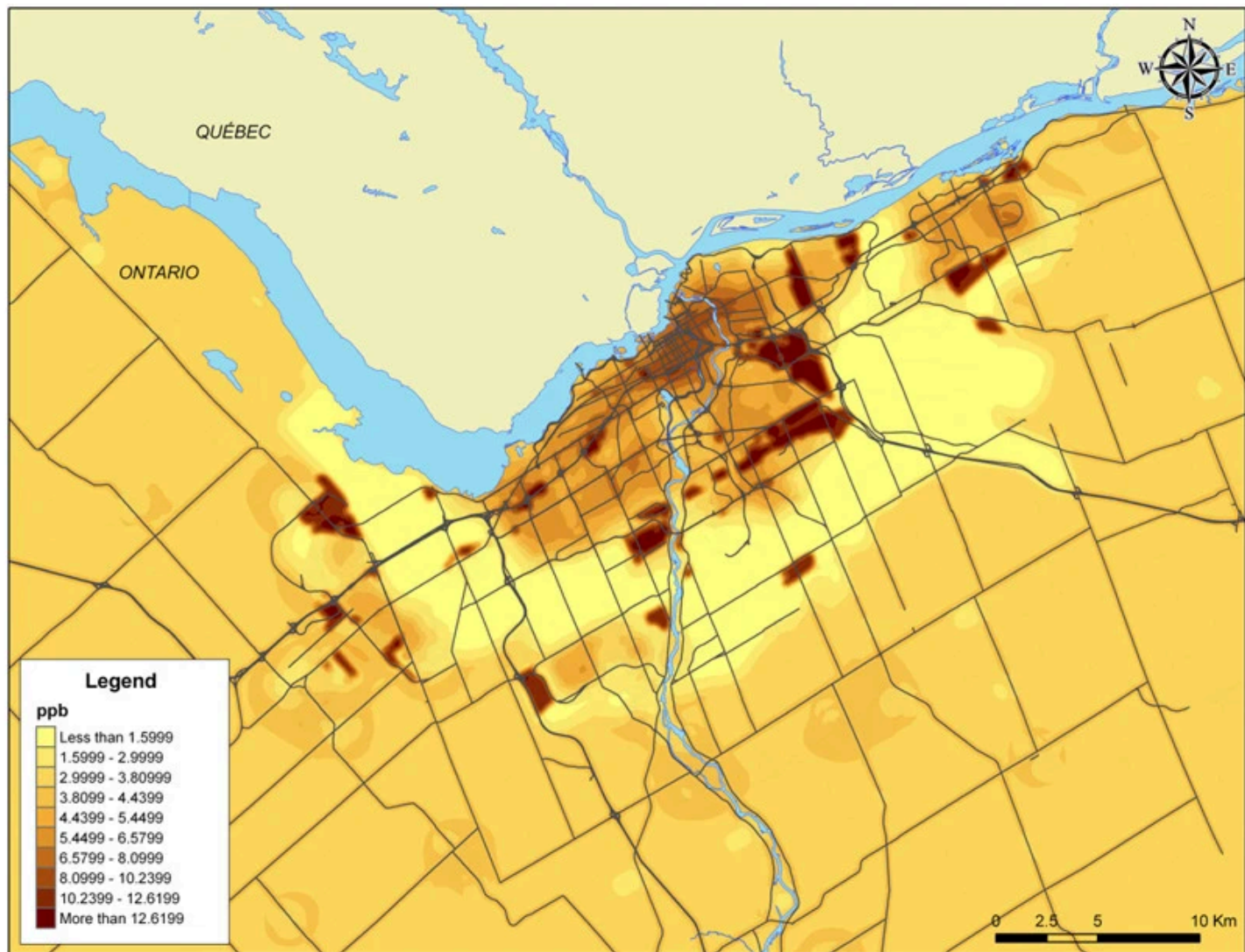
3. Small Areas (SAs), 237 in Study Area



Modifiable areal unit problem (MAUP)
Study Area: Tallaght, Dublin
Housing Vacancy Rate, 2011

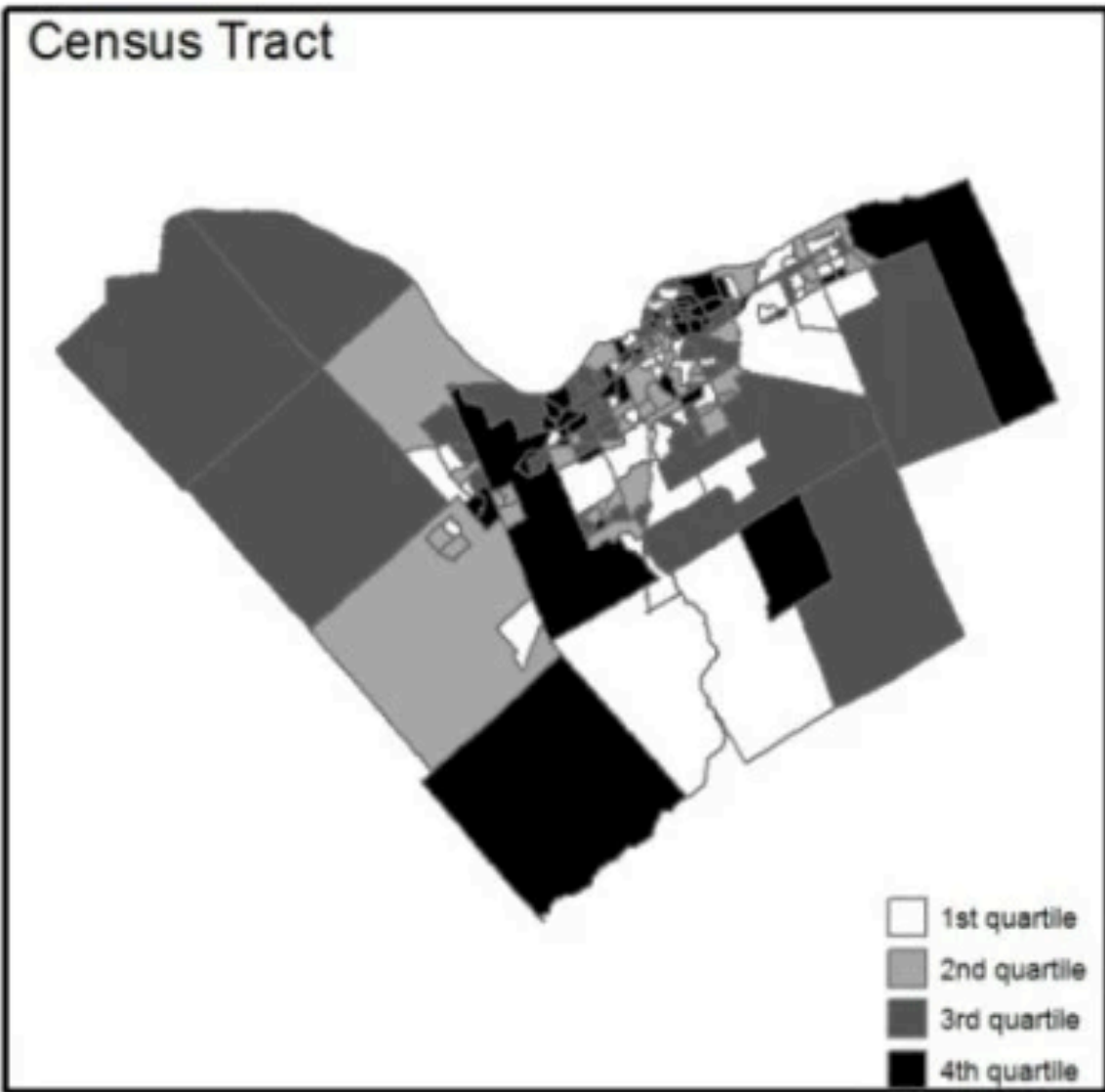
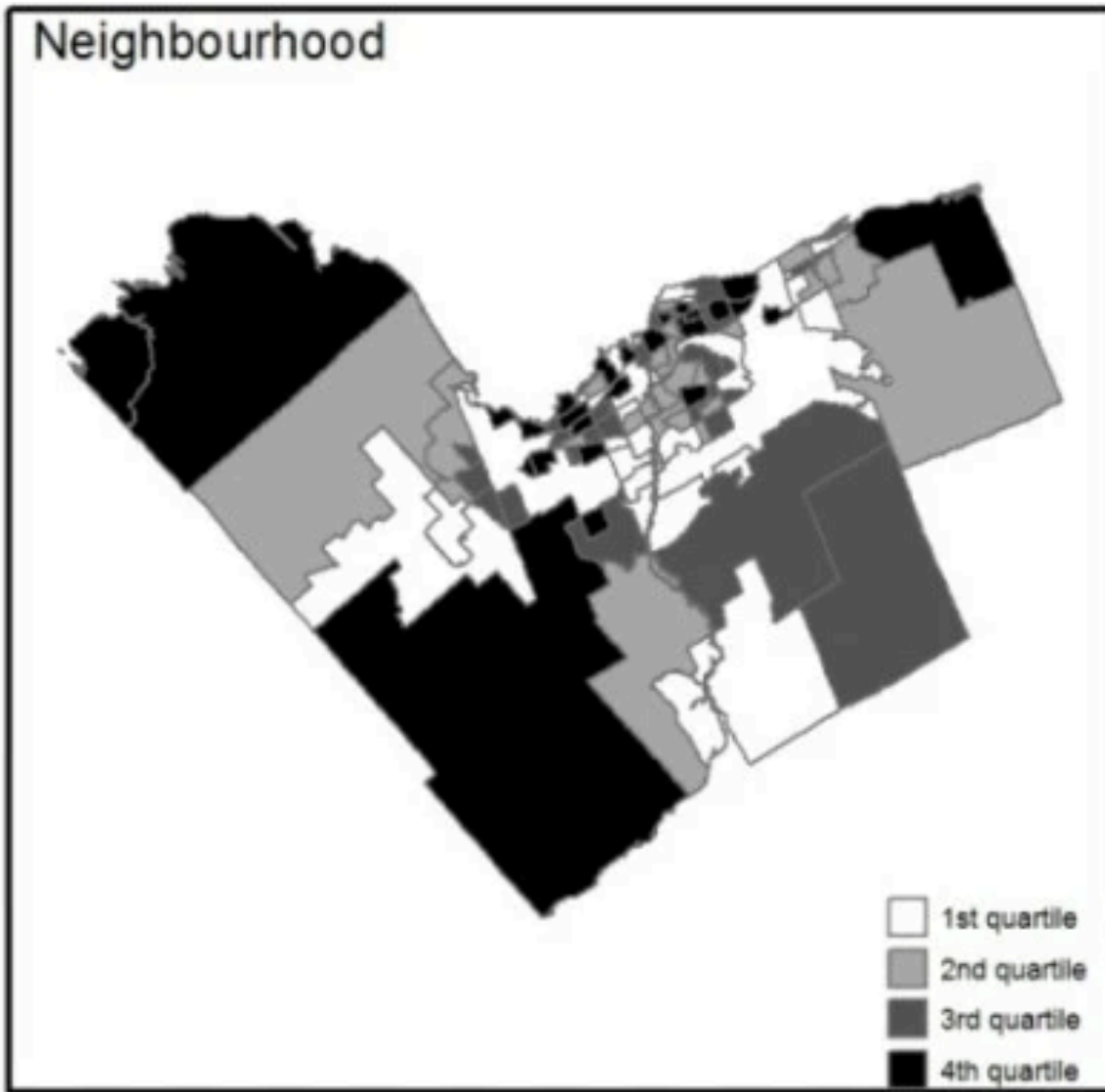
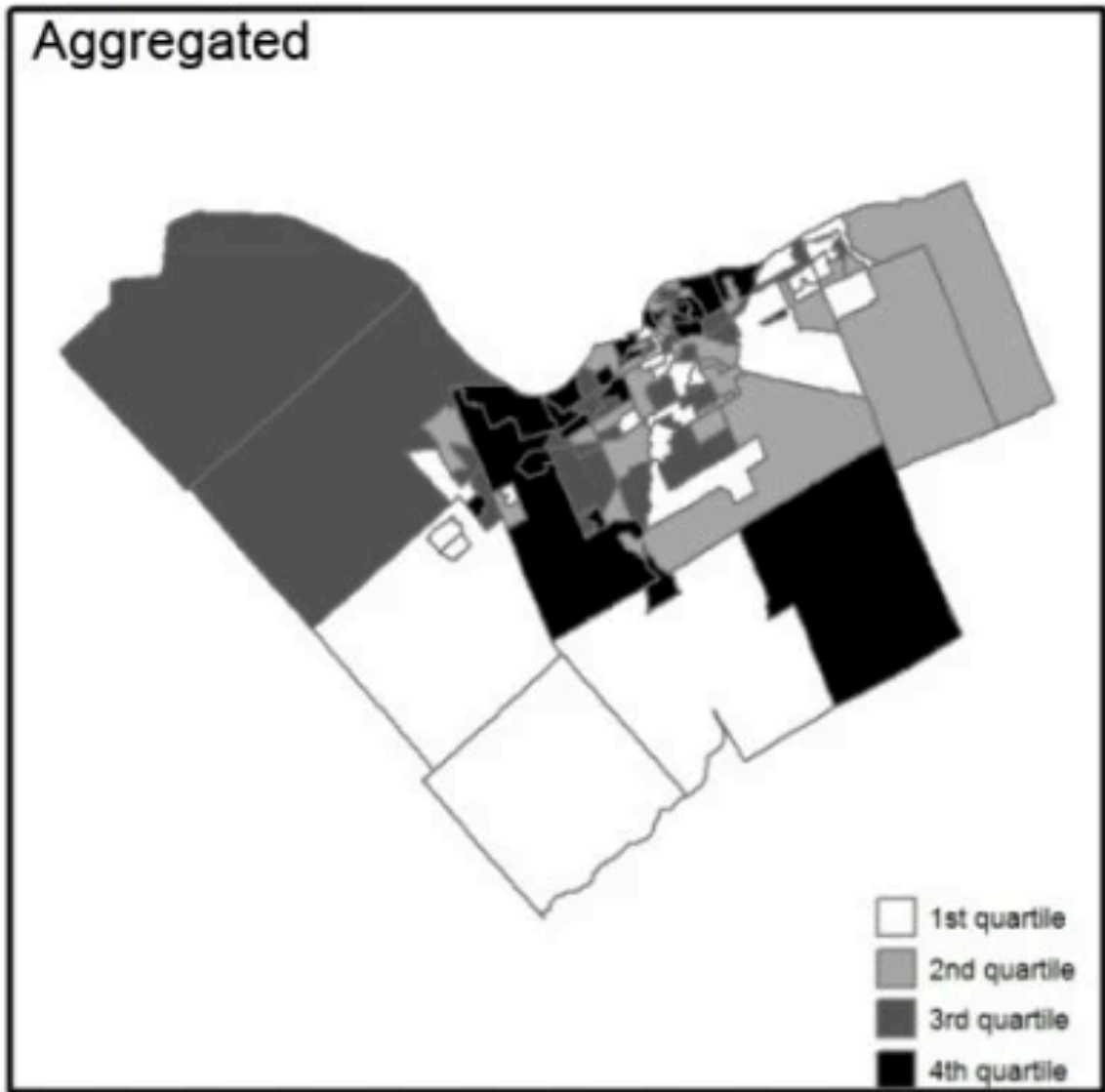


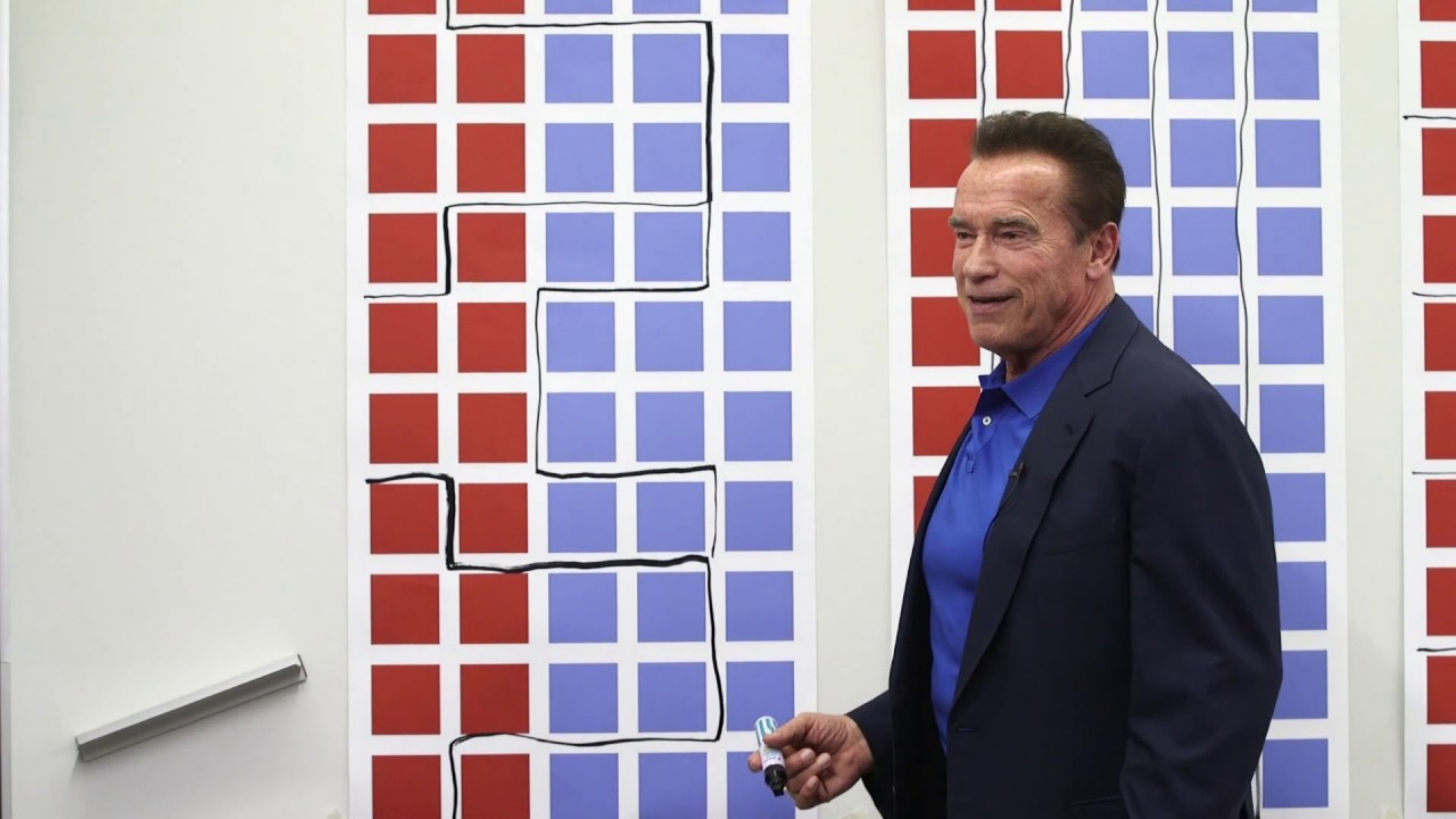
A common source of bias in spatial aggregation: MAUP



Pollution

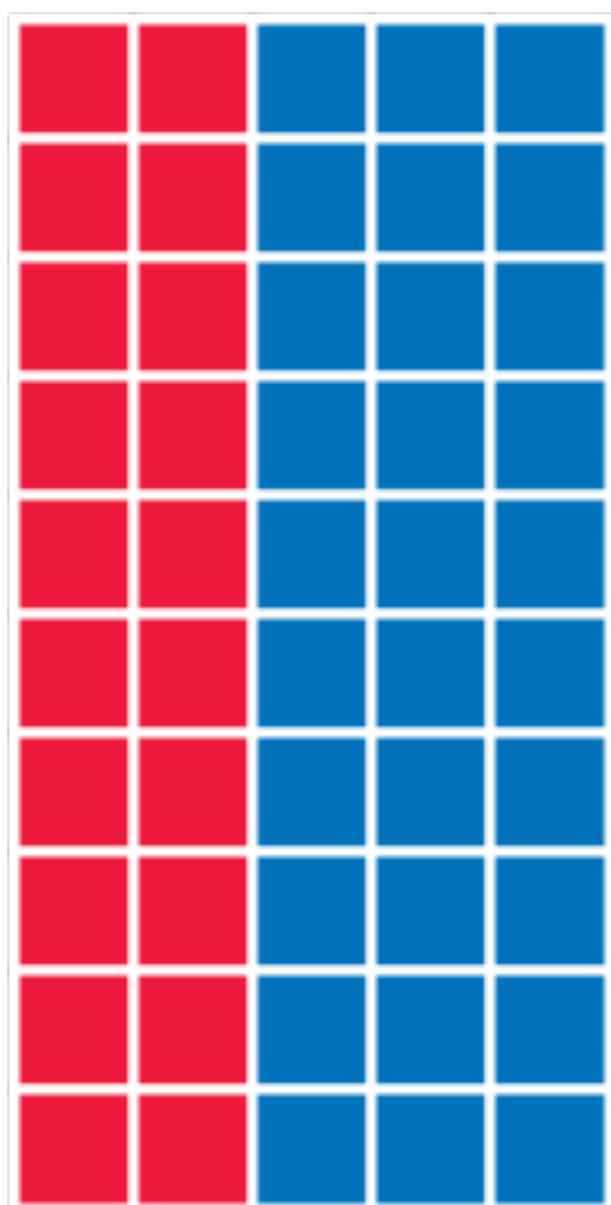
Respiratory Morbidity



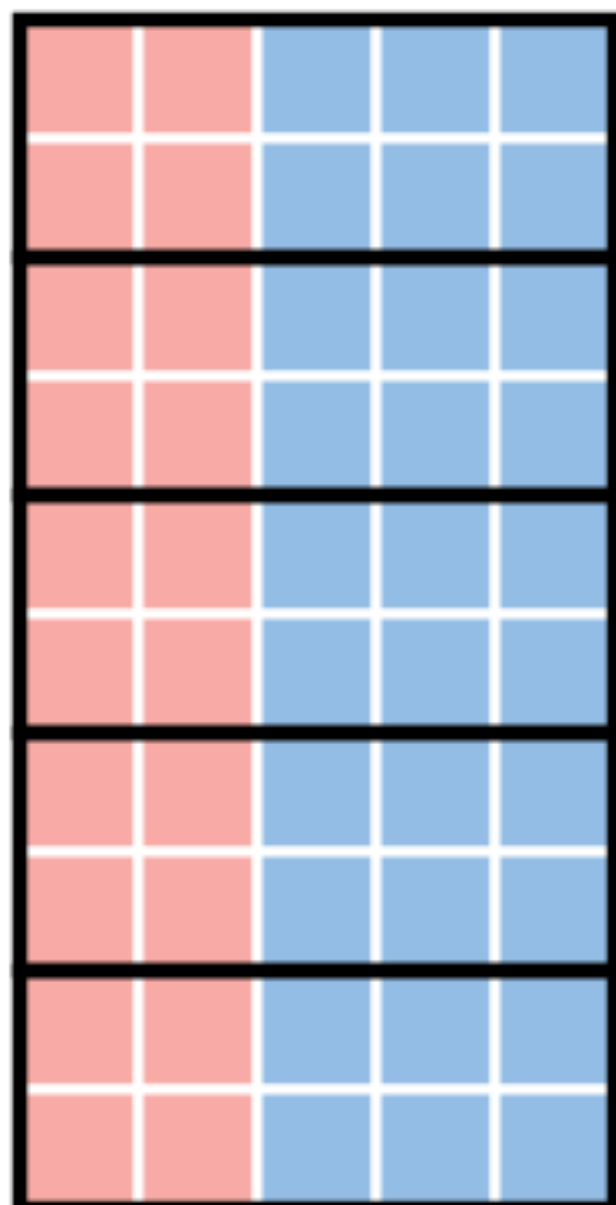


The MAUP is abused for Gerrymandering

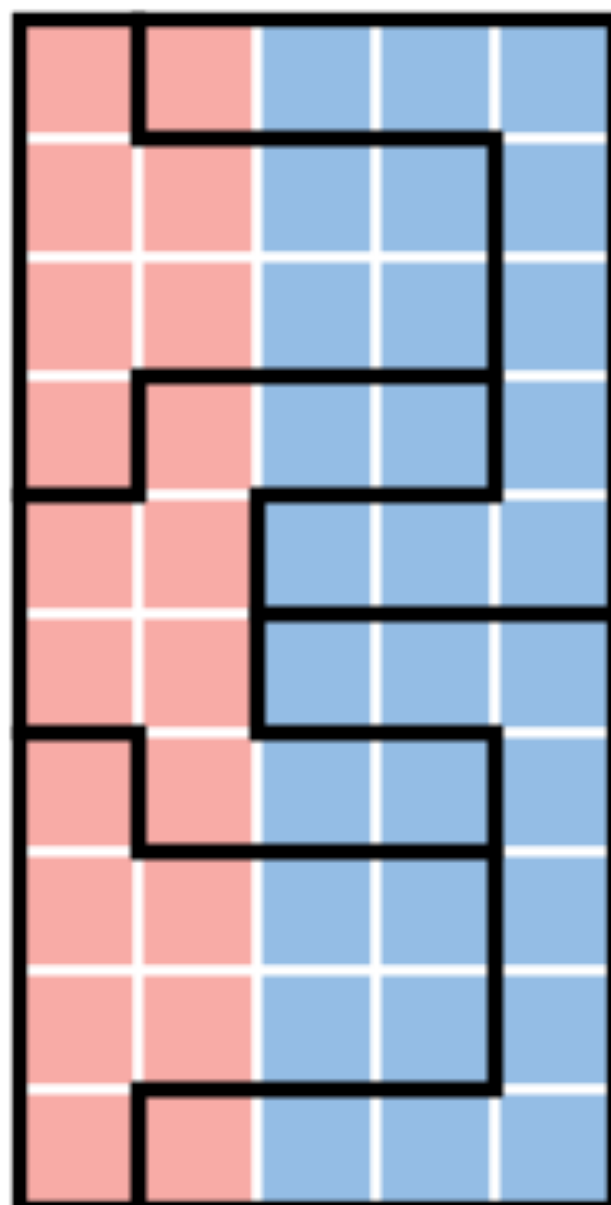
HOW TO STEAL AN ELECTION



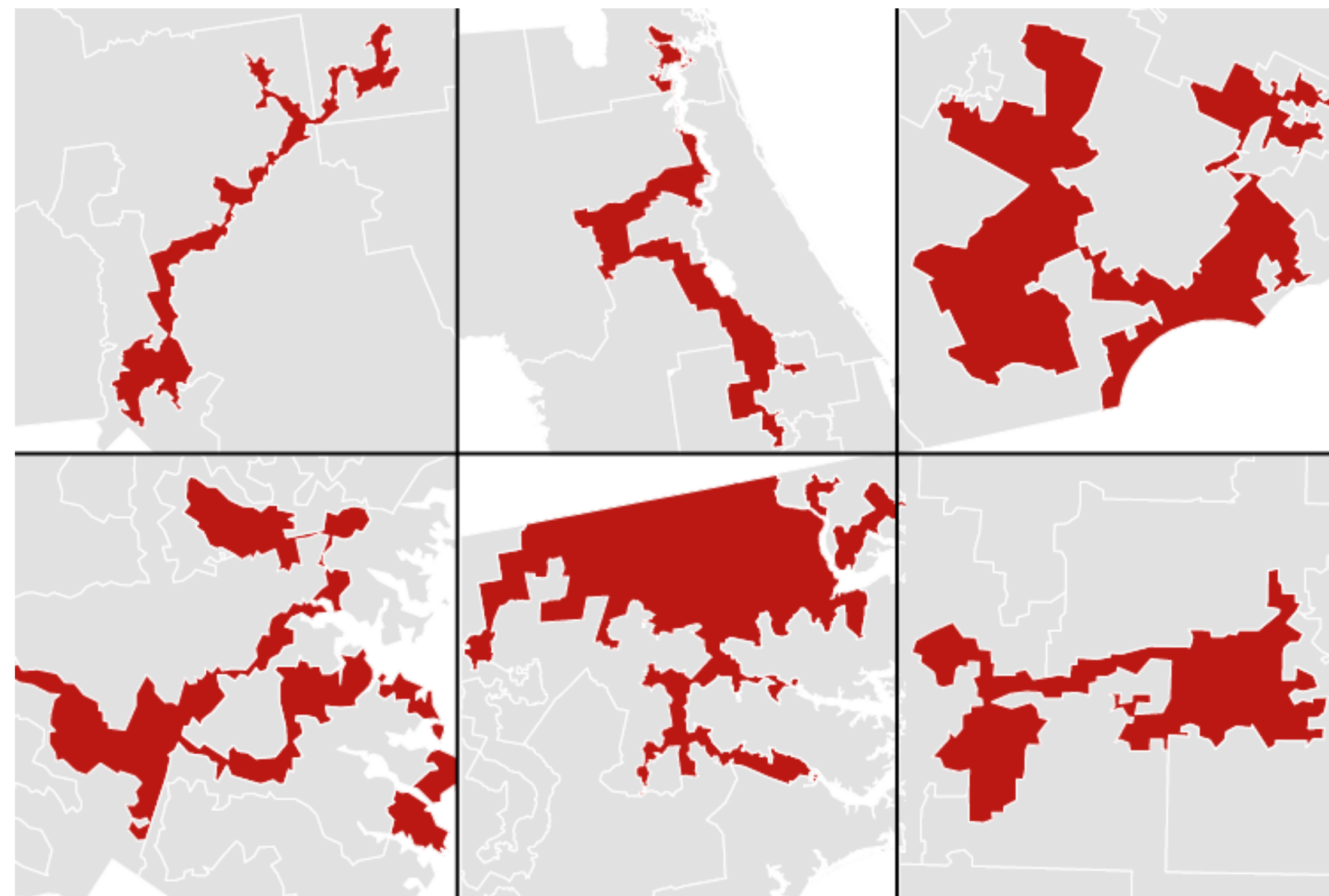
50 PRECINCTS
60% BLUE
40% RED



5 DISTRICTS
5 BLUE
0 RED
BLUE WINS



5 DISTRICTS
3 RED
2 BLUE
RED WINS





Can You Gerrymander Your Party to Power?

By [Ella Koeze](#), [Denise Lu](#) and [Charlie Smart](#)

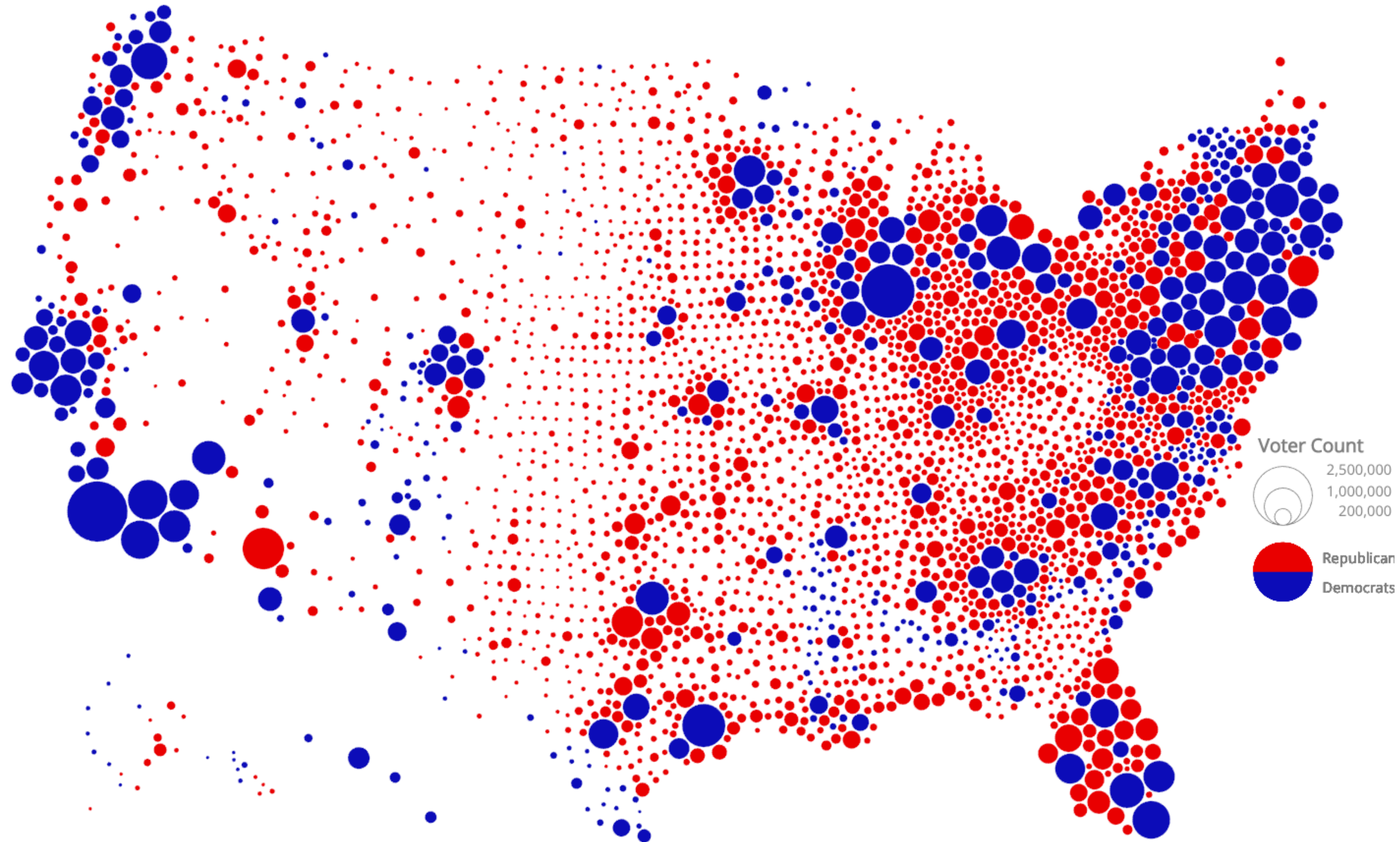
Gerrymandering is the [intentional distortion of political districts](#) to give one party an advantage, and it has been criticized for disenfranchising many voters and fueling deeper polarization.

To help you understand it better, we created an imaginary state called **Hexapolis**, where your only mission is to gerrymander your party to power.

Related bias: Area is not population



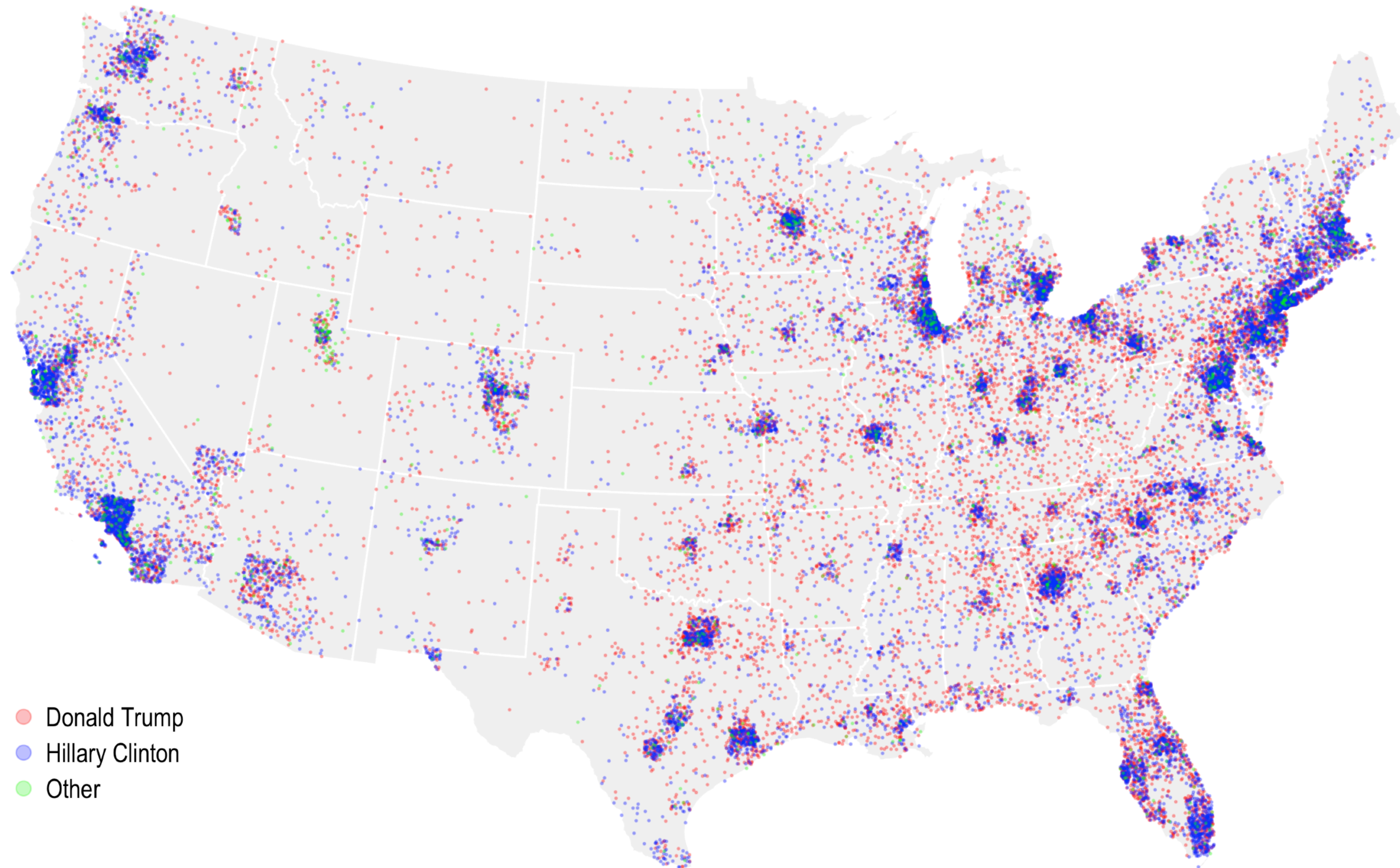
Related bias: Area is not population



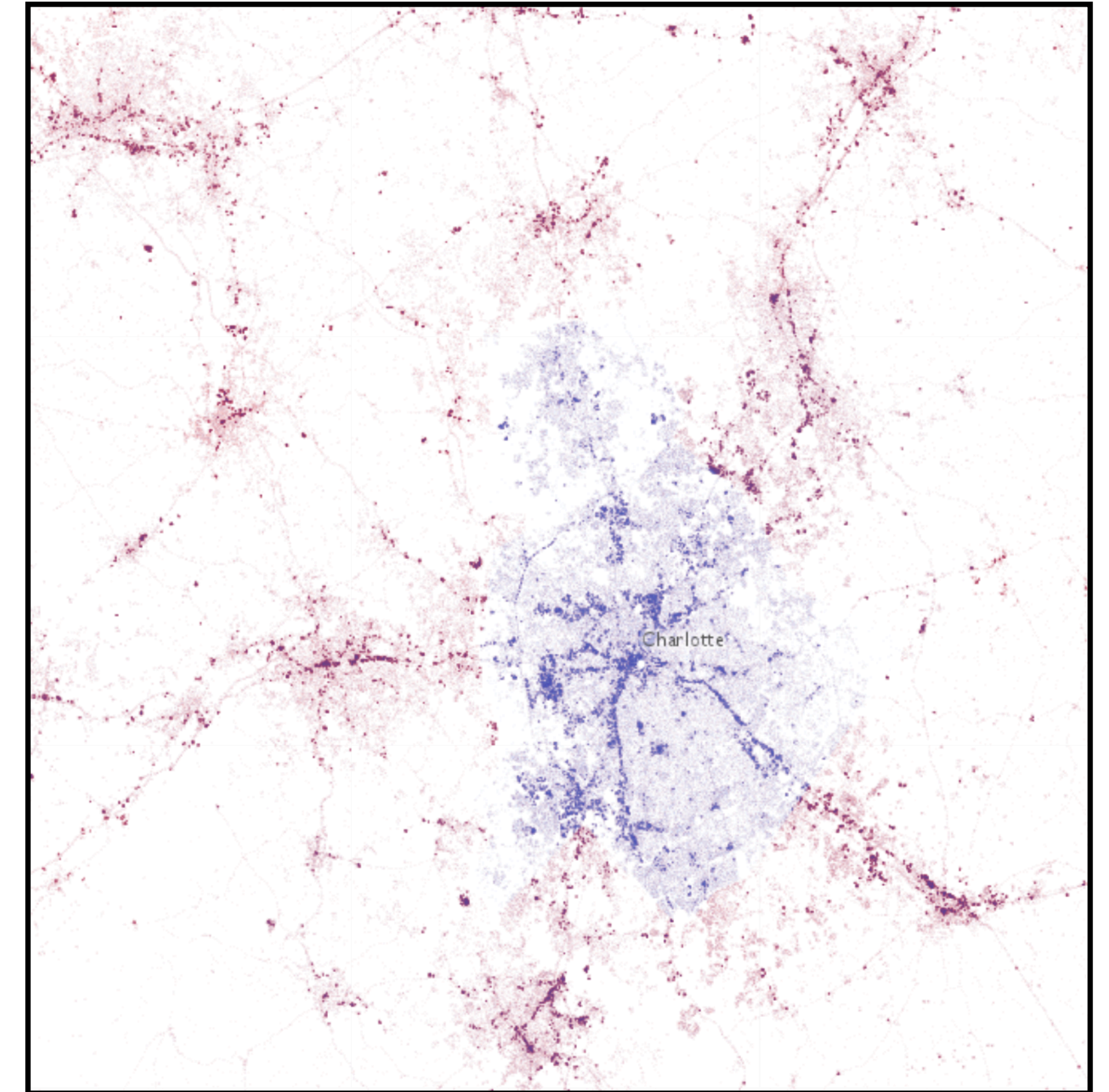
A dot density map can reduce the bias

US 2016 Presidential election results

Each dot represents 5,000 voters



- Donald Trump
- Hillary Clinton
- Other



<https://www.andybeger.com/2018/05/11/dot-density-map-of-the-2016-election/>
<https://www.maproomblog.com/2018/04/kenneth-fields-dot-density-election-map-redux/>

Always mind the MAUP
when exploring
aggregated data

Classification = Transforming continuous into categorical

Student ID	Year	Grade Point Average (GPA)	...
▶ 1034262	Senior	3.24	...
1052663	Sophomore	3.51	...
1082246	Freshman	3.62	...
⋮	⋮		

reject
accept
accept

How many categories should there be? If 2: **Binarization**
How should the values be mapped? What should be the thresholds?

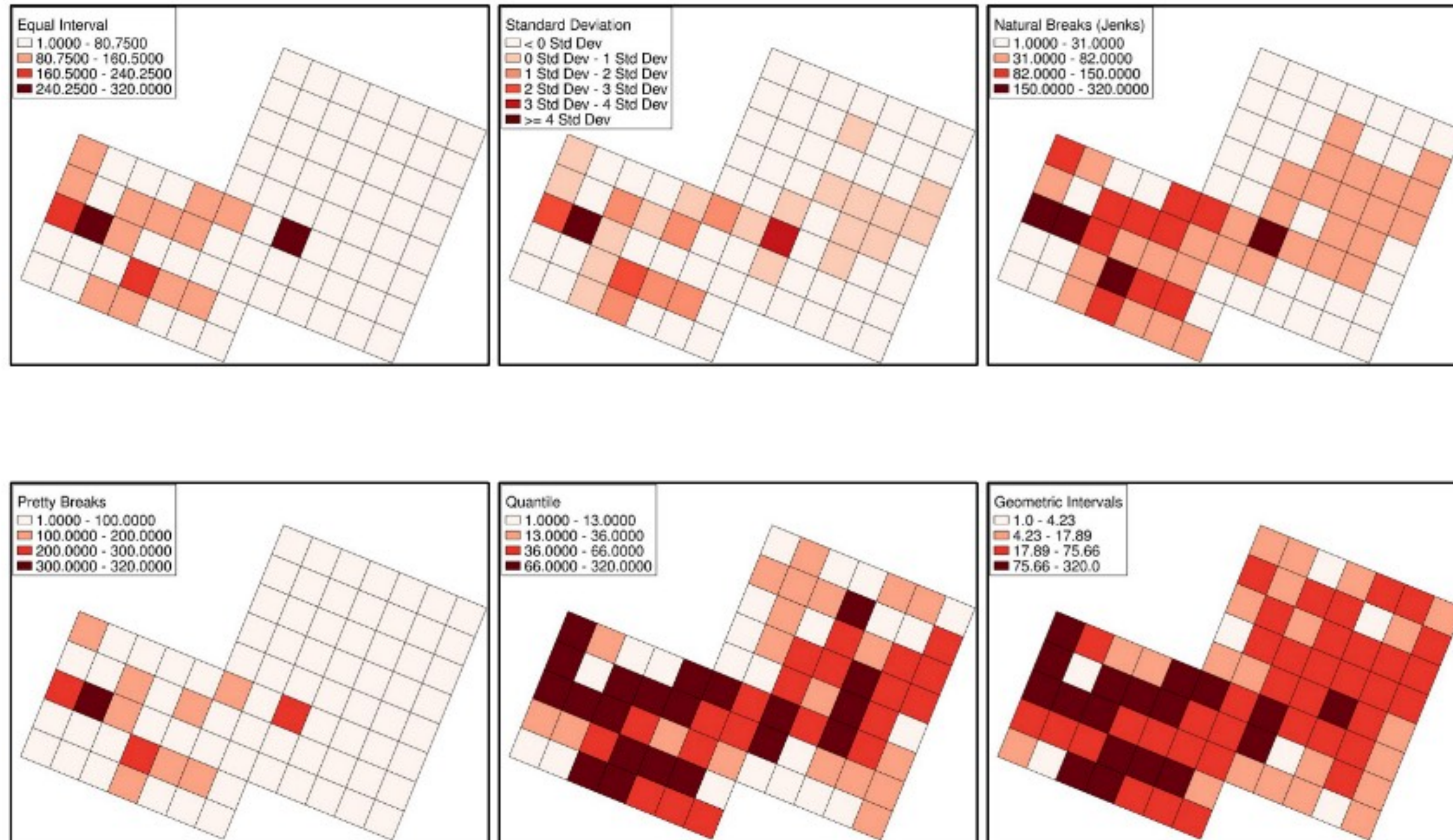
Number of finds in an excavation grid

Group experiment

Each of 3 groups describes their
picture in 1 sentence

Data can be classified very differently

Example: Weight of finds in an excavation grid



Same data, different split points

Classification: Grouping similar phenomena to gain relative simplicity in communication and interpretation. Also known as **binning**.

Formally, you must select thresholds to classify your data

The classification problem is to define class boundaries such that

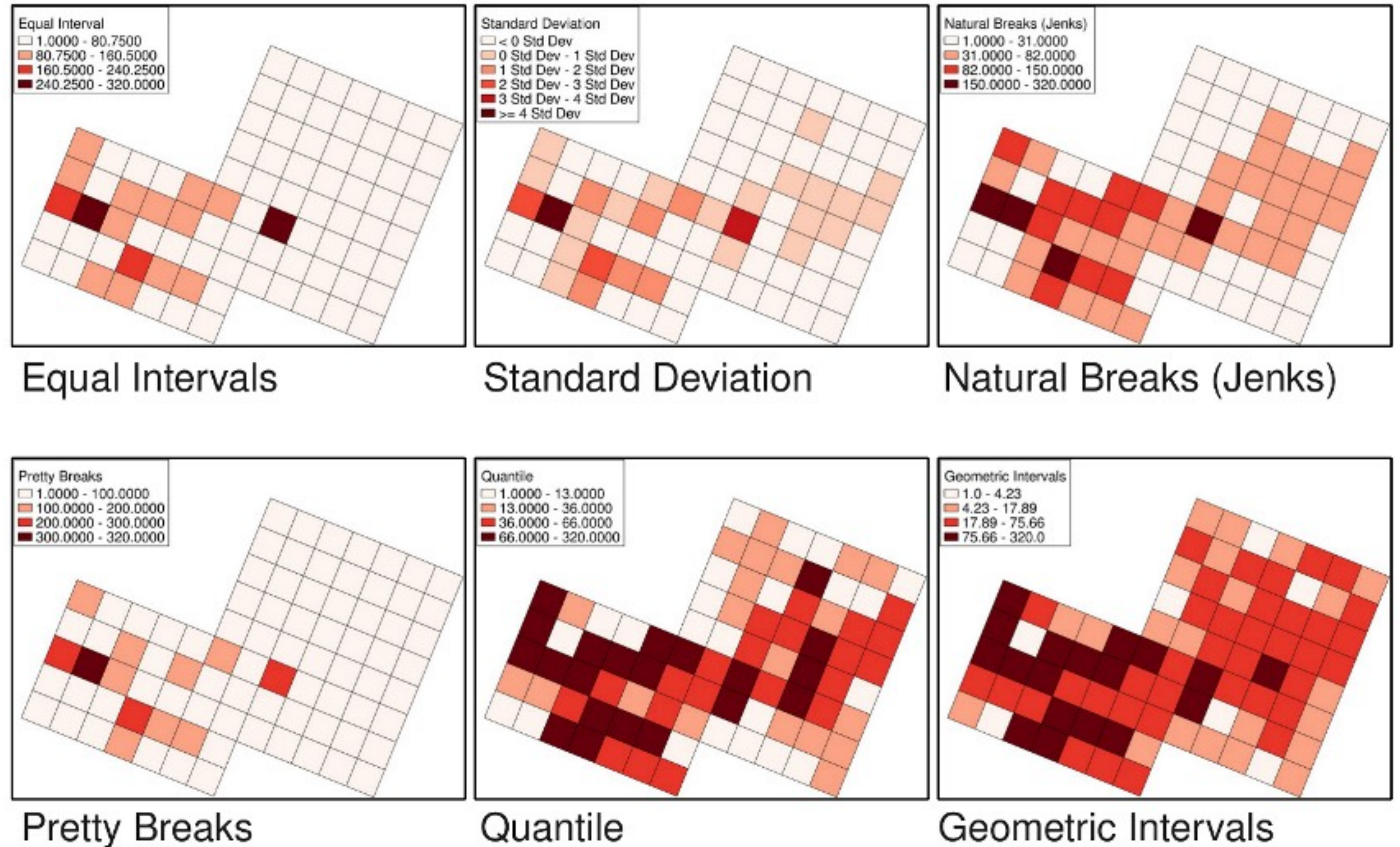
$$c_j < y_i \leq c_{j+1} \quad \forall y_i \in C_j$$

where y_i is the value of the attribute for spatial location i ,
 j is a class index,
 c_j represents the lower bound of interval j .

The choice of class boundaries defines the **classification scheme**

mapclassify:

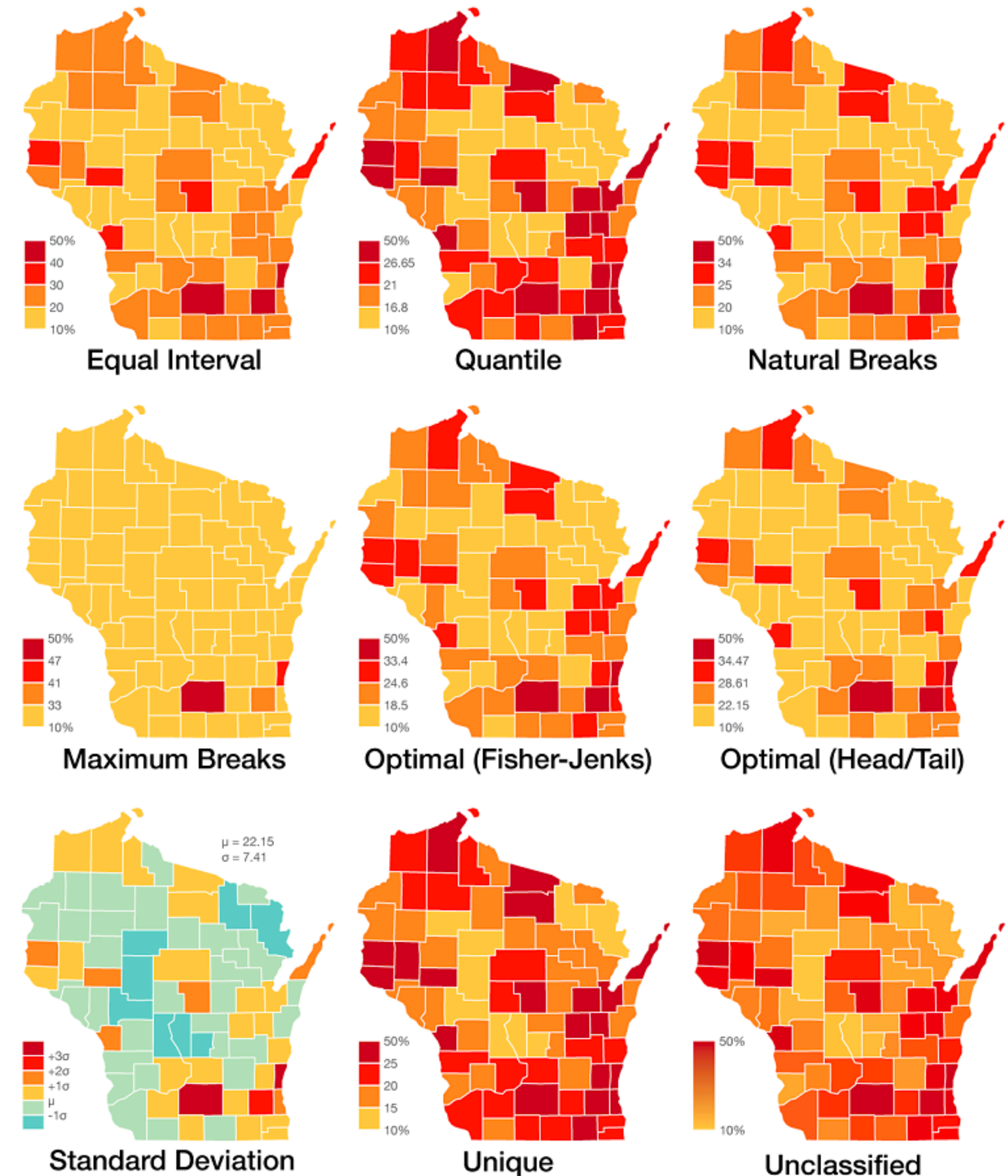
'Head-Tail Breaks',
'Fisher-Jenks',
'Maximum Breaks',
'Equal Interval',
'MaxP',
'Quantiles',
'Jenks Caspall',
'Mean-Standard Deviation'



Data can be classified very differently

Same data,
Same spatial units,
Same colors,
Different classification

Percentage of residents over 25 with a Bachelor's degree

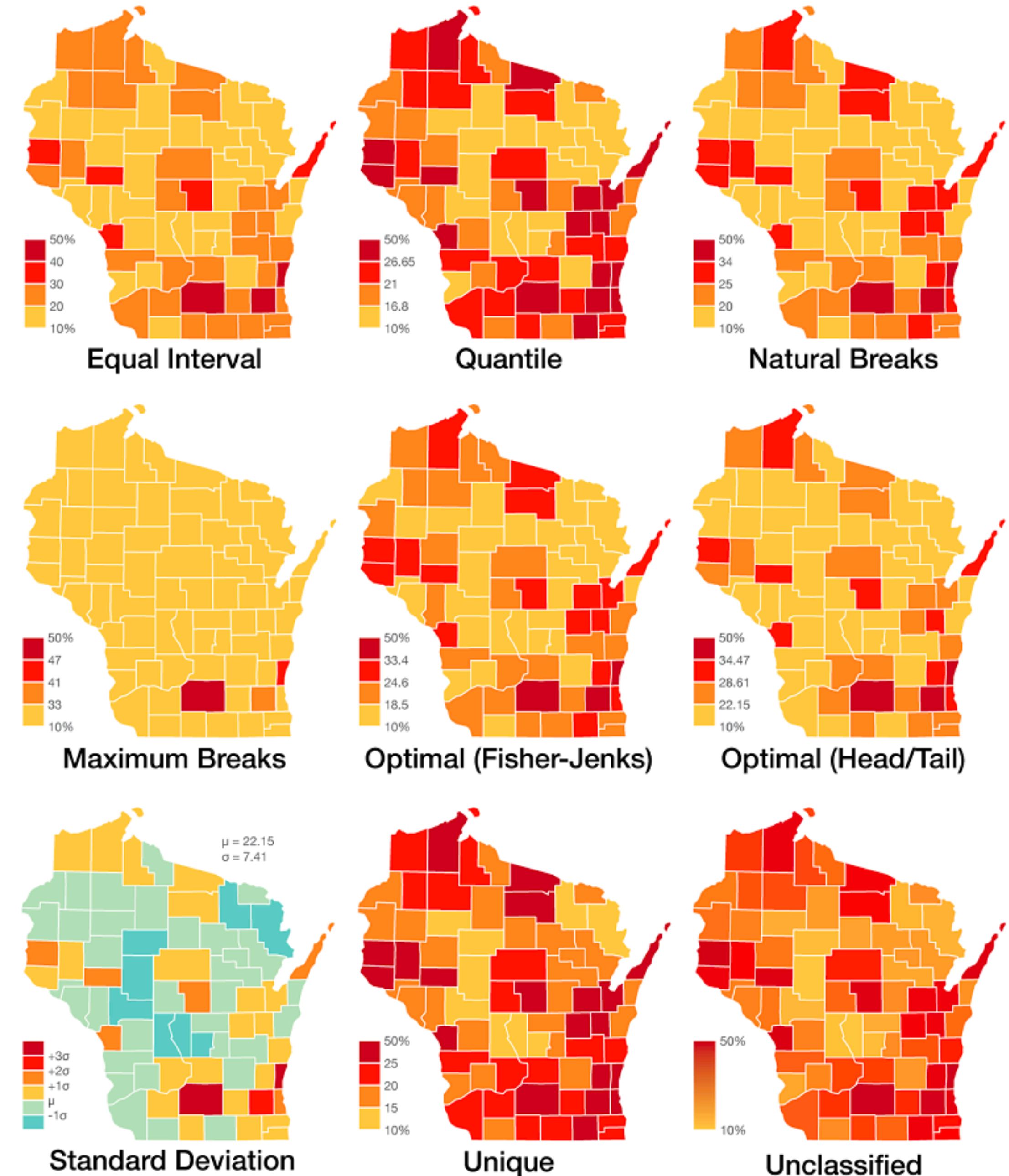


Data can be classified very differently

A data set does not have a “perfect” choropleth map

As with MAUP, you can create *arbitrary* interpretations of the same data!

Percentage of residents over 25 with a Bachelor's degree



The choice of class boundaries defines the **classification scheme**

mapclassify:

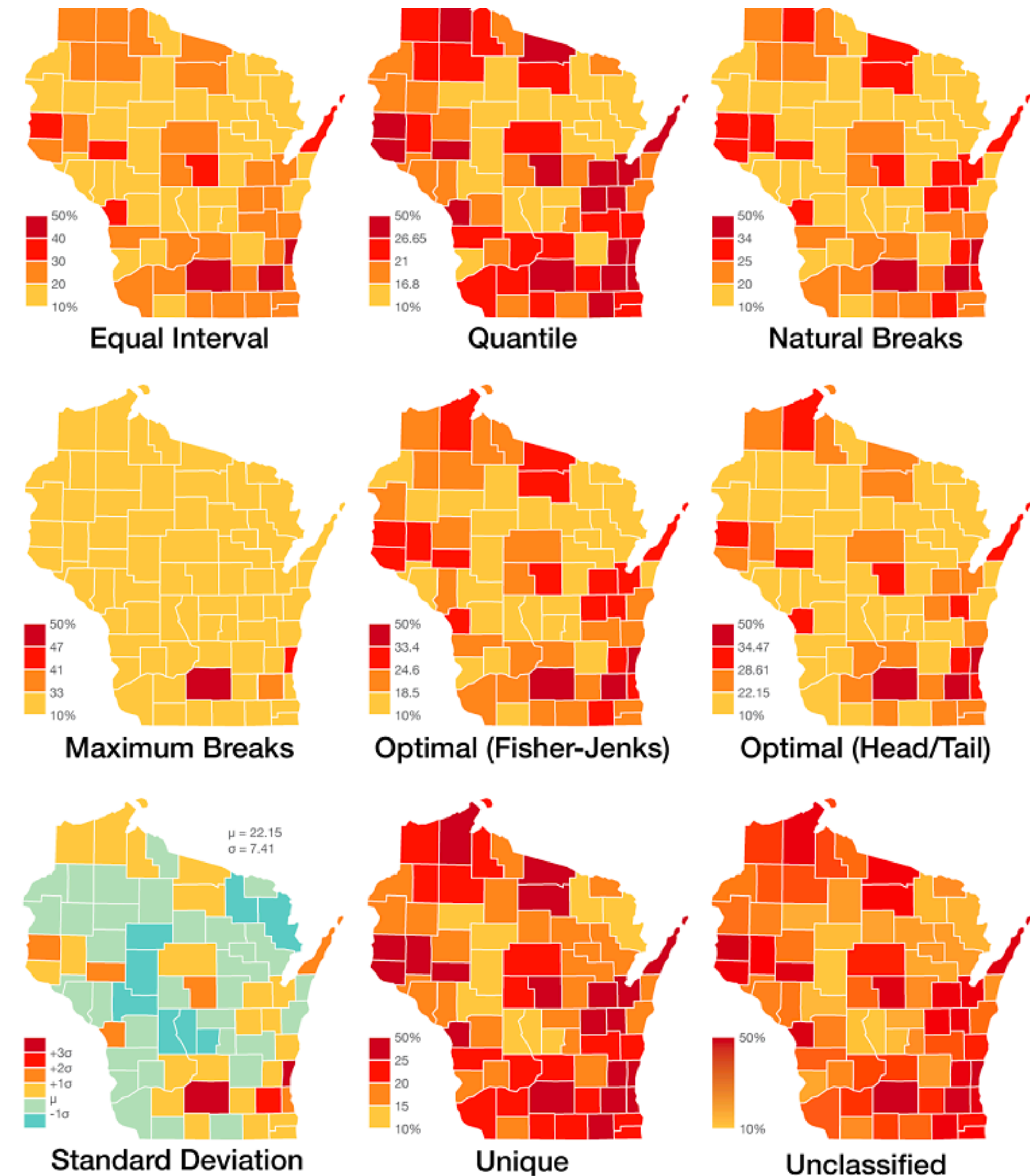
'EqualInterval',
'FisherJenks',

...
'HeadTailBreaks',
'JenksCaspall',

...
'MaximumBreaks',
'NaturalBreaks',
'Percentiles',

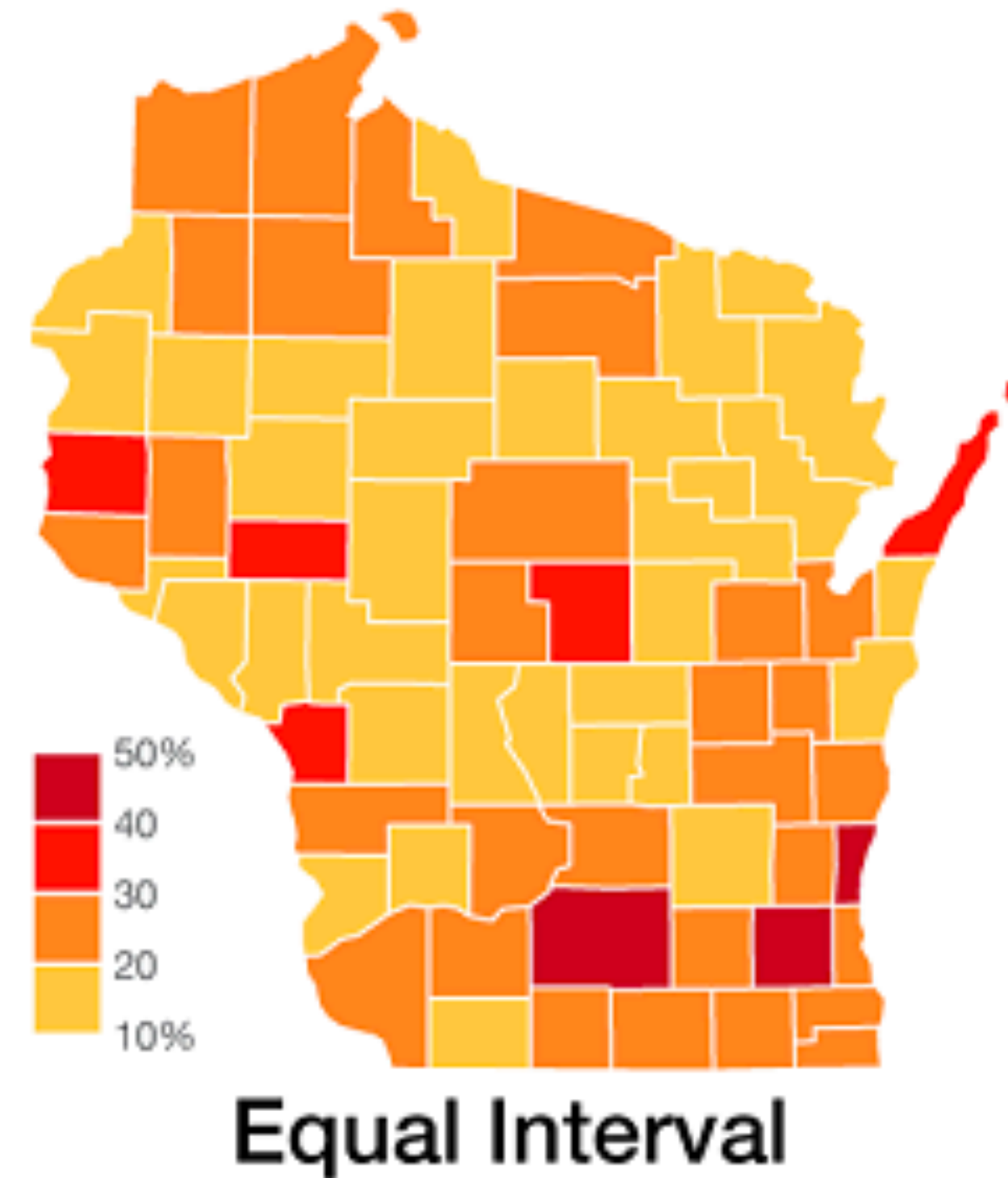
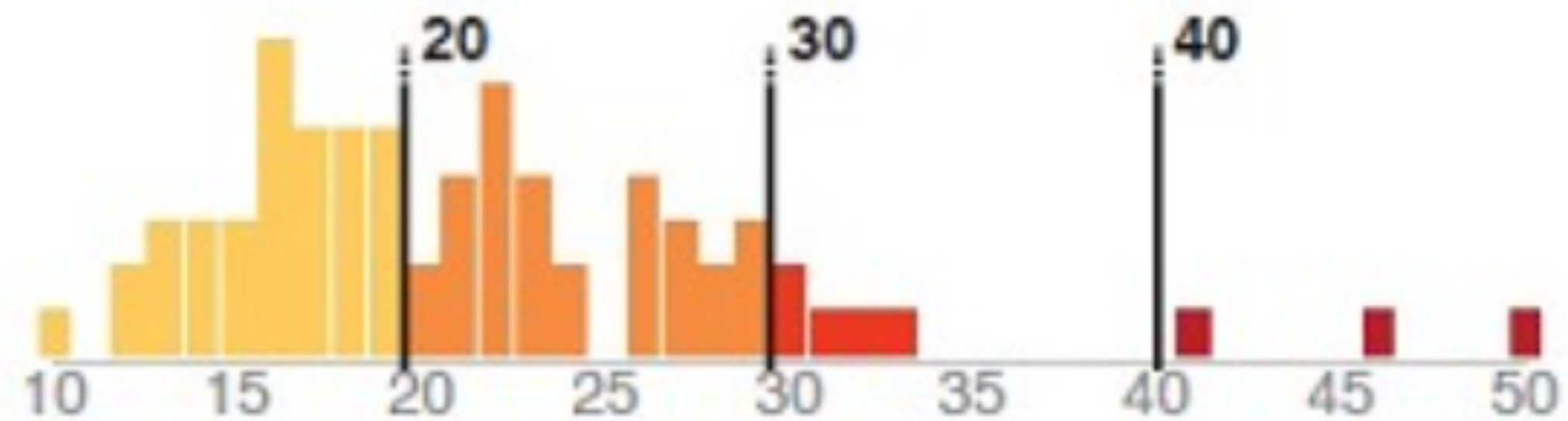
...
'Quantiles',
'StdMean',
'UserDefined',

Percentage of residents over 25 with a Bachelor's degree



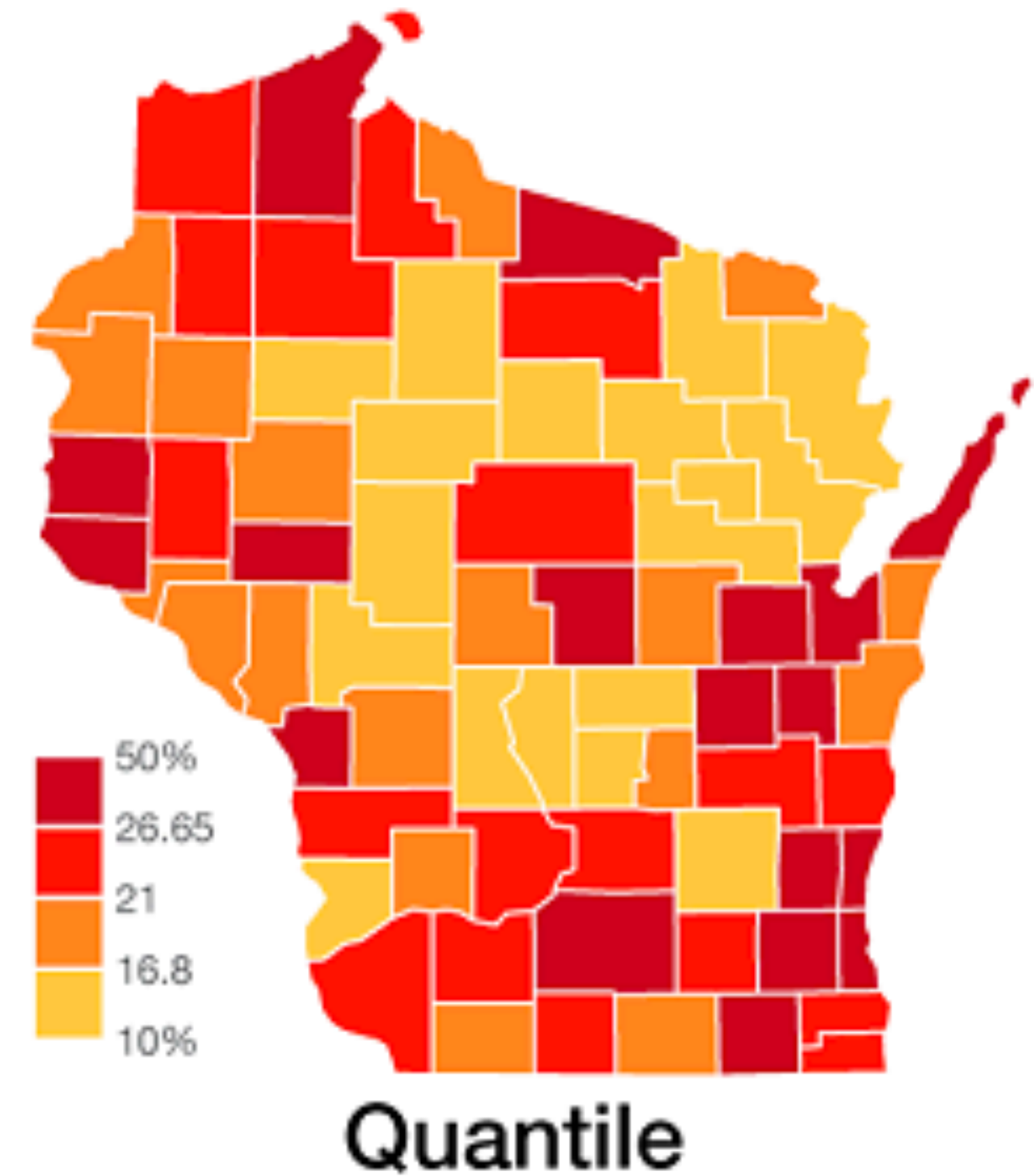
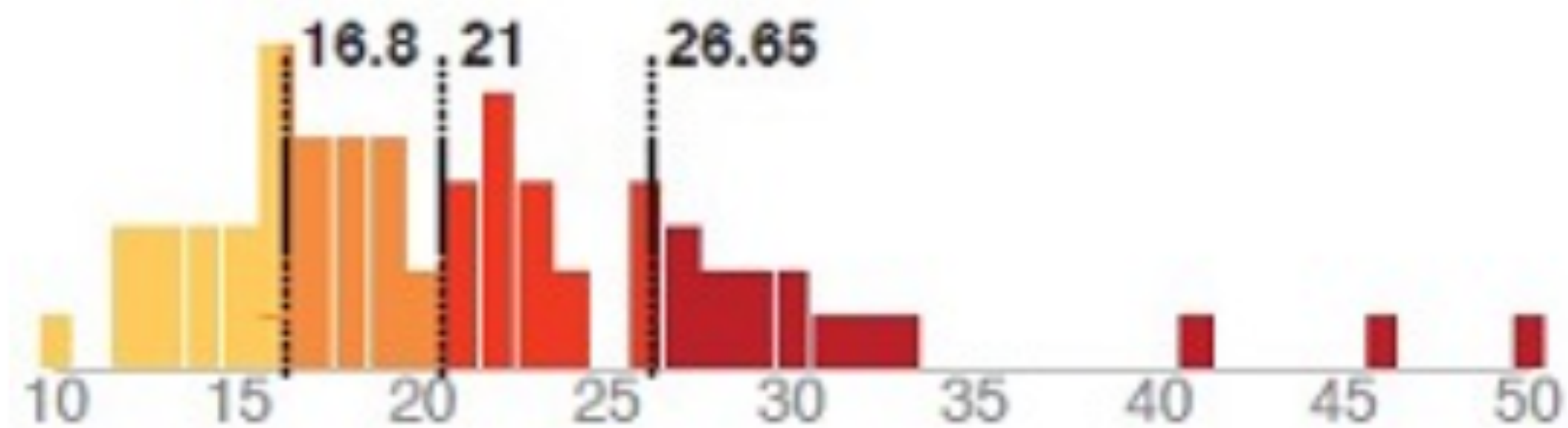
Equal Interval

For uniformly distributed data with familiar data ranges.



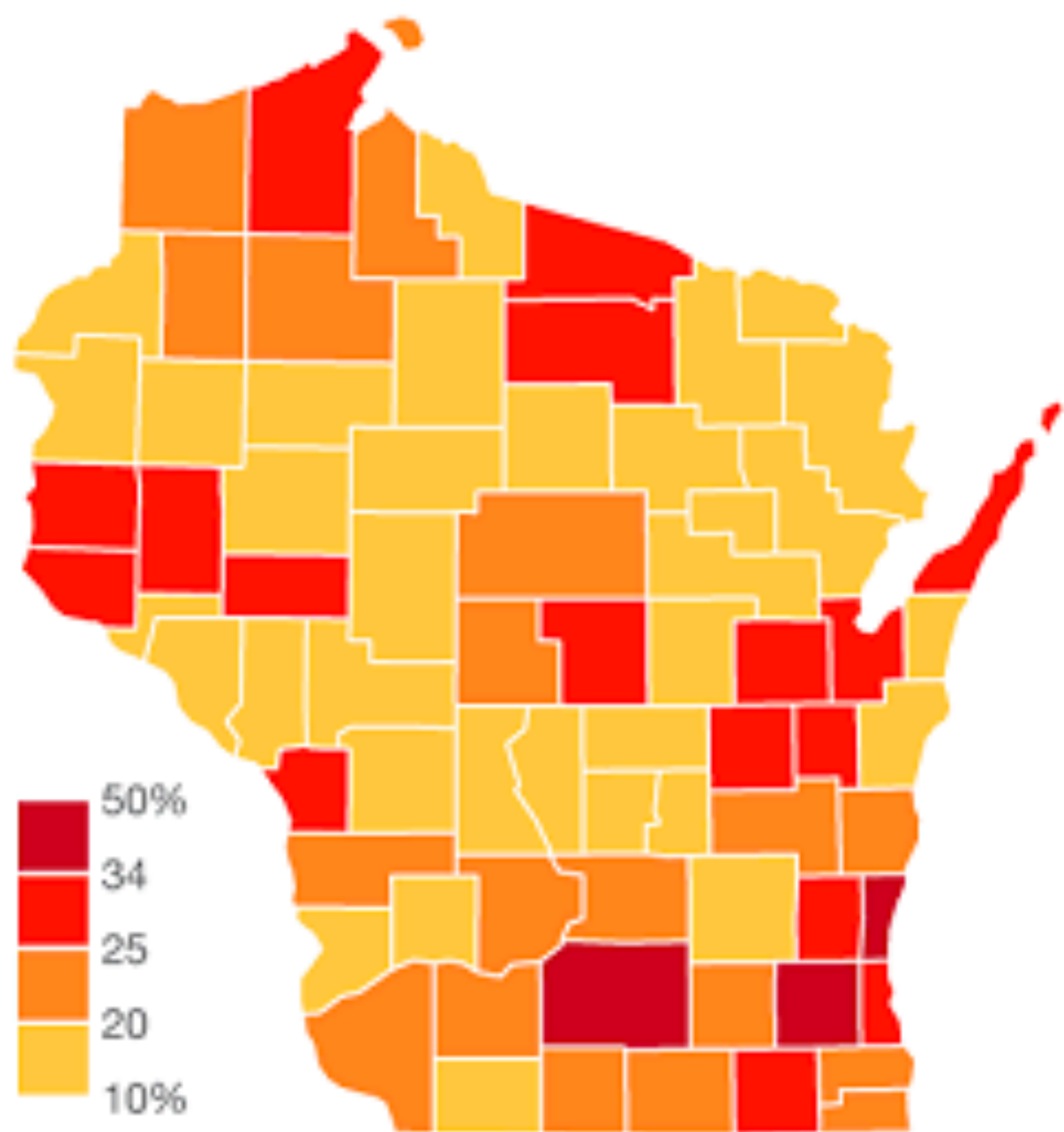
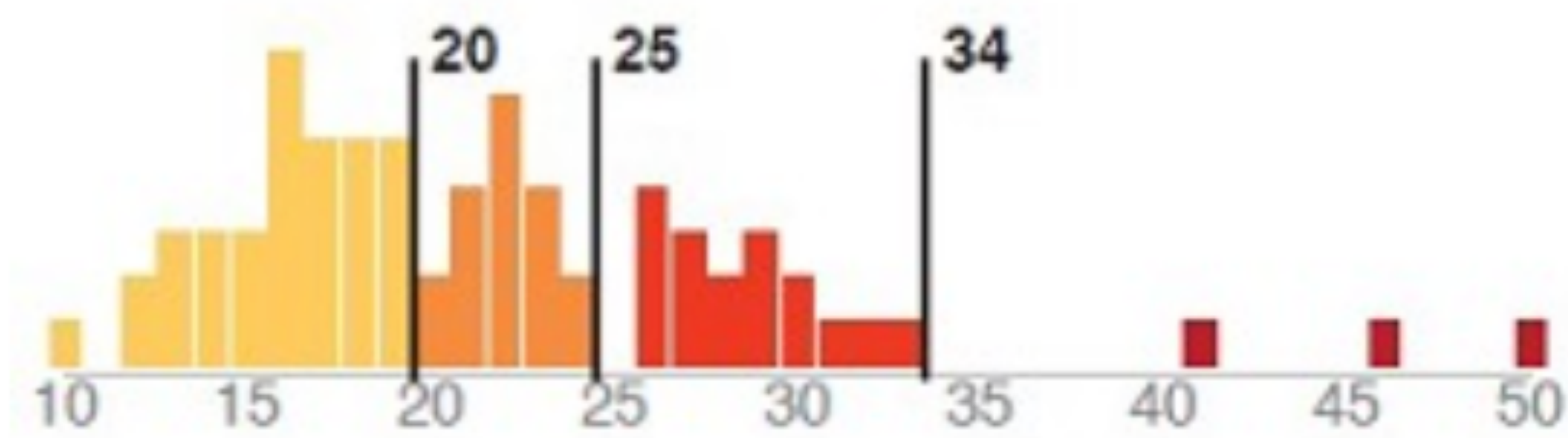
Quantiles (Equal Count)

For evenly distributed data and ordinal data



Natural Breaks

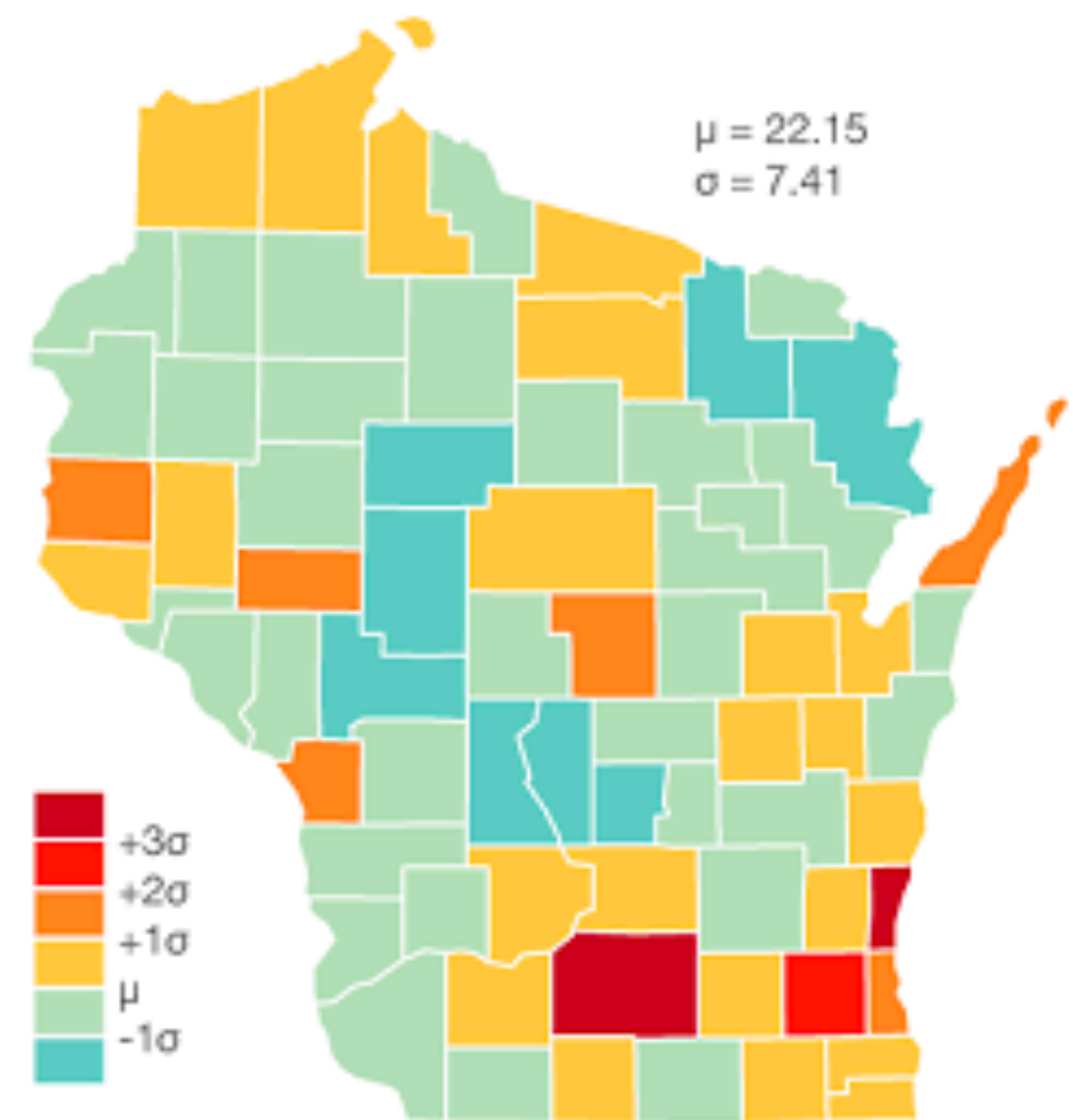
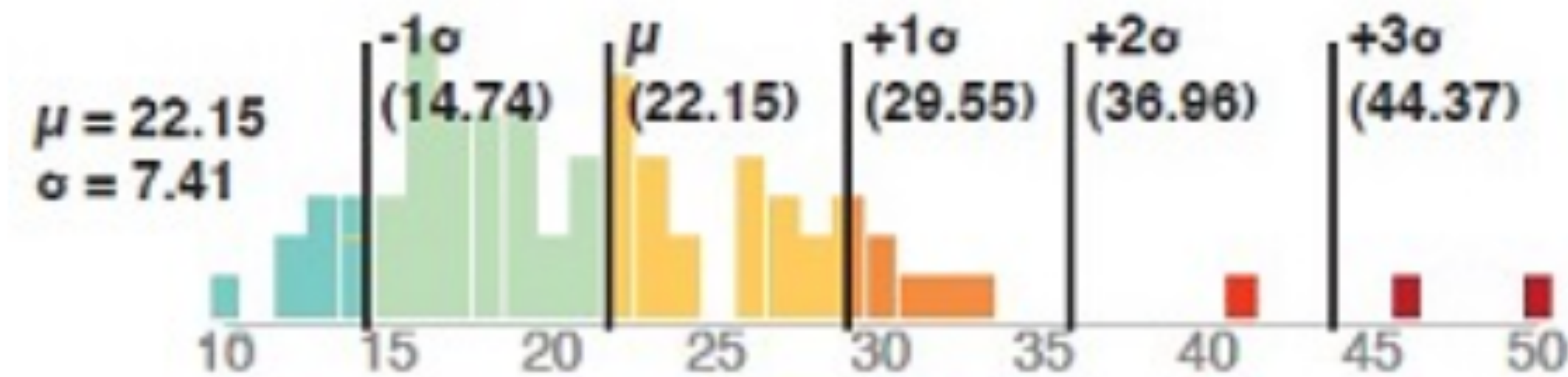
For clustered data



Natural Breaks

Mean-Standard Deviation

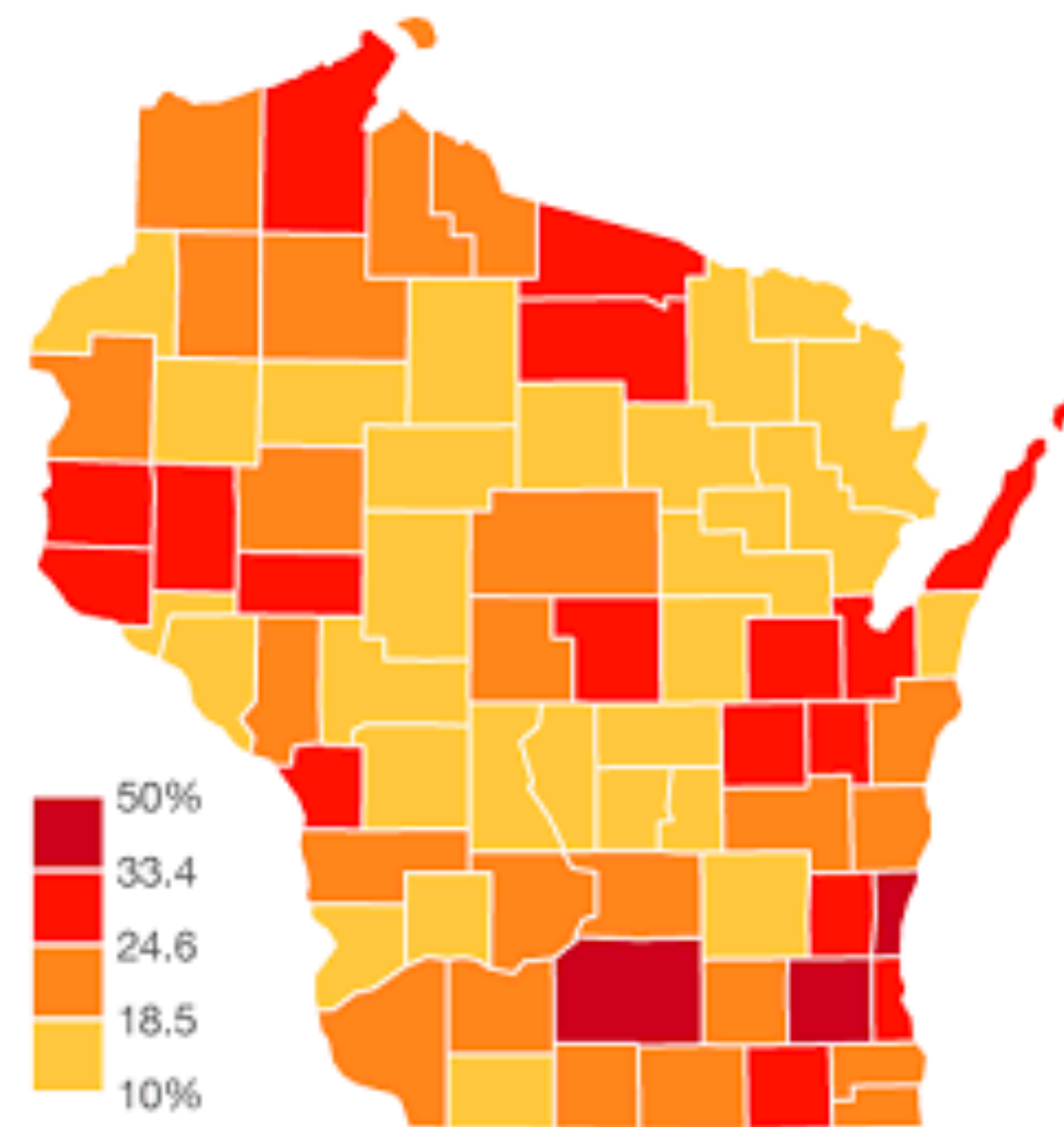
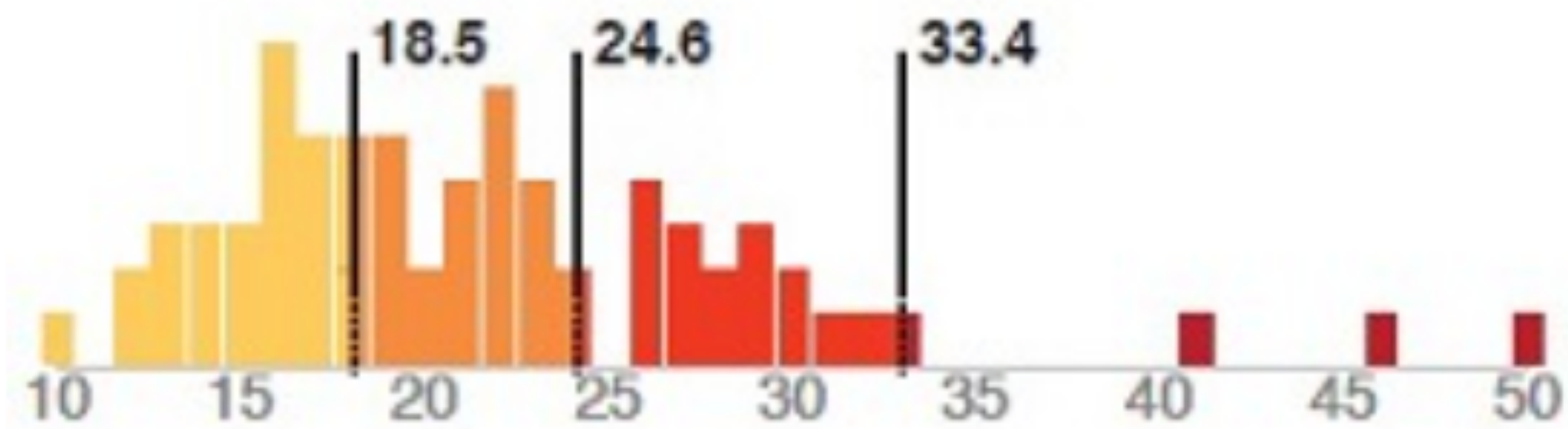
For normally distributed data



Standard Deviation

Jenks-Caspall & Fisher-Jenks

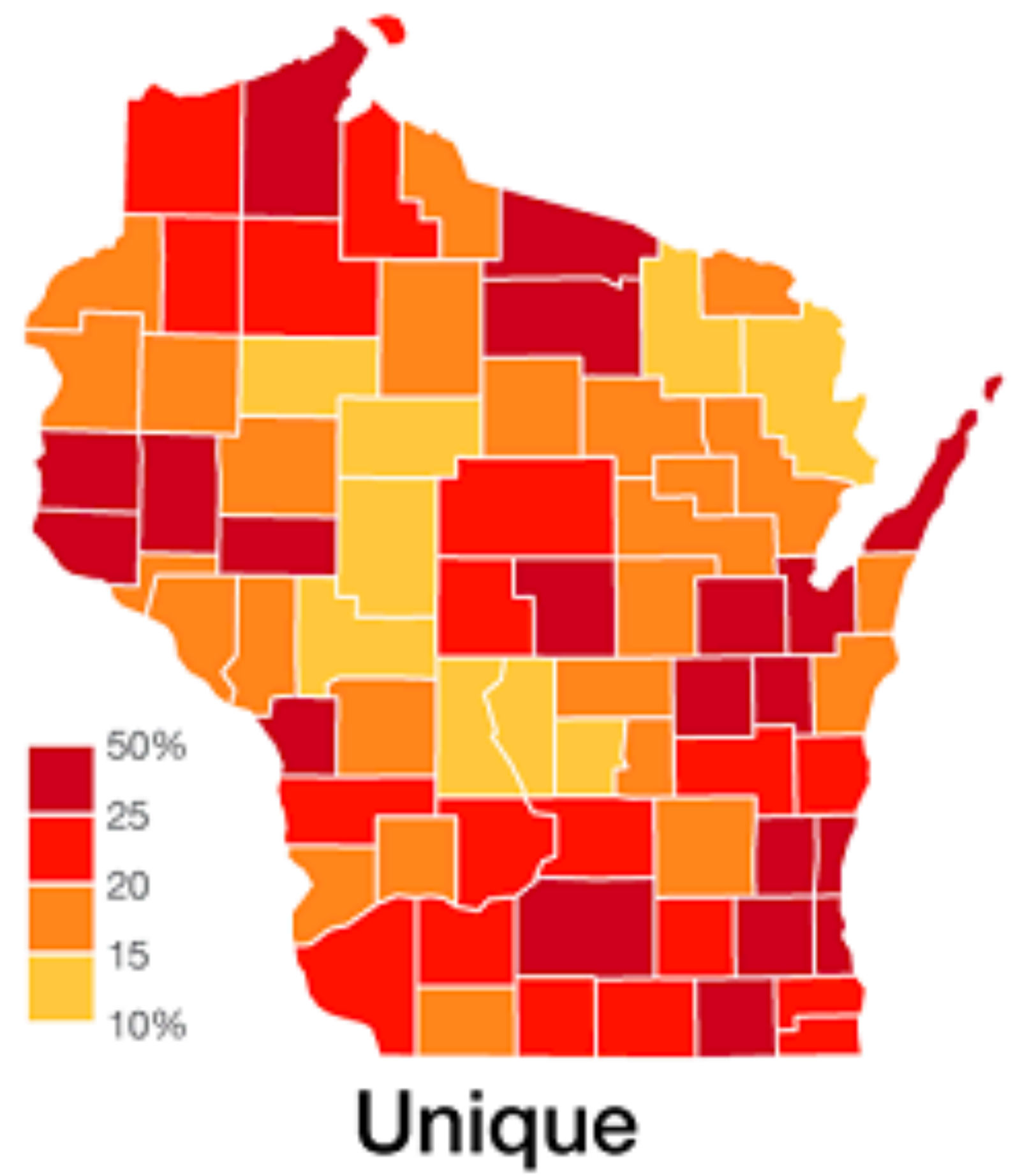
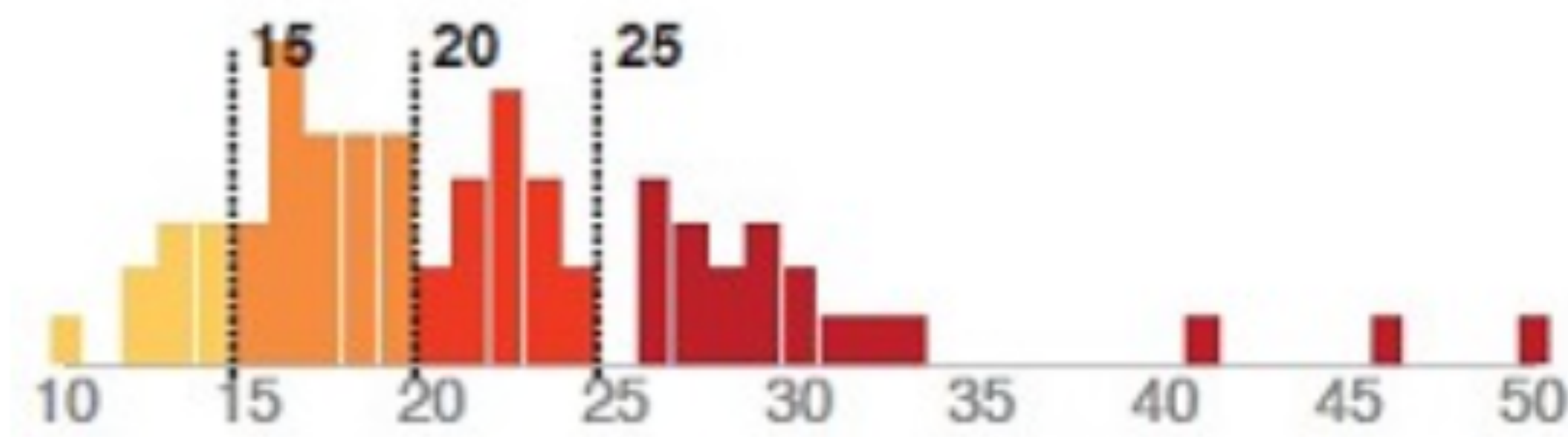
For clustered and skewed data



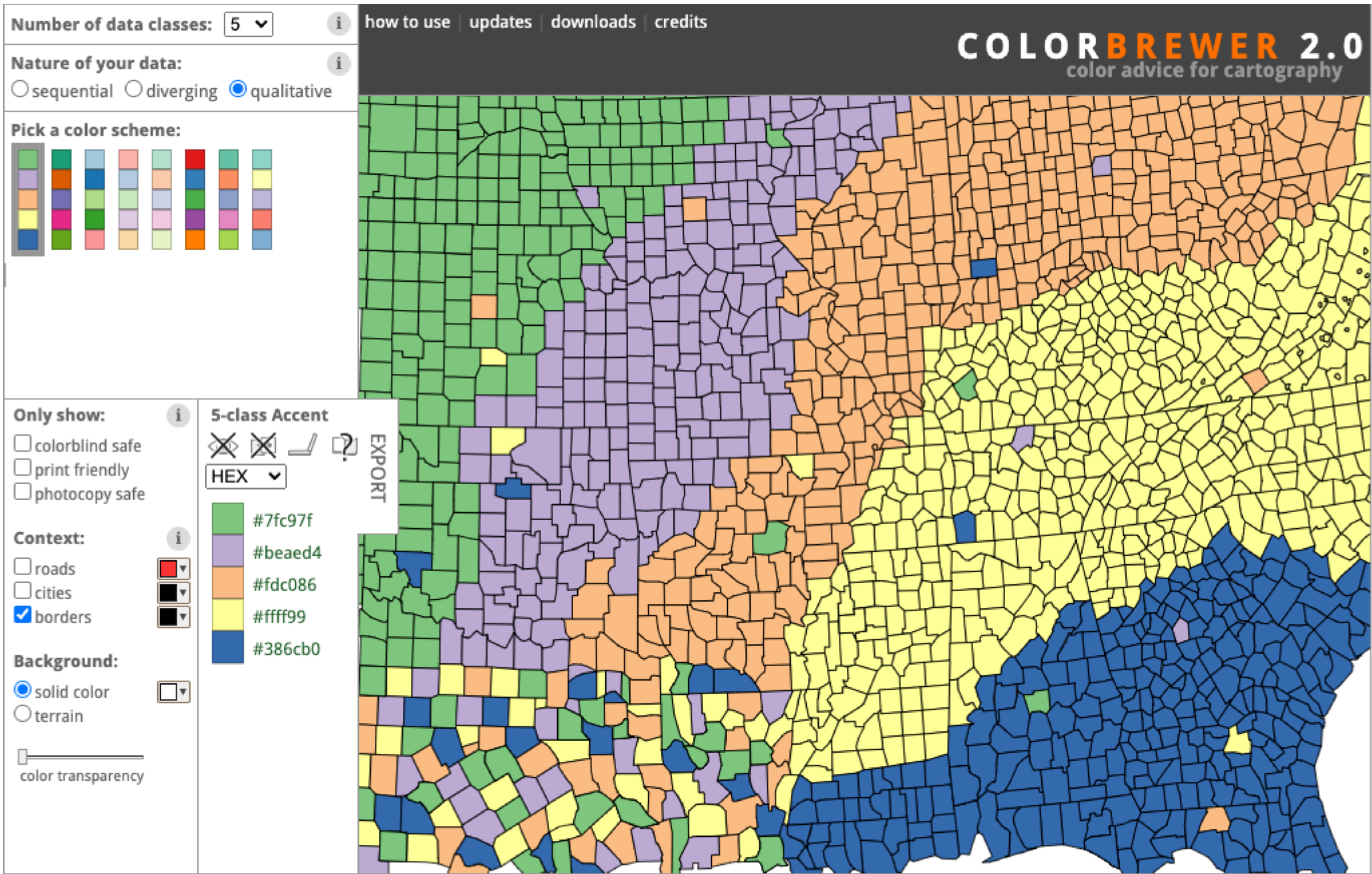
Optimal (Fisher-Jenks)

Unique/Manual

For data where key numbers are important



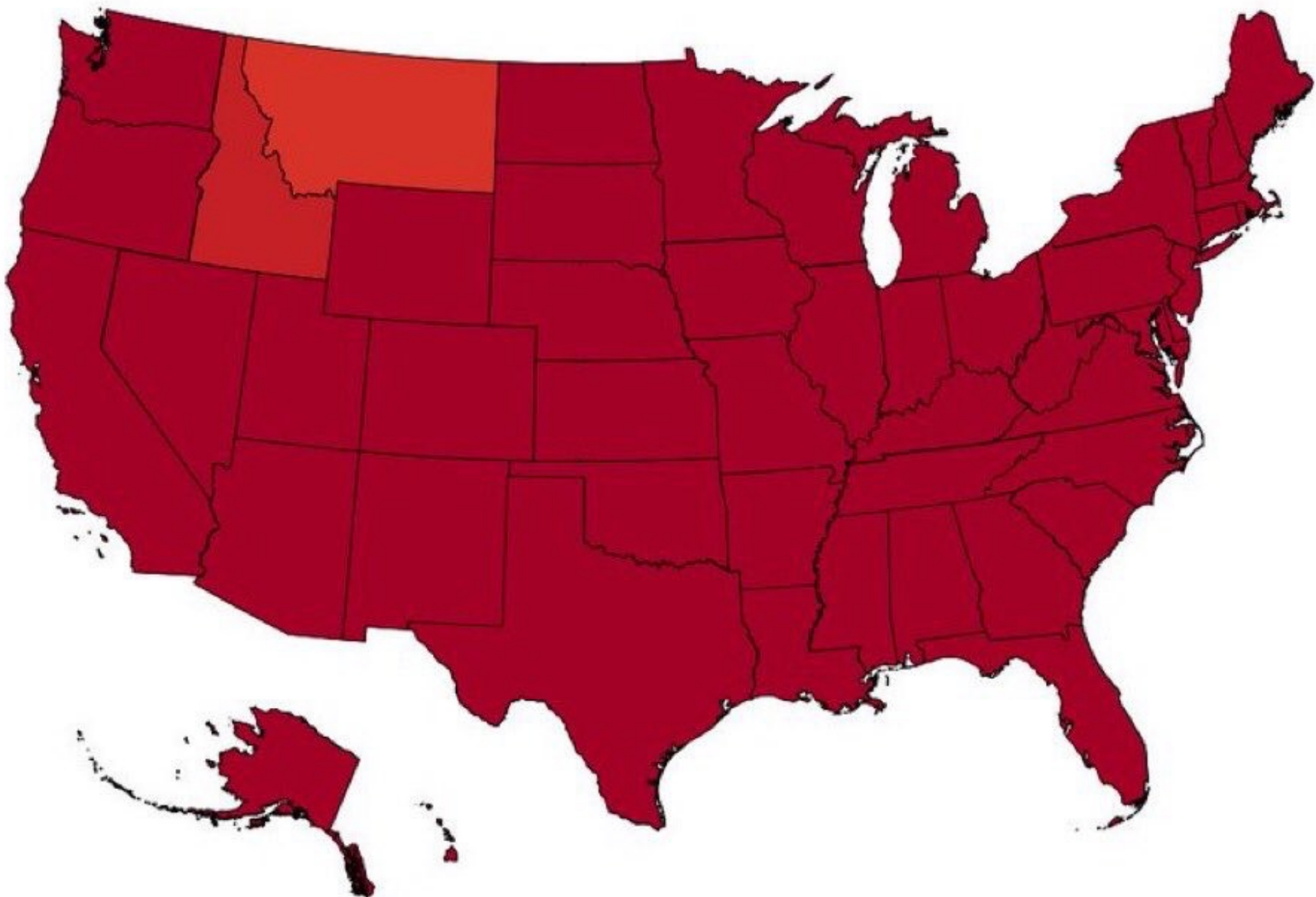
For choice of colors use tools like color brewer



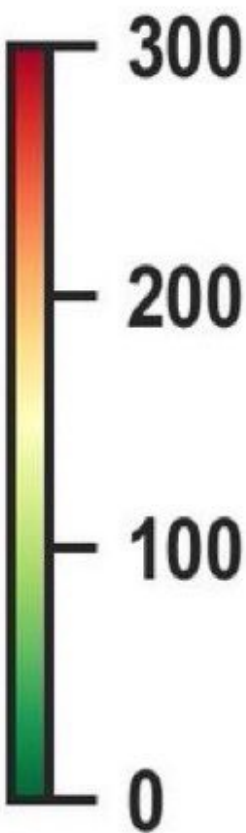
Example: Wrong use of colors and scales

How the year started

New COVID-19 cases per million population

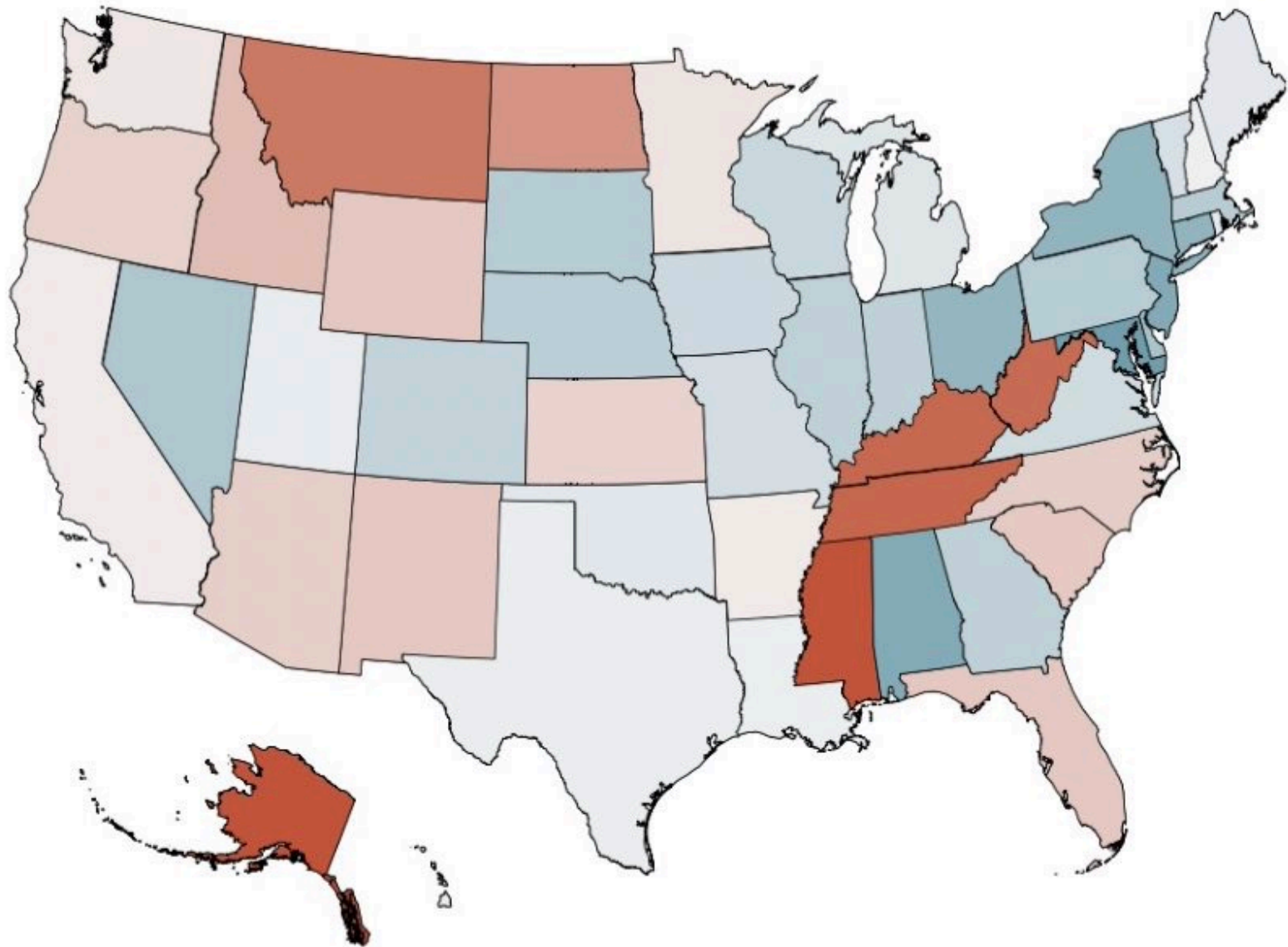


Source: Johns Hopkins, J. P. Morgan. As of December 31, 2021

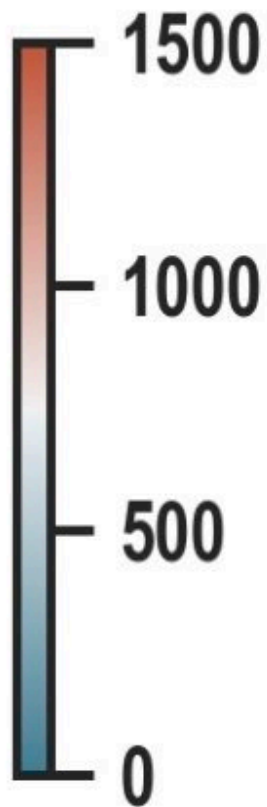


How it is going

New COVID-19 cases per million population

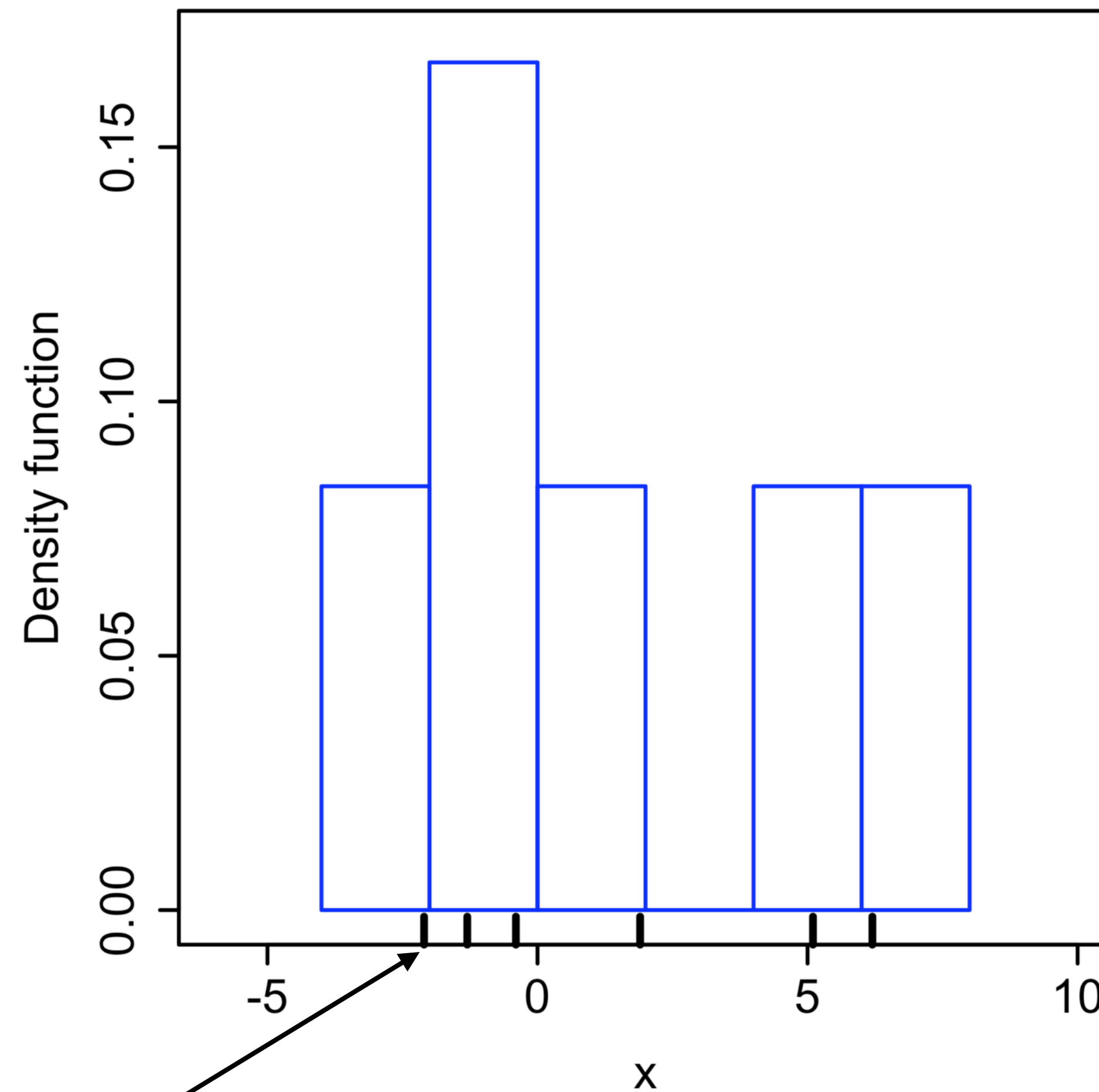


Source: Johns Hopkins, J. P. Morgan. As of February 08, 2022



Kernel density estimation (KDE)

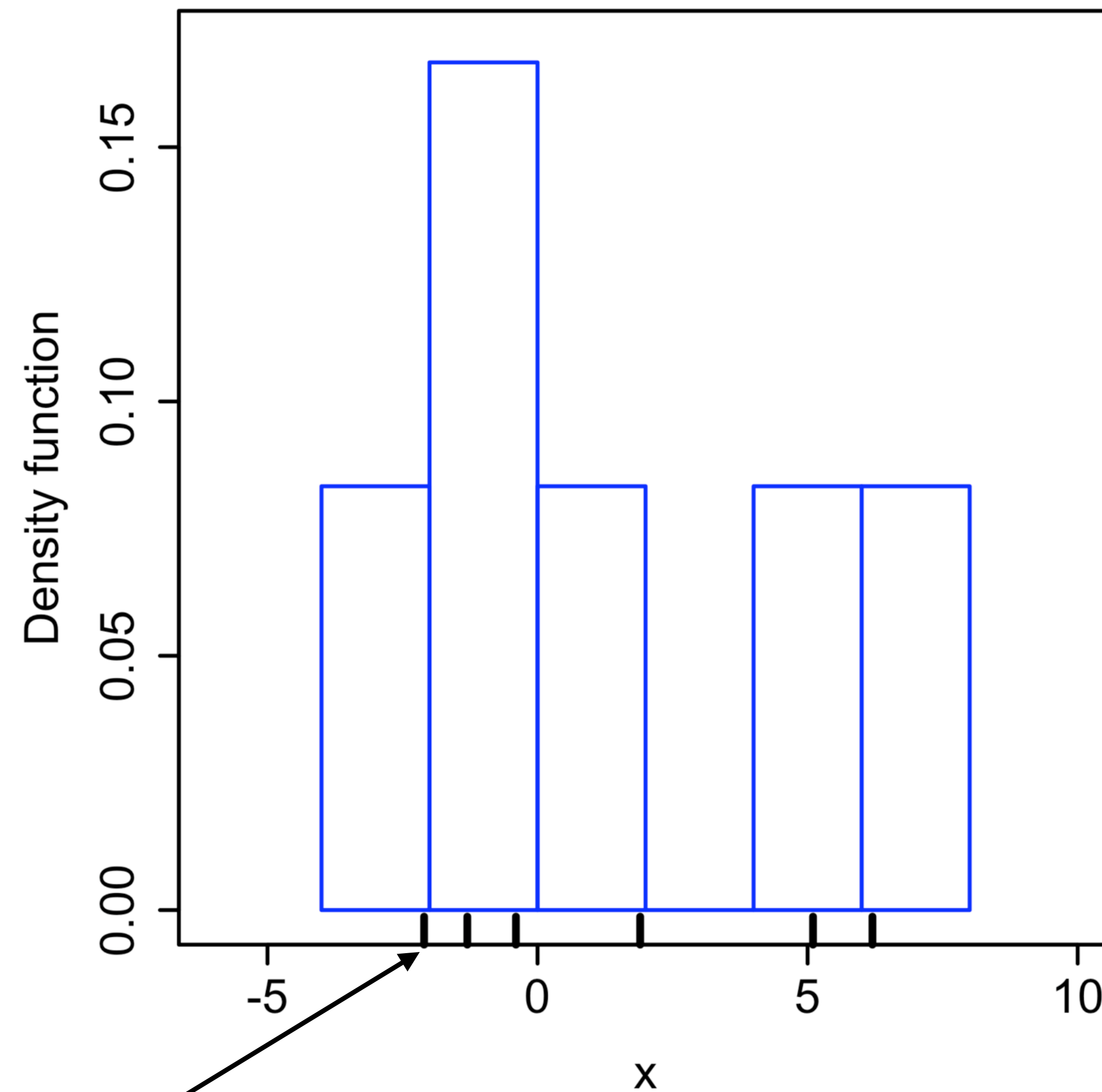
Histogram



Rug plot

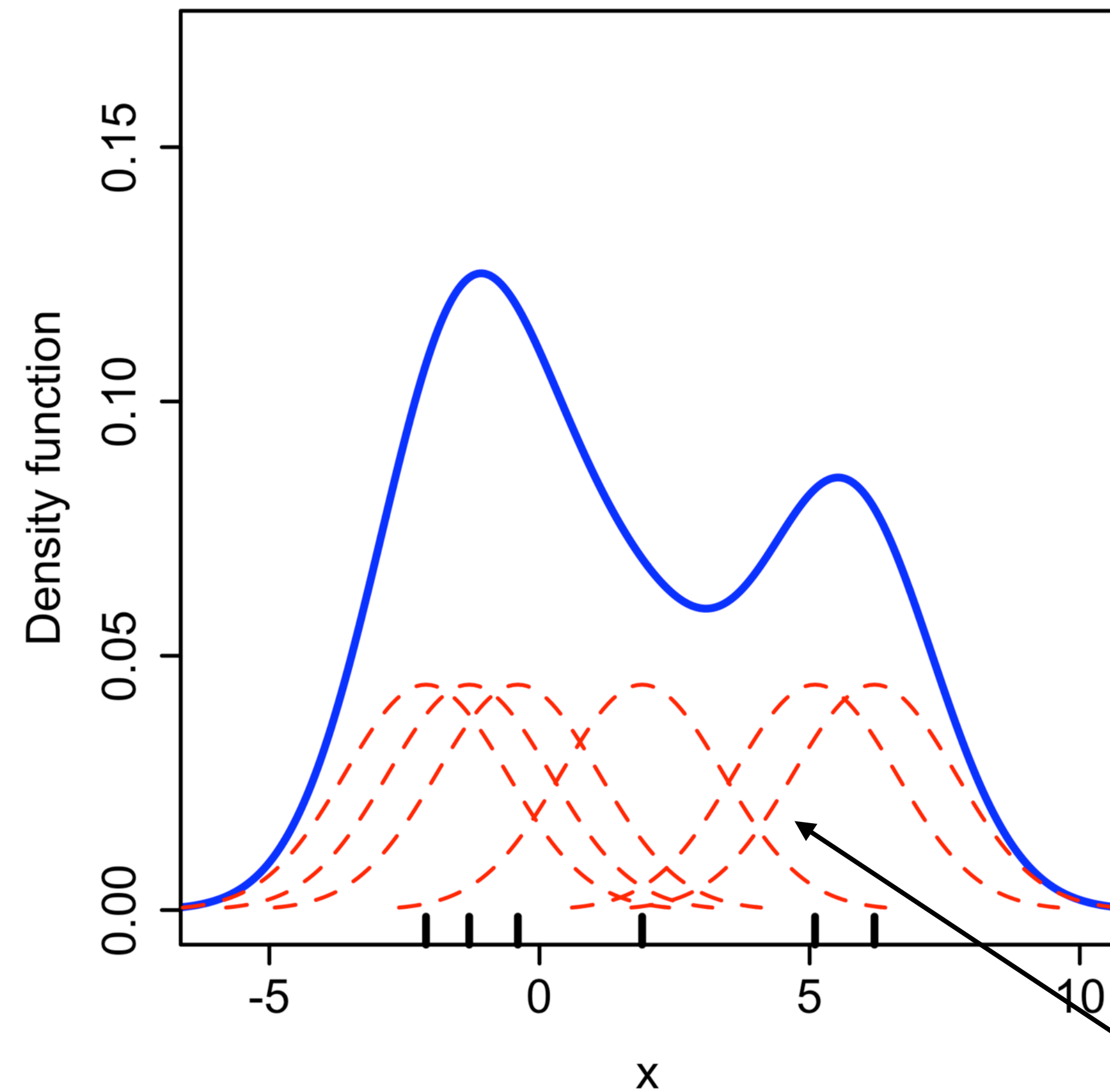
Kernel density estimation creates a smooth pdf to data points

Histogram



Rug plot

KDE



Kernel

Jupyter

Sources and further materials for today's class



Geographic Data Science with Python

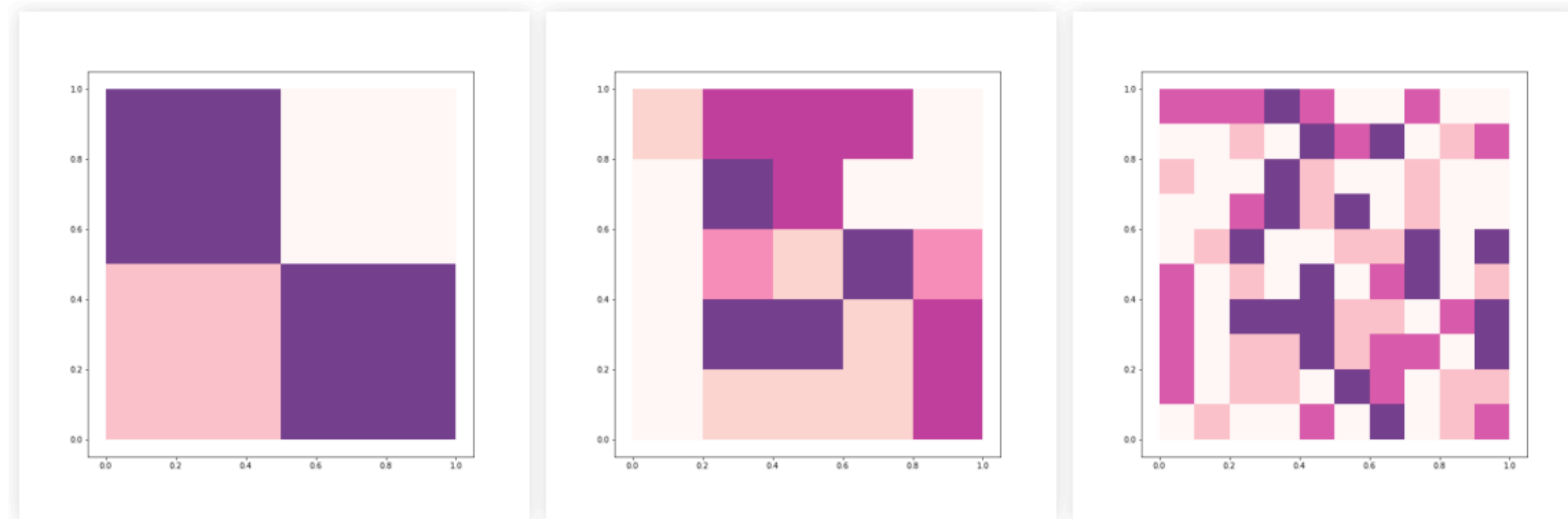


https://geographicdata.science/book/notebooks/05_choropleth.html

https://darribas.org/gds_course/content/bD/concepts_D.html

Take home messages for today

The choice of scale (MAUP) and classification scheme allows to create *arbitrary* interpretations



Aggregated data is simpler,
but can be very biased

Next week: Big Spatial Data

